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Standard Application SRS

Common Software Requirements Specification for Any Application

Application-agnostic edition · Version 1.9 · October 2026

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) — the “Standard Application SRS” — defines the functional and non-functional requirements that are common to every application, irrespective of the business domain it addresses, the team that builds it or the product family it belongs to. It specifies the platform layer of an application — tenancy, access control, configuration, governed AI, process design, generated documents, data entry, navigation, visual design and operations — and leaves the domain layer to each application.

This document is intended to serve as the baseline specification a new application is built against, and as the reference against which an existing application's platform layer (as distinct from its domain-specific modules) is assessed for conformance. It does not replace an application's own SRS: each application keeps its own specification for its domain-specific functional requirements — its principal record types and their lifecycle, its domain modules and its methodological content — and states there how it meets, extends or departs from this standard.

The standard was first derived by comparing the SRS documents of two applications serving different business domains and extracting the requirements, architecture patterns and governance mechanisms that recurred identically, or near-identically, across both. Each later revision added capabilities first built and verified in a reference implementation, or received as user feedback on one, and retained them only once they had been restated so that any application can meet them.

Revision 1.1 extends the standard with platform capabilities first built and verified in a reference implementation, where they proved independent of that application's business domain — for example the pinnable, repositionable navigation menu, internal benchmarking, and in-place upgrade and backup. Only requirements applicable to any application were retained; the reference implementation's own lifecycle and product-specific content remain in its own documentation. Appendix B lists every change made in this revision.

Revision 1.2 adds a second set of capabilities proven in a reference implementation that any application can reuse: the tenancy tree (Group Yes/No > Organization > principal records) and the provisioning of new organizations, the portfolio overview matrix, a stage checklist library, multi-format attachments, AI shown and run where it applies in the process design, the choice of the live AI model, a full-screen BPMN workspace, on-screen legends and a decision matrix for scored fields. Appendix C lists these changes.

Revision 1.3 integrates the application-agnostic requirements for the Vertical and SME layers — the vertical framework, vertical processes, data, compliance, integration and configuration, the SME mode with its tracks, complexity scoring, onboarding, packaging and pricing, and their combination — restated as application requirements. It also adds, for any application, a project template catalog (universal or per vertical, Full or SME mode) with full CRUD, three ways to create a project (from the catalog, manually, or with AI), a Checklist Library whose checklists may or may not be related to a gate, and a Gate Library to which checklists are associated; both are managed

with full CRUD and can be attached to any project phase. Appendix D lists the changes and Appendix E traces every source requirement family to its requirements here.

Revision 1.4 adds questionnaire and survey management, first specified in a change request (CR1, Scope of Work Definition & Questionnaire Management) and restated as a capability any application can reuse: versioned questionnaire templates tagged by focus, sector and language; elaboration by loading a template, by AI with mandatory human validation, or manually; RACSI allocation per section; distribution through Face-to-Face, Email, Hybrid and Custom channels with per-channel reminders; consent-gated, offline-capable response collection; and consolidation of complete responses into a tenant-scoped dataset that feeds the application’s downstream analysis.

Revision 1.5 adds capabilities first built and verified in a reference implementation of an integrated management system, together with product direction received with its user feedback, restated so that any application can reuse them: a global search reached from the navigation menu; full Create, Read, Update and Delete, with version history and the restore of any version, on every element of the process design (functions, end-to-end processes, phases, macro processes, tasks, steps, gates, checklists, forms and templates); an OBS made of units, roles and people, in which a role is linked to one or more functions and a person plays one or more roles; AI Use Cases mapped to the exact task or step they support, each with a prompt specification whose aspects (role, context, task, constraints, examples, format and others) are separate, versioned fields fully populated for that task or step; explicit naming and descriptions of process elements; typed step forms and a readiness checklist before a process starts; documented-information templates filled from the application's own data; versioned attachments; and an Organization-level language-model configuration. Appendix G lists every change.

Revision 1.6 adds what the second and third rounds of user feedback on that reference implementation established, restated for any application: generated documents that are consistent with the application's data and with each other, detailed and fine grained by document type, with full Create, Read, Update and Delete at every level (document, version, section, data-bound row, template and layout); data entry that is usable for real work (qualified field titles, controlled lists with a Custom value, a large row editor, scrollable dialogs, identifiers shown with their names, a search bar on every screen); records linked many to many, entered manually, from a library or by the AI, and propagated to registers and documents; graded audit and inspection findings; and transparent AI behavior (which engine answered and why, parameters adapted to each model). Appendix H lists every change.

Revision 1.7 aligns the standard with the AI Value Graphical Chart & Brand Guidelines, version 1.2 of 5 October 2026, which replaces the earlier product-family brand guide as the visual and interaction reference of every application. It turns the guidelines into verifiable requirements: design tokens as the single source of every color, type size, spacing, radius, shadow and motion value; a component library in which every state is designed; chart conventions with KPI tiles and chart export; multi-record data entry in inline editable tables; a focus contract that keeps the cursor where the user types; resizable layout panels; navigation and wayfinding; feedback and system states; motion; responsive breakpoints and density; right-to-left rules; and branded print, export and generated documents. Appendix I lists every change, Appendix J is the design token reference and Appendix K the UI release checklist.

Revision 1.7 also incorporates the fourth round of user feedback (1 October 2026), which established: project templates that are complete blueprints of a project with full CRUD at every level; document templates whose structure, formatting and pictures (including logos) are customizable; long lists shown by pages with a choice of page size; tables that scroll inside their own frame; inline multi-record entry with a keyboard map, duplication, undo, paste from a spreadsheet and autosave; resizable layout panels; a complete BPMN modeling palette; a focus contract and a page that never stays blank. These are reflected in Sections 4.33, 4.43, 4.49–4.53 and 5.13–5.14 of this document.

Revision 1.7 further absorbs the (New) SRS revision 1.7, which states the same revision-1.7 content under different section numbers (Section 4.0 and Appendix I.1 reconcile the two), and the Universal Field-Typing Fix, a whole-application remediation brief for typing-related focus-loss and input-integrity defects. The latter extends Section 5.14 with the complete catalog of the twenty-three known causes (C1 to C23), the fixes that are not accepted and the remediation audit an application undergoes, detailed in Appendix L.

Revision 1.8 makes the standard application agnostic. Every reference to a particular application, product family, reference build or source file has been removed or restated in general terms, so that any application, built by any team, can adopt the standard without change; no requirement was removed or renumbered and every requirement identifier keeps its meaning. Revision 1.8 also strengthens data entry in every Multi-Entry: a full-screen entry workspace with zoom controls (scroll, slider, zoom in and out, fit) in which several entries are typed, saved, changed and deleted with ease; the paste of complete records copied from Word, Excel or any other external file, as easily as a Word table is pasted into Excel; and a single action or the Escape key to return to the page. Appendix M lists every change.

Revision 1.9, this edition, comes from the acceptance of a reference implementation and keeps the standard application agnostic. It makes the AI Use Case Library fully governed and visible — a dedicated form and page for each use case, links to several process elements, the coverage of the process design, one governed path for every AI generation and a stated model precedence — and brings multi-record entry to large volumes and to real working conditions: validation only after a change, an issues list, undo and redo, browser drafts, edit conflicts, an offline queue, virtual rows, batched operations, a five-stage file import, export, a guard when leaving with unsaved rows and a phone layout. It also states rules learned in the same acceptance: language-independent references, a reopened step that reads as open everywhere, Office files that open without repair, data migrated in place on upgrade, and automated tests of multi-record entry.

1.2 Scope

In scope: the tenant and organizational hierarchy; identity and role-based access control; Configuration Management (Solution Packs, Add-Ons, Compliance & Security Standards); the governed AI Use Case Library; per-process/per-stage governance artifacts (Alerts, Business Rules, COSO-classified Controls, Risks & Opportunities, Custom KPIs, RACSI accountability); a Standards/Knowledge Base with RAG grounding; BPMN 2.0 process modeling; standard reports and data export; an in-app Notification Center; an AI Assistant / chatbot offering data-query and application-help modes; an audited, justification-governed change log; an in-app Help guide; template libraries; multi-language localization including Arabic RTL; and the non-functional

requirements (performance, security, reliability, usability, portability, maintainability, compatibility, data retention) these capabilities are held to.

Revision 1.1 adds to this scope: a personalizable navigation shell (a menu pinned open or sliding away, docked on any edge of the screen, with favorites that follow the user); internal benchmarking within an Organization and across the Organizations of one Group; and platform operations (scheduled backup with retention, additive startup migrations, startup guards, a health endpoint, a reproducible demonstration seed, and continuous integration across operating systems).

Revision 1.4 adds to this scope: questionnaire and survey management — questionnaire templates and their library, questionnaire elaboration, validation and RACSI allocation, multi-channel distribution and reminders, consent-based and offline response collection, completeness validation, consolidation and response reporting.

Revision 1.5 adds to this scope: global search; process design management with version history and restore; the OBS role model (functions, roles and people); AI Use Case mapping and prompt specification; process element naming and descriptions; typed step forms and process readiness; documented-information templates and generated documents; versioned attachments.

Revision 1.6 adds to this scope: consistency and fine-grained content of generated documents; full CRUD on documents, their versions, sections and data-bound content; data entry usability; linked records, choice libraries and register propagation; audit and inspection programmes with graded findings; AI engine transparency and model capability profiles.

Revision 1.7 adds to this scope: the visual design system (design tokens, color, typography, spacing, shape, elevation, iconography and components); charts and data visualization; multi-record data entry; focus and input integrity; resizable layout panels; navigation and wayfinding; feedback and system states; motion; responsive breakpoints and density; branded print, export and generated documents. Brand assets and channel set-ups that are not part of an application (logo artwork, email signatures, Outlook, Teams, SharePoint, Power BI and Figma configuration) remain the authority of the AI Value Graphical Chart; this standard requires only what an application and the documents it generates must show.

Revision 1.7 also adds to this scope, from the (New) SRS revision 1.7: project template blueprints; document formatting and pictures; data tables and pagination; inline multi-record entry; resizable layout panels and navigation robustness; the BPMN modeling palette; design tokens and the focus contract of the brand graphical chart. From the Universal Field-Typing Fix it adds the input-integrity defect catalog and the remediation audit of typeable fields.

Revision 1.8 adds to this scope: the full-screen entry workspace of every Multi-Entry, with its zoom controls and its return to the page; the paste of complete records from word-processing, spreadsheet and other external files, with a paste preview and column mapping; and the copy of records out to those files.

Revision 1.9 adds to this scope: the AI Use Case page, form, links and coverage view, and the single governed path of every AI generation with its model precedence; for multi-record entry, error finding, the save-state and draft model, browser drafts and edit conflicts, the offline queue, virtual rows and optional pages, batched bulk operations, file import and export, the unsaved-changes

guard, readability without horizontal scrolling and the phone layout; step forms chosen by the meaning of the step; language-independent references; valid Office files; in-place data migration on upgrade.

Out of scope: any requirement specific to a single application's business domain — its principal record types and their lifecycle, its domain modules, its methodological frameworks and its analysis or root-cause tooling. Those remain the authority of that application's own SRS.

Also out of scope: external benchmarking — any comparison with an organization outside the requester's Group, or with third-party or industry data.

1.3 Intended Audience

Product owners defining a new application; architects and developers building or rebuilding an application's platform layer; quality assurance engineers verifying platform-level conformance; and auditors assessing governance, security, and compliance posture across the applications of an organization.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
Application	Any software application, whatever its business domain, built to the platform pattern described in this document. Also called a conforming application.
Solution Pack	One of three commercial tiers defining an Organization's feature and quota entitlement matrix.
Add-On	A priced, independently toggleable capability that grants a feature to a Pack not already including it standard.
Compliance & Security Standard	An independently priced/toggleable module (e.g. GDPR, ISO/IEC 27001, SOC 2) that seeds GRC scaffolding; explicitly not a certification.
AI Use Case	A governed, versioned catalog entry describing one AI-assisted capability, tiered Assistive or Augmented (never Autonomous).
RAG	Retrieval-Augmented Generation — search over a knowledge corpus (TF-IDF/cosine-similarity or equivalent) grounding an AI suggestion or chatbot answer in the tenant's own data or a static reference corpus, without requiring a live external LLM call.
OBS	Organizational Breakdown Structure — the site/department/team or resourcing tree other governance modules reference for ownership and assignment.
RACSI	Responsible, Accountable, Consulted, Support, Informed — a five-role accountability model; exactly one Accountable per activity.
COSO	Committee of Sponsoring Organizations of the Treadway Commission — the internal-control framework used to classify Controls.
BPMN	Business Process Model and Notation — the OMG 2.0 standard used to model process workflows in-browser.

Term	Definition
RBAC	Role-Based Access Control — access rights determined by one or more assigned roles, mediated by a runtime-editable Permission Matrix.
Justification Governance	The stage-then-justify pattern: a practitioner stages a new value, records a justification note, and only then commits the change, appended atomically to an audit/change log.
Multi-Tenant	Architecture in which multiple Organizations share one deployment with isolated data.
Work Breakdown Structure (WBS)	A hierarchical decomposition of any grouping of Actions into a tree of WBS nodes, used as the basis for Gantt scheduling.
Gantt Chart	A timeline rendering of a WBS, plotting each node/Action's planned dates and dependencies.
Version Management	The platform's generic service for versioning any registered entity type: immutable history, no destructive edits, and revert-by-duplication to a new Current version.
Multi-Channel Communication	Delivery of a notification or alert through one or more channels (in-app, email, SMS, push, webhook) behind a single shared dispatch interface.
Mobile Compatibility	A responsive layout, meeting WCAG 2.1 AA touch-target sizing, that keeps an application's core workflows fully usable on a mobile viewport without a separate native application.
Return on Experience (REX)	A structured lessons-learned entry captured at the closure of a Project or an Action — what went well, what didn't, root cause, recommendation — indexed for future retrieval.
External Integration	A registered, independently toggleable connection from the platform to an external application, configured through the External Integration Registry (Section 3.3).
Navigation Shell	The frame shared by every authenticated screen: the primary menu, the header bar (context switcher, language, notifications, AI Assistant, user menu), and the content area.
Pinned / Sliding Menu	The two menu modes. Pinned, the menu stays open and the content reflows beside it. Unpinned, it collapses to an edge handle and slides over the content when pointed at, focused, or toggled.
Dock Position	The screen edge — left, right, top, or bottom — a user chooses for the primary menu; a per-user preference stored server-side.
Internal Benchmarking	Comparison of standard metrics between segments of one Organization, or between Organizations of the same Group, using aggregates only. External benchmarking is out of scope.
Minimum Sample Size	The smallest number of records a benchmark segment or Organization must contain to be ranked; smaller samples are shown as not comparable.
Additive Migration	A change applied at startup that only adds tables, columns, permission codes, or default grants, and never drops or rewrites existing data or an administrator's edits.

Term	Definition
Tenancy Tree	The view Group (Yes/No) > Organization > principal records, limited to what the viewer may see; an Organization with no Group appears as Independent.
Provisioning	Everything a new Organization receives on creation: OBS skeleton, standard governance catalog, AI Use Case Library, checklist and record templates, and optionally a starting team.
Portfolio Overview	A matrix of principal records (rows) against process stages (columns) whose cells show the status of each stage, for a chosen scope.
Checklist Template	A reusable, versioned list of checklist items kept in the Checklist Library. It stands on its own and may be associated with zero, one or several gates of the Gate Library; when associated, it is copied into the gate checklist when the phase opens.
Vertical	An industry sector (for example Automotive, Food & Beverage, Public Education) defined as a first-class, versioned configuration entity that activates its own macro processes, E2E processes, data extensions, compliance mappings, checklists and gate criteria.
Vertical Instance	The content of one vertical — its prefix, macro and E2E processes, data model, compliance, integration, configuration and governance — delivered as configuration data, not as platform code.
SME	A small or medium-sized enterprise, by default 1–250 employees (EU) or up to 500 (US SBA), thresholds configurable per region, segmented into Micro (1–10), Small (10–50) and Medium (50–250).
Mode (Full / SME)	The rigor profile of a project: Full mode runs the complete process set; SME mode runs a right-sized SME track.
SME Track	One of two to four configurable levels of process rigor for SME mode, each defining its macro and E2E processes, gates, checklist items per gate and typical duration.
Complexity Score	A weighted score of configurable criteria (weights summing to 100 %), including vertical-specific drivers, that recommends the track of a project.
Project Template Catalog	The governed, versioned library of project templates, each classified by scope (universal or one vertical) and mode (Full or SME, with its track).
Creation Mode	The way a project is created: from the catalog, manually, or with AI (an AI-drafted project reviewed and accepted by a person).
Phase	A stage of a project's process chain (for example an E2E process or a lifecycle stage) to which gates and checklists are attached.
Gate Definition	A reusable decision point — name, criteria, approvers, decision options Go / No-Go / Hold / Recycle — kept in the Gate Library, with zero or more associated checklists, and attachable to any phase.
Pack / Bundle	A priced commercial package: an SME pack, a vertical pack, or a bundle of both.

Term	Definition
Questionnaire Template	A versioned, reusable questionnaire — sections and questions — tagged by focus, sector and language, from which a questionnaire is loaded.
Focus	The thematic lens of a questionnaire or engagement: All needs, Digital, AI, or a named Custom focus.
Distribution Channel	The way a questionnaire reaches a respondent: Face-to-Face (with offline capture), Email, Hybrid, or a Custom channel.
Stakeholder	A person registered as a respondent for an engagement, with role, function, decision level, preferred channel and consent status.
Function	A business function of the Organization — for example Quality, Health & Safety, Finance, Human Resources, Operations — to which roles, processes and governance items are linked. Functions are a versioned catalog per Organization.
OBS Role	A position defined in an OBS unit (for example Quality Manager of the head office), linked to one or more functions and played by one or more people. An OBS Role describes what a person does in the organization; it is distinct from an access role of RBAC.
Role Assignment	The fact that a person plays an OBS Role, with a start date, an optional end date, an allocation percentage and a holder type (holder, deputy or acting). A person may hold several Role Assignments at once.
Process Design	The set of typed elements that define how the application's work is done: functions, end-to-end processes and their phases, macro processes, tasks, steps, gates, checklists, step forms, templates and the AI Use Cases linked to them.
Process Element	Any one item of the Process Design, held as data, with a stable identifier and a version history.
Design Release	A named, immutable snapshot of the Process Design, or of one function or process within it, that can be published, compared with another release and restored.
Current Version	The version of a versioned entity that the application uses for new work. Every earlier version stays viewable and can be restored, which creates a new Current Version with its content.
Prompt Specification	The structured prompt of an AI Use Case: Role (persona), Context, Task (instruction), Inputs, Knowledge sources, Constraints, Examples, Output format, Tone and language, Quality criteria, Human checkpoint and Model parameters, each held in its own field and versioned.
Step Form Kind	The type of input form a step uses — for example record table, assessment matrix, decision, objectives, plan, monitoring, review, assignment, document — which decides its fields, its checks and the records its completion creates.

Term	Definition
Readiness Checklist	An optional list, shown before a process starts, of what should be in place: inputs, predecessor processes, owner and RACSI, templates, KPIs. Items the system can verify are checked automatically; the list never blocks the start.
Documented Information Template	A versioned template for a controlled document or record (policy, procedure, register, report), made of typed sections — text, table, approval block — each filled from a named data source of the application.
Global Search	One search entry, in the navigation menu, that finds every record, process element, governance item, document and Help topic the user may open.
Document Profile	The minimum content a generated document of a given type must carry — identification block, purpose and scope, method or rating scales, data sections, analysis, actions, revision history, approval — and the checks applied before it is published.
Data Source Binding	The link between a section of a template or document and a named data source of the application (a register, a list of records, the values of a step), with its filters and sort order.
Consistency Check	An automatic verification, run on a generated document and across the documents of a project, that every value, identifier and reference agrees with the application's records and with the other documents.
Section Override	Text or a value typed in a document section that replaces, for that document only, what the data source provides; it is marked as such and kept across regenerations.
Edit-through	Changing a row of a data-bound table from within a document, the change being written to the source record (with its rights, checks and audit) rather than to the document.
Controlled List	A list of proposed values for a field (for example categories, types, parties), managed as configuration data, always offering a Custom value typed by the user.
Choice Library	The values of a Controlled List together with the values users have already typed in the Organization, offered again the next time the field is filled.
Entry Origin	How a record was entered: typed by the user (Manual), taken from a library (Library) or suggested by the AI and accepted by a person (AI).
Record Link	A many-to-many link between two records (for example a need and the interested parties it concerns), each end chosen from the records already identified or typed as a new value.
Register Propagation	The creation or update of register entries from the rows of a completed step, each row keeping the identifier of the entry it produced.
Row Editor	A large window that edits one row of a record table field by field, with scrolling, previous and next row, and a return to the table.

Term	Definition
Engine Transparency	The display, with every AI answer, of the engine that produced it (live model or built-in engine) and, when the live model was not used, the reason.
Design Token	A named design value (color, type, spacing, radius, shadow, focus ring, motion, breakpoint, layer) defined once, prefixed --aiv-, and referenced by every component instead of a raw value. Appendix J lists them.
Status Scale	The five semantic status colors, always shown in the order Critical, Emerging, Developing, Managed, Optimized (red to green), used only for maturity, health, audit and severity, never for decoration.
Component State	One of the designed appearances of an interactive component: default, hover, focus-visible, active, disabled, loading, error and success.
Inline Editable Table	A record table in which one row is one record and each cell is edited in place, with keyboard navigation, row status, row actions and per-row autosave; the default pattern for records of two to eight fields.
Master–Detail	A list of records beside an editor pane for the selected record; the pattern for records of more than eight fields.
Focus Contract	The rule that, while a user types in or operates a field, keyboard focus stays in that field whatever the application does in the background (validation, saving, re-rendering, panel change).
Layout Panel	One of the three horizontal regions of the application shell: navigation (start), content (center) and context (end). The navigation and context panels are resizable; the content panel fills the remaining width. Also called Resizable Panel.
Density Mode	The spacing profile of a table or data-heavy view: Comfortable (56-pixel rows) or Compact (40-pixel rows), chosen and kept per user. Also called Density in the (New) SRS.
Command Palette	The keyboard-first overlay opened with Ctrl+K (Cmd+K on macOS) that runs the global search and jumps to screens and actions.
KPI Tile	A card showing one indicator: label, value, change since the previous period and an optional trend line.
System State	A designed screen or component appearance for loading, empty, error, success, offline or permission situations, with the message and the action the user needs.
Project Template Blueprint	Everything a project created from a template contains: its end-to-end processes, macro processes, tasks and steps, gates, business rules, controls, risks and opportunities, alerts, KPIs and reporting plan.
Custom Element	A phase, macro process or step a template adds to the reference process design; it exists only in projects created from that template.
Reporting Plan	The reports a project produces — name, audience, frequency, format and owner — defined by its template.

Term	Definition
Document Formatting	Fonts and sizes, alignment, orientation, margins, cover page, table of contents, numbering, page numbers, colors, logo and its position, header and footer of a document template.
Section Formatting	Alignment, text size and color, background (callout), bold, italic, page break before and hidden title of one document section.
Picture Section	A section showing an uploaded picture or logo, with a caption, a width as a percentage of the page and an alignment.
Page Size	The number of records a long list shows at a time: 20 by default, 50, 100 or all.
Active Draft	The value a user is editing in a field, row or form; it takes precedence over saved, derived or server state until an explicit lifecycle boundary (leaving the field, save-and-close, cancel, discard).
Input-Integrity Defect	A defect that takes a field away from the user while typing: the cursor jumps, characters are dropped, a value reverts or resets, a row disappears or is replaced, a dialog, drawer or form is re-created, or focus is lost on a control. Its known causes are C1 to C23 (Appendix L).
Typeable Field Inventory	The list, built at the start of a remediation audit, of every screen and shared component in which a user types or selects a value, including custom and third-party controls.
Multi-Entry	Any component in which a user creates, edits, duplicates or deletes several records of the same kind on one screen: an Inline Editable Table, a Master–Detail list, the record table of a step form, a register, a checklist and its items, a RACSI editor, a decision or assessment matrix, the questions of a questionnaire, the lists of a project template blueprint, or a data-bound table edited through a document.
Full-Screen Entry Workspace	A view of one Multi-Entry that fills the whole screen, with a fixed header, Zoom Controls and every entry action of the page, opened by one action and left by one action or the Escape key. Also called the entry workspace.
Zoom Controls	The controls that scale the records area of the Full-Screen Entry Workspace: Ctrl/Cmd + scroll wheel or pinch, a zoom slider, Zoom in, Zoom out, Fit and Reset to 100 %, with the current level shown as a percentage.
Structured Paste	The paste of one or several complete records copied from a word-processing document, a spreadsheet, a CSV or text file, a web page or another application, each copied row becoming one record and each copied column one field, as when a Word table is pasted into Excel.
Paste Preview	The view shown before a Structured Paste is applied when its columns, header row or overwritten values need confirmation: the column-to-field mapping, the rows to add and to overwrite and the invalid cells, with Apply, Adjust and Cancel.

Term	Definition
AI Use Case Link	The link of an AI Use Case to a process element it serves: its primary step, or a further step, task, macro process or end-to-end process. A use case serves every step reached through its links.
Coverage of the Process Design	The list of every task of the process design with the AI Use Cases that support it, showing the tasks that have none.
Model Precedence	The fixed order in which the model of an AI generation is chosen: the model of the AI Use Case, the Organization-level live model, the user's personal connection, then the built-in engine.
Draft Backup	The copy, kept in the user's browser, of the rows of a Multi-Entry not yet saved, with the time it was taken and the server stamp each row was based on.
Row Stamp	The server version marker of a saved row (for example its last update time), sent back with a change so that a change based on an older version is detected.
Edit Conflict	A change based on an older version of a row than the one stored, or a restored draft row whose server record changed or disappeared since the draft was taken.
Batch Operation	A bulk paste, import, update or delete sent in successive batches of rows, with its progress, cancellation and resumption.
Import Session	The five stages of a file import — template, upload, column mapping, error repair, summary — from the chosen file to the committed rows.
Leave Guard	The question asked when the user leaves a page that holds rows not saved: save and leave, discard and leave, or stay.

1.5 References

- AI Value — Graphical Chart & Brand Guidelines, Version 1.2, 5 October 2026 (Sections 3 to 25). The visual and interaction reference of this standard from revision 1.7.
- Application-Agnostic Requirements for Verticals and SME — Vertical and SME layers of any PLM or PSLM platform (29 verticals, SME framework, Vertical × SME intersection, non-functional requirements, traceability and requirements governance), 2026.
- Change request CR1 — Addition of Scope of Work Definition & Questionnaire Management, September 2026.
- Universal Field-Typing Fix — Whole Application: remediation brief for typing-related focus-loss and input-integrity defects (causes C1 to C23), October 2026.
- Standard Application SRS, revision 1.7, and its alternate-numbering edition, the (New) SRS revision 1.7 — predecessors of this edition. Section 4.0 and Appendix I.1 map the section numbers of the latter.
- Source application SRS documents, reference implementations and user-feedback rounds (August to October 2026) from which revisions 1.0 to 1.7 were derived. They are not normative: this edition restates everything it retained from them in application-agnostic terms.

- ISO/IEC/IEEE 29148 — Systems and software engineering — Life cycle processes — Requirements engineering.
- Object Management Group — Business Process Model and Notation (BPMN), Version 2.0.
- W3C — Web Content Accessibility Guidelines (WCAG) 2.1 and 2.2.
- COSO — Internal Control — Integrated Framework.

1.6 Document Overview

Section 2 describes a conforming application at a high level and states the design principles every application is expected to follow. Section 3 defines the common platform architecture. Section 4 enumerates the common functional requirements, grouped by shared capability and individually identified (FR-DA-xxx) for traceability. Section 5 defines common non-functional requirements (NFR-DA-xxx). Section 6 summarizes the common data-model entities. Section 7 defines common external interface conventions. Appendix A holds the conformance record each application keeps against this standard, so that a gap is explicit rather than silently absorbed, and the rules that resolve differences between this standard and the Graphical Chart.

Revision 1.1 adds Sections 3.11 to 3.13 and 4.19 to 4.21, adds requirements to existing sections under new, never-reused IDs, and adds Appendix B, the change log of this revision. No 1.0 requirement was removed or renumbered.

Revision 1.2 adds Sections 4.22 to 4.24, new requirements in Sections 4.1, 4.4, 4.7, 4.12, 4.19, 4.21 and 5.5, and Appendix C, the change log of revision 1.2. As in 1.1, nothing was removed or renumbered.

Revision 1.3 adds Section 3.14, Sections 4.25 to 4.35, new requirements in Sections 4.13, 5.1 to 5.6, 5.8 and 5.10, and Appendices D (change log) and E (traceability to the Verticals and SME source). It renames Section 4.23 Checklist Library and rewords FR-DA-CHK-01 to 05 so that a checklist no longer has to belong to a gate; Section 4.35, the Gate Library, holds the gates and their associated checklists. Nothing was removed or renumbered.

Revision 1.4 adds Section 4.36 Questionnaire & Survey Management and Section 5.12 Questionnaire Data, new requirements in Sections 4.13 and 4.17, five entities in Section 6, 4 terms in Section 1.4, and Appendix F, the change log of revision 1.4. Nothing was removed or renumbered.

Revision 1.5 adds Sections 3.15 to 3.18 and 4.37 to 4.43, new requirements in Sections 4.4, 4.6, 4.16, 4.19, 4.24, 5.1, 5.2, 5.3, 5.5 and 5.8, 12 terms in Section 1.4, 16 entities in Section 6, and Appendix G, the change log of revision 1.5. Nothing was removed or renumbered. NFR-DA-SEC-07 still governs a personal live-model connection; NFR-DA-SEC-17 adds the rules for an Organization-level connection stored encrypted on the server (FR-DA-AI-16).

Revision 1.6 adds Sections 4.44 to 4.48, new requirements in Sections 4.4, 4.19, 4.40, 5.1, 5.2, 5.3, 5.5 and 5.8, 12 terms in Section 1.4, 12 entities in Section 6, rows in Appendix A and Appendix H, the change log of revision 1.6. Nothing was removed or renumbered; FR-DA-DOC-01 to 11 (Section 4.43) remain the base that Sections 4.44 and 4.45 detail.

Revision 1.7 adds Sections 4.49 to 4.53, 5.13 and 5.14, new requirements in Sections 4.19, 5.5 and 5.6, 11 terms in Section 1.4, a design principle in Section 2.2, 2 entities in Section 6, rows in

Appendix A, Appendix I, the change log of revision 1.7, Appendix J, the design token reference, and Appendix K, the UI release checklist. The document itself now follows the AI Value Graphical Chart (Montserrat headings, Open Sans body, Navy and Ink text on light surfaces, tables with a Navy header row) and opens with a generated, linked table of contents. Nothing was removed or renumbered; NFR-DA-UX-01 remains the base requirement that Section 5.13 details.

Revision 1.7 also absorbs the (New) SRS revision 1.7, whose revision-1.7 content is renumbered as follows: its 4.49 → 4.33/4.49 here; 4.50 → 4.43/4.50; 4.51 → 4.49/4.51; 4.52 → 4.49; 4.53 → 4.50. The absorption adds Section 4.0, FR-DA-PTC-07 and 08 and FR-DA-PTB-01 to 10 in Section 4.33, FR-DA-DFP-01 to 08 in Section 4.43, FR-DA-TBL-01 to 07 in Section 4.51, the alias Section 4.53, NFR-DA-PERF-12, NFR-DA-SEC-20 and NFR-DA-REL-09 and 10, 7 terms in Section 1.4, 9 entities and an entity-alias table in Section 6, and Appendix I.1, the numbering reconciliation. The Universal Field-Typing Fix adds NFR-DA-FOC-06 to 13 in Section 5.14, 3 terms in Section 1.4 and Appendix L, the input-integrity defect catalog and remediation audit. Nothing was removed or renumbered.

Revision 1.8 is editorial for applicability and additive for multi-record entry. It removes every reference to a particular application or product family (cover, Sections 1 to 7, Appendices A to I and K), rewrites Appendix A as an application-neutral conformance record, formalizes the multi-currency statement of Section 4.14 as FR-DA-I18N-09, and adds FR-DA-MRE-15 to 23 in Section 4.49, NFR-DA-PERF-13, NFR-DA-SEC-21 and NFR-DA-UX-20, 5 terms in Section 1.4, 1 product function in Section 2.3, 1 entity in Section 6, rows in Table 4.49-1 and Appendix K, and Appendix M, the change log of revision 1.8. Nothing was removed or renumbered.

Revision 1.9 adds FR-DA-AI-21 to 26, FR-DA-MRE-24 to 35, FR-DA-SFM-08 and 09, FR-DA-I18N-10, FR-DA-DOC-12, NFR-DA-PERF-14, NFR-DA-SEC-22, NFR-DA-REL-11, NFR-DA-UX-21, NFR-DA-I18N-07, NFR-DA-MAINT-11 and NFR-DA-PORT-05; amends FR-DA-AI-01, 07 and 13, FR-DA-AIP-07, FR-DA-MRE-05, 08, 11, 14, 18 and 22, NFR-DA-REL-10 and NFR-DA-MAINT-04, and Table 4.49-1; adds 9 terms in Section 1.4, 3 entities and 2 amended entities in Section 6, one paragraph in Sections 3.4 and 7.1, 2 interface families in Section 7.2, rows in Appendices A and K, and Appendix N, the change log of revision 1.9. Nothing was removed or renumbered.

Requirement identifiers keep the FR-DA- and NFR-DA- prefixes of earlier revisions, so that every conformance record and traceability link written against an earlier revision remains valid. The prefix is a frozen identifier; it names no application.

2. Overall Description

2.1 Product Perspective

An application built to this standard is a self-contained, installable web application — a browser-based single-page frontend paired with a REST API backend — that can run standalone or, per NFR-DA-MAINT-01, expose its composable modules (Action/Task Register, Alert & Notification Engine, RBAC layer, KPI/Reporting engine, Audit Trail Service) for reuse by sibling applications behind a shared API contract. Every application is positioned as data- and AI-led: its deliverables are evidence-based records, structured workflows, and governed AI-assisted suggestions, never autonomous decisions.

2.2 Design Principles

- Explainable AI only. Every AI capability is tiered Assistive or Augmented; a third, Autonomous tier — an AI that takes an irreversible action without a human decision in the loop — is out of scope for every application.
- Tenant isolation by construction. Every data-bearing record is scoped to an Organization; no query, export, or AI retrieval may return another tenant's data.
- Commercial gating as a second, independent axis. Solution Pack and Add-On/Compliance entitlement gate feature and quota access alongside, never instead of, RBAC.
- Governance is structured, not narrative. Business rules, controls, risks, and accountability are captured as typed, queryable records — tagged back to a process or stage — rather than left in prose documents.
- One shared visual and interaction language. Every application, and every document it generates, conforms to the common visual identity (Section 5.13) (color palette, typography, iconography) across every module, output format, and supported language.
- Multilingual and RTL-first. English, French, and Arabic (full right-to-left layout) are first-class from the outset, not a later localization pass.
- One navigation shell, personal to each user. Every application shares one navigation shell; each user may pin the menu or let it slide away, dock it on any screen edge, and mark favorites, and these choices follow the user across sessions and devices.
- Compare within the family, never outside it. Benchmarking compares an Organization with itself and with its own Group, from aggregates only; it never exposes another tenant's records and never reaches beyond the Group.
- Upgrade and recover in place. A new release upgrades an existing installation at startup without data loss, and the data is backed up on a schedule the operator does not have to remember.
- One process design, versioned. Every element of the process design is data with a full history: it can be created, changed, compared and restored to any earlier version, and a running record keeps the version it started with.
- Accountability through roles, not names. Work is assigned to OBS roles, linked to functions and resolved to the people who play them, so a change of person never breaks an assignment.
- Explicit prompts. Every AI Use Case carries its full prompt as separate, reviewable, versioned fields, populated for the exact task or step it serves.

- One truth, many documents. A generated document never holds data of its own that contradicts the application: every value comes from a record, is traceable to it, and is the same in every screen and every document that shows it.
- Usable before complete. A field names what it expects, offers a list with a Custom value, gives room to type, and explains what is computed; the user never meets a generic "Item" or a box too small to read.
- Evidence-first, light and precise. Every screen and every document leads with the finding or the number, takes its colors, type, spacing and shapes from the shared design tokens rather than raw values, stays light and spacious, and never trades legibility, keyboard use or accessibility for decoration.

2.3 Product Functions (Summary)

- Register, resolve/manage, and evidence a structured record type (for example a case, a project, an assessment or a request) through a defined multi-stage lifecycle.
- Provide one or more rule-based AI agents grounded in a retrieval corpus, plus a chat-style AI Assistant offering RBAC-scoped data queries and application-help queries.
- Maintain a governed AI Use Case Library — a seeded catalog plus organization-authored custom entries, versioned, with Organization-level and Project-level activation.
- Maintain per-process/per-stage governance: Alerts, Business Rules, COSO-classified Controls, Risks & Opportunities, Custom KPIs, and a RACSI accountability matrix.
- Model process workflows as real BPMN 2.0 diagrams, importable and exportable as standard XML.
- Provide Configuration Management: Solution Packs, priced Add-Ons, Compliance & Security Standards, deployment options, and support tiers, each enforced wherever the platform gates a feature.
- Provide full role-based access control with a runtime-editable Permission Matrix, and an organizational hierarchy (Group/Organization/Project) with an OBS for people/department scoping.
- Compute standard KPIs plus Custom KPIs, and standard Reports exportable to PDF, Excel, and Word; support CSV export on high-traffic tables.
- Operate multi-tenant with data isolation, and multi-language with full Arabic RTL support.
- Provide a searchable, multi-language in-app Help guide covering every module.
- Provide a personalizable navigation shell: menu pinned or sliding, docked left, right, top, or bottom, with favorites, persisted per user.
- Provide internal benchmarking of standard metrics between segments of one Organization and between Organizations of the same Group.
- Protect and upgrade data in place: on-demand and daily backups with retention, and additive migrations at startup.
- Activate verticals and an SME mode by configuration, score project complexity to choose the right track, and start projects from a governed template catalog, by hand, or with AI, with gates and checklists attached to any phase.
- Design questionnaires from a versioned template library, by hand or with AI under human validation, distribute them across channels with reminders, and collect, validate and consolidate consented responses into a tenant-scoped dataset.

- Find any record, process element, document or Help topic the user may open from one search in the navigation menu.
- Manage the process design — functions, processes, phases, tasks, steps, gates, checklists, forms and templates — with full history, comparison and restore of any version.
- Model the organization as units, roles and people, with roles linked to functions and people playing one or more roles.
- Generate controlled documents and records from versioned templates filled with the application's own data, downloadable in Word, PDF and Excel.
- Generate detailed, consistent documents from the application's data, checked before publication and editable at every level — document, version, section and data-bound row — with full history.
- Capture records with qualified fields, controlled lists with Custom, large row editing and many-to-many links, entered manually, from a library or with the AI, and propagate them to registers and documents.
- Let users enter many records at once in a full-screen workspace with zoom controls, paste complete records copied from Word, Excel or any other file, and return to the page with one action or the Escape key.

2.4 User Classes and Characteristics

Every application defines its own domain-specific role names, but each role maps onto one of the following six baseline classes, which the common Permission Matrix (Section 4.2) is built around. The example role names are illustrative only.

Baseline Class	Characteristics	Example Mapping
Platform / Organization Administrator	Full configuration authority: Configuration Management, RBAC, hierarchy, AI Use Case governance, Compliance Standards.	Administrator; Super Admin; Group Admin; Organization Admin
Process Owner / Manager	Owns the governed process end-to-end: validates records, configures thresholds, owns Controls/Business Rules, full governance CRUD.	Process Owner; Quality Manager; Improvement Pilot; Change Manager
Team Contributor	Creates/edits records and actions within an assigned scope; completes assigned workflow stages.	Team Member; Practitioner; Contributor
Action Owner / Evaluator	Executes or verifies a specific assigned action; owner and evaluator of the same action must always differ.	Action Owner; Action Evaluator
Scoped Viewer / Sponsor	Department- or team-scoped visibility into records, dashboards, and governance items relevant to that scope.	Department Head; Sponsor; People Manager
Reporter / Read-Only / Auditor	Either submission-only creation with no post-submission edit rights, or full read-only access including the audit trail.	Reporter; Employee; Executive; Read-Only; Auditor

2.5 Operating Environment

- Server: a standard Node.js runtime (Node.js 18+ minimum; Node.js 22.13+ recommended — the first long-term-support line whose built-in SQLite module works without an experimental flag — so the platform can use the runtime's own SQLite module rather than a native-compiled dependency). The required version shall be declared in the package manifest.
- Client: any evergreen desktop, tablet, or mobile web browser, with a responsive layout adapting to each; no legacy-browser support is in scope.
- No mandatory external service dependency: an application must remain fully functional with no live third-party call, with the sole opt-in exception of a user-configured Real LLM Provider Connection (Section 4.3).
- Report-export and persistence libraries must be pure JavaScript / native-runtime, so installation succeeds without a native C/C++ build toolchain, a required container orchestrator, or an external database server.

2.6 Design and Implementation Constraints

- Every data access must be scoped by tenant (organization_id); no query may return another tenant's data.
- All AI agent behavior must be explainable and logged; no black-box autonomous decision-making.
- The UI, and every generated document, must conform to the common visual identity (Section 5.13) across every module, output format, and supported language.
- An action's Owner and its Evaluator, wherever the platform models a two-party verification step, must always be different users, enforced server-side.
- Solution Pack entitlement is a second, independent gating axis alongside RBAC, and every Pack quota must be checked server-side at the point of creation/activation.
- Every user-facing string must exist in a single translation dictionary for every supported language; no hardcoded UI string is acceptable outside it.
- Benchmark data leaving an Organization, even towards its own Group, must be aggregated; no record, name, or identifier of another tenant may be returned.

2.7 Assumptions and Dependencies

- Organizations are responsible for their own user provisioning within the tenant they administer.
- Rule-based AI agent implementations are a functional, explainable stand-in that a future LLM/vector-database-backed implementation can replace behind the same internal interface, without changing calling code.
- Compliance & Security Standard modules provide GRC scaffolding only; an Organization remains responsible for its own compliance program and any external certification.
- Demo/seed data is fictional and is not a substitute for real organizational data before go-live.
- Off-site copies of backups, and the hosting platform's own disaster-recovery arrangements, remain the responsibility of the operator of each deployment.

3. Common Platform Architecture

3.1 Tenancy & Hierarchy

Every application is built on the same three-to-four-level hierarchy: an optional Group above the Organization (tenant) level, the Organization itself as the hard data-isolation boundary, and one or more Projects below it. An Organizational Breakdown Structure (OBS) — a site/department/service/team tree, or an equivalent per-project resourcing roster — supplies the shared 'who/where' backbone that governance modules (Business Rules, Controls, Risks & Opportunities, RACSI, BPMN diagrams) point back to.

Group membership is also the only boundary across which Organizations may compare performance (Section 3.12). Every hierarchy screen shows whether an Organization belongs to a Group (Yes/No) and which one; an Organization with no Group is shown as Independent.

3.2 Identity & Access Control

An application recognizes a fixed catalog of standard roles, defined by each application, and supports assigning one or more roles per user. Every write/manage capability check reads from a runtime-editable Permission Matrix — a role × capability grid, seeded with sensible defaults, editable in place by a Platform/Organization Administrator, and effective immediately, platform-wide, without a code change or restart. The matrix is a fail-safe override on top of an underlying default model, never a bypass of it.

3.3 Configuration Management & Commercial Gating

A single Configuration Management surface records, per Organization, exactly one active Solution Pack (defining a stock feature/module list, AI tier, and quota ceilings), any number of independently priced Add-On toggles, and any number of independently priced Compliance & Security Standard toggles. Every Pack-gated route and every quota check must be enforced at the server/API boundary, independently of and in addition to RBAC — client-side navigation gating is a UX convenience only, never the security boundary.

3.4 AI Governance Architecture

A governed AI Use Case Library is the single designated integration point for any live LLM call. Every use case is versioned, tiered Assistive or Augmented, activatable per Organization and overridable per Project (Inherit/On/Off), and logged to an append-only AI Usage Log capturing each suggestion's outcome (Accepted/Edited/Rejected). A Real LLM Provider Connection, where offered, is configured and stored client-side only, is forwarded once per generation request, and is never persisted server-side, logged, or included in any export. Every AI Use Case computes a deterministic, explainable result and falls back to it automatically if a live call fails, so a misconfigured or unreachable connection never blocks a suggestion.

Revision 1.9: every AI generation, in every module, goes through the AI Use Case linked to the place where it is shown, so its activation, tier, Prompt Specification, logging and engine transparency always apply; a place served by no active use case offers no AI. The model of a

generation is resolved in one documented order — the use case's model, the Organization-level model, the user's personal connection, the built-in engine — and the answer names the engine that produced it.

3.5 Retrieval-Augmented Generation (RAG) Architecture

AI-assisted suggestions, an AI Assistant's application-help mode, and any 'ask the application' chatbot are grounded by retrieval over one or more corpora — the tenant's own historical records, a Standards/Knowledge Base, the tenant's own Return on Experience (REX) entries (Section 3.10), and/or a static feature/FAQ corpus — using a TF-IDF/cosine-similarity search or an equivalent lightweight mechanism that runs fully offline, with no network call or API key, so it is reproducible and independently verifiable. RBAC visibility rules applied to retrieval must be the same rule the rest of the application enforces, never a second, hand-maintained copy liable to drift out of sync.

3.6 Internationalization Architecture

A single, versioned string dictionary keyed by string id holds English, French, and Arabic values for every user-facing string. Language precedence resolves, on every sign-in and every Organization switch, in this order: the signed-in user's own explicit preference, then the Organization's configured default language, then English as the platform fallback. Arabic renders full right-to-left — navigation, tables, and charts mirror correctly, not merely the text direction.

3.7 Justification Governance Architecture

Every score or state change to a governed record follows a stage-then-justify pattern: the user stages the new value, writes a justification note, then saves — at which point the new value and its justification are appended together, in one atomic update, to that record's audit/change log. A platform-wide toggle determines whether the justification note is a hard requirement before Save enables, or merely offered and recorded when given.

3.8 Version Management Architecture

Any governed entity type may opt into the platform's generic Version Management service rather than implementing its own ad-hoc history mechanism: on every edit, the service duplicates the entity's current state into an immutable, timestamped version record, advances the entity's Current pointer to the new version, and never rewrites or deletes a prior version. Reverting to a historical version duplicates that version's content forward into a brand-new Current version — the same operation already defined for the AI Use Case Library (Section 4.4) and Template Libraries (Section 4.13) — so the same mechanism backs any entity type an application registers with it, rather than being re-implemented per entity.

3.9 Multi-Channel Communication Architecture

The in-app Notification Center (Section 4.9) is one of several delivery channels a Communication Dispatch service can route a message through — in-app, email, SMS, push, and webhook — behind one shared abstraction, so a new channel can be added without changing the code that raises the underlying alert or message. Every message is composed once, in the recipient's own

display language (Section 3.6), and dispatched to every channel the recipient has enabled for that message category; a channel failure is logged and retried on its own schedule and never blocks delivery on another channel.

3.10 Return on Experience (REX) Architecture

Any closable entity — at minimum the application's principal record (for example a Project, a case or an assessment) and an individual Action — can carry a Return on Experience (REX) capture prompted at closure/completion, structured rather than left in free-form prose, and, once saved, indexed into the same retrieval corpus the Standards/Knowledge Base uses (Section 3.5) so a future AI suggestion or AI Assistant answer can be grounded in a prior REX entry exactly as it would be grounded in a Standard or the tenant's own historical records. A REX entry is itself a versioned entity under the generic Version Management service (Section 3.8), so an edited REX entry keeps its full history like any other governed record.

3.11 Navigation Shell & Personalization Architecture

Every authenticated screen renders inside one shared navigation shell: a primary menu, a header bar carrying the Organization/Project context switcher, language switcher, notification indicator, AI Assistant entry point and user menu, and the content area. The menu is generated from a single navigation model in which each item declares its module group, route, icon, and the view permission and Solution Pack entitlement it requires — the same codes the server enforces on the route — so an item a user cannot open is never shown, and a new module appears in navigation by being registered once.

Each user chooses where the menu is docked (left, right, top, or bottom) and whether it is pinned open or slides away. Unpinned, the menu collapses to an edge handle and slides over the content when pointed at, focused, or toggled. These choices, the user's favorite items, and collapsed menu groups are stored server-side as user preferences, so they follow the user to any browser or device. Left and right are logical sides: under Arabic RTL they mirror. Below a mobile breakpoint the shell switches to a drawer, whatever the stored dock position.

3.12 Internal Benchmarking Architecture

Internal benchmarking computes a catalog of comparable metrics — each defined once with its unit, its direction of good (higher or lower is better), and its computation from the platform's own records, including the audit trail where history matters — and aggregates them by segment. Two scopes exist. Within an Organization, segments are dimensions of the Organization's own data (for example record or project type, process track, or owning department), and authorized users keep full drill-down to the underlying records. Across a Group, each participating Organization contributes only per-metric aggregates; no record, name, or identifier leaves its tenant.

Every segment or Organization below the minimum sample size is flagged as not comparable and excluded from ranks and medians. An Organization Administrator can withdraw the Organization from Group sharing at any time, after which its figures are no longer shown to the rest of the Group. Comparison with organizations outside the Group, or with external or third-party data, is out of scope.

3.13 Platform Operations Architecture

Operational robustness belongs to the platform layer, not to each deployment. On every start the server applies additive migrations — creating missing tables and columns and adding new permission codes with their default role grants — without dropping data or overwriting an administrator's edits, so an existing installation upgrades in place. Startup guards report an unsupported runtime, a port already in use, or an empty database, each as a plain-language message naming the command that fixes it, rather than a stack trace.

The persistence layer takes consistent online backups on demand and on a daily schedule, keeps them for a configurable retention period, and lists them to the platform administrator. A short-lived, per-user read cache, invalidated by any write in the same Organization, keeps read latency within target under concurrent load without ever serving one user's or one tenant's data to another.

3.14 Vertical & SME Layer Architecture

Requirements are organized in five layers, each built on the one below: Layer 1 Platform (tenancy, security, operations), Layer 2 Core (the common requirements of this standard), Layer 3 Vertical (sector content), Layer 4 SME Track (right-sized rigor for small and medium enterprises) and Layer 5 Packaging & Pricing. A vertical and an SME track are configuration data interpreted by the platform: adding, versioning, activating, rolling back or retiring them never requires a change to platform code or downtime.

Activation is additive and ordered. An Organization activates zero or more verticals; each activates its macro processes (three to five) and E2E processes (two to four). When an SME selects a vertical, the vertical processes are activated first and the SME track processes second. A project then takes a mode (Full or SME), a track recommended by its complexity score, and a template from the catalog; the template brings the phases, the Gate Library supplies the gates of each phase, and each gate brings the checklists associated with it from the Checklist Library; a checklist can also be attached to a phase without any gate.

3.15 Process Design & Versioning Architecture

The process design is a hierarchy of typed Process Elements held as data: functions, end-to-end processes and their phases, macro processes, tasks and steps, the gates and checklists attached to phases, the form of each step, the templates (project, checklist, document, questionnaire) and the AI Use Cases linked to tasks and steps. Every element is registered with the generic Version Management service (Section 3.8): an edit creates a new version, no version is ever rewritten, and restoring any earlier version duplicates it forward as the new Current Version.

Elements can be grouped into a Design Release, a named, immutable snapshot of the whole design or of one function or process, which is published, compared and restored as one unit. A running record is bound to the element versions that were current when it started, or when its phase opened; a later version applies to new records, and to running ones only when an authorized user migrates them with a justification. Completed work is never altered by a change of design.

3.16 OBS Role Model Architecture

The OBS has three layers: units (sites, departments, services, teams, project teams), roles defined within units, and people. A role is linked to one or more functions and carries its responsibilities; a person plays one or more roles through dated Role Assignments. Ownership, RACSI letters, approvers, gate approvers and notification targets are assigned to roles and resolved, at the time of use, to the people who currently play them, so a change of holder moves open work without re-assigning it item by item.

OBS Roles describe what people do in the organization. They are distinct from the access roles of RBAC (Section 3.2): a role may propose default access roles to its holders, but access is always decided by the Permission Matrix.

3.17 AI Prompt Specification Architecture

Every AI Use Case is linked to the exact tasks or steps it supports and carries a Prompt Specification made of separate fields. At run time the platform assembles the prompt from these fields in a fixed order, resolves its input variables from the task or step and the record, adds the retrieved references (Section 3.5), and sends it to the configured live model or to the built-in engine. The specification is versioned as a whole and field by field, under Section 3.8, and the version used is written to the AI Usage Log with every suggestion.

3.18 Global Search Architecture

One search service indexes, per tenant, the principal records, the process design, governance items, registers, documents, attachments metadata and Help topics, in every supported language. It applies the same RBAC and Solution Pack rules as the screens, through the same code (Section 3.5), so a result, a count or a snippet never reveals something the user could not open. The navigation menu carries its entry point in every dock position.

4. Common Functional Requirements

Each requirement below is common to every application, stated in platform vocabulary and in “the system shall” form for verifiability. Where the source documents of a revision diverged on a requirement's implementation depth, the stronger (more complete) formulation was taken as the standard; Appendix A is where each application records how far its current build meets it.

Requirements added in revisions 1.1, 1.2, 1.5, 1.6 and 1.7 are generalized from reference implementations and from their user feedback; they carry new IDs following the last ID of their section, never reuse an earlier one, and are listed in Appendices B, C, G, H and I.

Requirements added in revision 1.3 in Sections 4.25 to 4.32 restate the Application-Agnostic Requirements for Verticals and SME source in platform vocabulary; duplicates in the source (for example the vertical approval workflow and audit trail stated twice) are merged, and Appendix E gives the mapping. Sections 4.33 to 4.35 come from product direction.

Requirements added in revision 1.8 — FR-DA-I18N-09 and FR-DA-MRE-15 to 23 — are listed in Appendix M.

Requirements added or amended in revision 1.9 are listed in Appendix N.

4.0 Section-Number Reconciliation with the (New) SRS revision 1.7

The (New) SRS revision 1.7 states the revision-1.7 content of this standard under different section numbers. Table 4.0-1 maps each of its sections to the section of this document where the requirements are specified.

Where the two documents differ in section number, Old's numbering governs; New's IDs remain valid as aliases. In this standard, “Old” is this document and “New” is the (New) SRS revision 1.7; no requirement ID is ever reused in either document.

Table 4.0-1 — Section-number reconciliation with the (New) SRS revision 1.7

New SRS section	Topic	Section in this document	Requirement IDs
4.49	Project Template Blueprints	4.33 (cross-reference at 4.49)	FR-DA-PTC-07, 08; FR-DA-PTB-01 – 10
4.50	Document Formatting & Pictures	4.43 (cross-reference at 4.50)	FR-DA-DFP-01 – 08 (with FR-DA-DOC-06, 07)
4.51	Data Tables & Pagination	4.51, with 4.49	FR-DA-TBL-01 – 07 (with FR-DA-MRE-11 – 13)
4.52	Inline Multi-Record Entry	4.49	FR-DA-MRE-01 – 14
4.53	Layout Panels & Navigation Robustness	4.50 (alias Section 4.53)	FR-DA-PNL-01 – 08; FR-DA-NAV-16 – 20

4.1 Tenant & Organizational Hierarchy

- **FR-DA-TEN-01:** The system shall support full Create, Read, Update, and Delete on Group, Organization, and Project records, gated by a hierarchy-management capability.

- **FR-DA-TEN-02:** The system shall allow an Organization to optionally belong to a Group; an Organization not assigned to a Group shall operate as fully independent.
- **FR-DA-TEN-03:** Deleting a Group shall not delete its member Organizations; deleting an Organization shall cascade-delete its Projects and directly-scoped user accounts and AI activation/override state.
- **FR-DA-TEN-04:** The system shall support full Create, Read, Update, and Delete on Organizational Breakdown Structure (OBS) entries, each typed or free-text, optionally parented to another OBS entry, forming a per-Organization or per-Project tree.
- **FR-DA-TEN-05:** The system shall prevent deleting an OBS node from breaking referential integrity, either by cascading or by re-parenting/reassigning dependent records.
- **FR-DA-TEN-06:** The system shall display, for every OBS node, a live count of linked item types, so accountability and activity concentration are visible at a glance.
- **FR-DA-TEN-07:** The system shall enforce the active Solution Pack's maximum-Projects and maximum-OBS-node quotas at creation time, server-side, rejecting an over-quota request with HTTP 409.
- **FR-DA-TEN-08:** The system shall ensure that no Organization can enumerate, query, or infer the existence of another Organization through any API response, error message, or shared identifier sequence.
- **FR-DA-TEN-09:** The system shall display, on every Organization list and detail screen, whether the Organization belongs to a Group (Yes/No) and, if so, which Group; an Organization with no Group shall be labeled Independent.
- **FR-DA-TEN-10:** The system shall reject, server-side, an OBS parent assignment that crosses trees — an Organization-level node under a Project-level node, or a node under another Project's tree — and an assignment that makes a node its own ancestor.
- **FR-DA-TEN-11:** The system shall identify records by values that reveal nothing about other tenants' activity — opaque identifiers (e.g. UUIDs) or per-tenant sequences — and shall answer a request for another tenant's record exactly as for a record that does not exist (HTTP 404), in support of FR-DA-TEN-08.
- **FR-DA-TEN-12:** The system shall present a tenancy tree, Group (Yes/No) > Organization > principal records, and the same content as a table (Group Yes/No, Group, Organization, sector, records), limited to the Organizations the viewer may see.
- **FR-DA-TEN-13:** The system shall let an authorized user choose the Organization a new principal record belongs to before creating it, showing whether that Organization belongs to a Group; a platform administrator may choose any Organization, other users only their own.
- **FR-DA-TEN-14:** The system shall provision every new Organization on creation with the OBS skeleton, the standard governance catalog, the AI Use Case Library (FR-DA-AI-02), the checklist and record templates, and, when requested, a starting team of one account per standard role under the Organization's e-mail domain with an initial password; a domain already used by another Organization shall be refused.
- **FR-DA-TEN-15:** The system shall let a dedicated permission grant read-only visibility of the principal records of the other Organizations of the viewer's own Group; records of Organizations outside the Group shall never be visible, and records of another Organization shall never be editable from the viewer's Organization.

- **FR-DA-TEN-16:** The system shall let an OBS node list the people placed in it, each with a standard role and the role played in that node (for example "Engineering lead" in a project team), and shall let an administrator add and remove them.

4.2 Identity & Role-Based Access Control

- **FR-DA-RBAC-01:** The system shall support full Create, Read, Update, and Delete on user accounts, including name, email/username, one or more assigned roles, and language preference.
- **FR-DA-RBAC-02:** The system shall provide a Permission Matrix screen displaying every permission/capability code against every role, editable by users holding a dedicated permission-management capability, with edits taking effect immediately and platform-wide without a code change.
- **FR-DA-RBAC-03:** The system shall enforce every API endpoint, server-side, against the authenticated user's resolved permission set (the union across all assigned roles); a request lacking the required permission shall be rejected with HTTP 403.
- **FR-DA-RBAC-04:** The system shall enforce that the Owner and the Evaluator of any two-party verification step are never the same user, both in the UI and at the server.
- **FR-DA-RBAC-05:** The system shall provide a minimal user directory (id, name, email) accessible to any authenticated user for assignment purposes, independent of the full user-management permission.
- **FR-DA-RBAC-06:** The system shall issue a signed, time-boxed authentication token on successful login, required on every subsequent API call, and shall never expose a stored password hash or credential via any API response.
- **FR-DA-RBAC-07:** The system shall expose a single platform-wide justification-required toggle (Section 3.7), defaulting to on, gated by a hierarchy/governance-management capability.
- **FR-DA-RBAC-08:** The system shall, when a user opens a screen or deep link that their roles or the Organization's configuration do not allow, show an in-app access notice explaining why and whom to ask, never a blank page or a raw error.
- **FR-DA-RBAC-09:** The system shall provide a dedicated permission for viewing the content of individual evaluation or verification verdicts; other users, except the parties to that verification, shall see that an evaluation exists but not its content.
- **FR-DA-RBAC-10:** The system shall, at startup, add any permission code introduced by a new release to the catalog together with its default role grants, without overwriting grants an administrator has changed (Section 3.13).

4.3 Configuration Management (Solution Packs, Add-Ons, Compliance Standards)

- **FR-DA-CFG-01:** The system shall record, per Organization, exactly one Solution Pack as the single source of truth for that Organization's feature and quota entitlements, defined centrally in one configuration module and consumed by every enforcing route, so entitlements are never duplicated or able to drift between routes.

- **FR-DA-CFG-02:** The system shall define, for each Solution Pack, at minimum: which modules/features are enabled, the AI tier available (Assistive-only vs. Assistive+Augmented), and numeric ceilings on Projects, OBS nodes, and AI Use Cases.
- **FR-DA-CFG-03:** The system shall record a set of priced Add-Ons, each an independent toggle that grants its associated feature to a Pack that does not already include it standard.
- **FR-DA-CFG-04:** The system shall record a set of priced Compliance & Security Standards modules (at minimum GDPR, ISO/IEC 27001, and SOC 2), each an independent toggle available regardless of Solution Pack.
- **FR-DA-CFG-05:** The system shall, on activating a Compliance & Security Standard, idempotently seed a starting Controls scaffold tagged to that framework, without duplicating it on repeated activation, and shall not delete any Control tagged to a Standard that is later deactivated.
- **FR-DA-CFG-06:** The system shall enforce every Pack-gated feature and every Add-On/Compliance-gated feature server-side via dedicated middleware on the corresponding API route, independently of and in addition to RBAC permission checks; a request for a feature the active configuration does not include shall be rejected with HTTP 403, naming the missing feature.
- **FR-DA-CFG-07:** The system shall expose the Organization's effective configuration and the static Pack/Add-On/Compliance catalog to every authenticated user regardless of role, so client-side navigation can correctly show or hide gated menu items; this read is a UX convenience, not the enforcement boundary.
- **FR-DA-CFG-08:** The system shall expose current usage against every quota dimension, computed live from current data, and shall visually flag a quota that meets or exceeds its allotment.
- **FR-DA-CFG-09:** The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security Standard constitutes a certification, an external audit, or a legal attestation of compliance.
- **FR-DA-CFG-10:** The system shall allow an authorized Organization administrator to register, configure, enable, and disable a connection to an external application through the External Integration Registry (Section 3.3), independently of any other registered integration.
- **FR-DA-CFG-11:** The system shall store every external integration's credentials using the same client-side-only or encrypted-server-side pattern already required for the Real LLM Provider Connection (NFR-DA-SEC-07), never in plaintext, and never logged.
- **FR-DA-CFG-12:** The system shall verify an external integration's connection health on demand and shall record the outcome (success, authentication failure, unreachable) with a timestamp, without exposing provider-side error detail that could aid an unrelated attacker.
- **FR-DA-CFG-13:** The system shall allow an authorized administrator to map fields between an external application's records and the platform's own entities, and shall log every inbound or outbound integration transaction (record, timestamp, direction, and result) for audit (Section 4.11).
- **FR-DA-CFG-14:** The system shall accept inbound events from an external application only through the Multi-Channel Communication webhook channel (Section 4.17) or an equivalent authenticated endpoint, never an unauthenticated route, and shall scope every inbound event to the Organization that registered the originating integration.

4.4 AI Use Case Library & Governance

- **FR-DA-AI-01:** The system shall support full Create, Read, Update, Delete, Duplicate and version-revert on an AI Use Case catalog, each entry storing an id, name, tier (Assistive/Augmented), owning module, trigger, expected output, a human-checkpoint description, its links to process elements (FR-DA-AI-13) and versioned prompt-template content, gated by a dedicated AI-use-case-management capability. The owning module, trigger and expected output shall be derived from the primary linked step when the author leaves them empty, and stored with the entry, never computed only for display. Entries shall be created and edited in a dedicated form — the primary step chosen from the process design, every text written in each configured language, a justification required to save a change — never in a generic record editor.
- **FR-DA-AI-02:** The system shall seed every Organization with the same library of AI Use Cases on creation, and shall allow an Organization to additionally author custom AI Use Cases up to its active Solution Pack's custom-use-case quota.
- **FR-DA-AI-03:** The system shall create Version 1 of an AI Use Case automatically on creation, mark it Default/Current, and shall retain the full version history indefinitely; reverting to a historical version shall duplicate its content into a brand-new Current version, never rewrite or delete history.
- **FR-DA-AI-04:** The system shall allow an Organization Administrator to activate or deactivate any AI Use Case for the whole Organization, and shall allow an authorized Project-level user to set a tri-state (Inherit/On/Off) override per Project; effective activation shall resolve the Project override when set, else the Organization's activation state.
- **FR-DA-AI-05:** The system shall block, server-side, any new AI suggestion from a deactivated AI Use Case (by its effective state) immediately on deactivation, while never retroactively hiding or removing content a human already reviewed and approved before deactivation.
- **FR-DA-AI-06:** The system shall visually label every AI-generated output, wherever shown, as AI-generated and requiring human review, alongside its tier and its generation source (deterministic built-in engine or a configured live model).
- **FR-DA-AI-07:** The system shall write an append-only AI Usage Log entry for every AI suggestion’s outcome (Proposed, Accepted, Edited, or Rejected), capturing the use case, Organization, Project, the record it applied to, the acting user, the specification version, the engine and model, and a timestamp; when a suggestion holds several items (for example several rows), each item accepted, edited or rejected — one by one or all at once — shall be logged, an item changed before its acceptance being logged as Edited; the log shall expose no edit or delete route.
- **FR-DA-AI-08:** The system shall allow a user to configure a Real LLM Provider Connection (provider, model, endpoint, API key) that is persisted client-side only, is forwarded once per generation request, is never returned in any API response, never written to any log, never cached or persisted server-side beyond that single request, and is never included in any data export.
- **FR-DA-AI-09:** The system shall, on any failure of a live LLM call, fall back automatically to the AI Use Case's deterministic built-in generator, so a misconfigured or unreachable connection never blocks a suggestion from being produced.

- **FR-DA-AI-10:** The system shall ensure an AI Use Case's activation, tier, or generation never grants a user visibility into data beyond what their RBAC role and permissions already allow.
- **FR-DA-AI-11:** The system shall, before generating any suggestion, ground it in retrieved reference content (Section 4.5), instruct the underlying model to stay within the application's own vocabulary, and state no number, score, or name not present in the retrieved references or the record's own data; the retrieved references shall be disclosed to the user alongside the output.
- **FR-DA-AI-12:** The system shall show every active AI Use Case, labelled with its tier (Assistive AI or Augmented AI) by a distinct icon and wording, on the processes, macro processes, tasks and steps it is linked to, and shall let an authorized user run it there and accept, modify or reject the suggestion (FR-DA-AI-07).
- **FR-DA-AI-13:** The system shall let an AI Use Case be linked to one primary step and to further steps, tasks, macro processes or end-to-end processes (AI Use Case Link), the use case serving every step reached through its links; and shall show the Coverage of the Process Design — every task with the AI Use Cases that support it, filterable by tier and by covered or not covered — with, for a task that has none, the suggestion of FR-DA-AIP-11.
- **FR-DA-AI-14:** The system shall let an authorized user delete a custom AI Use Case, keeping its usage-log entries; seeded catalog entries shall be deactivated instead of deleted.
- **FR-DA-AI-15:** The system shall offer, wherever the live model is configured (personal settings and administration), a provider list and a model list with a "custom model" choice and an optional custom endpoint, and a connection test that reports only the outcome category (works, authentication failure, model not found, rate limited, unreachable).
- **FR-DA-AI-16:** The system shall let an Organization Administrator configure an Organization-level live model — provider (standard providers or a custom endpoint compatible with a common API), model, temperature and maximum answer length — as an alternative to the personal connection of FR-DA-AI-08; its key shall be stored encrypted under NFR-DA-SEC-17, shown masked, and never returned in full.
- **FR-DA-AI-17:** The system shall let an authorized user choose, per AI Use Case, a model of the configured provider other than the Organization default.
- **FR-DA-AI-18:** The system shall let a user view, before and after a run, the exact prompt the AI Use Case sends — assembled from its Prompt Specification and resolved for the current task or step — together with the engine that answered (live model or built-in engine) and any fallback warning.
- **FR-DA-AI-19:** The system shall keep a capability profile per model — the request parameters it accepts (for example sampling parameters), whether it reasons before answering, its maximum output — and send each request with only the parameters the model accepts, reserving enough output length for its reasoning.
- **FR-DA-AI-20:** The system shall offer a connection test for the configured model that reports, in plain language, whether the key was refused, a parameter was rejected, the model declined the request or the model answered, and record the result.
- **FR-DA-AI-21:** The system shall open each AI Use Case, from its row in the library, on one page showing its whole structure: identification (id, name, tier, module, primary step and task, trigger, expected output, human checkpoint, model task type, risk, approval, scope, origin), its links with the input fields it reads and the output fields it proposes, its activation for the

Organization and the current Project, the twelve fields of its Prompt Specification with their completeness, the assembled prompt with a test run, the versions of the use case and of its specification with revert, and its usage log with the counts of each outcome.

- **FR-DA-AI-22:** The system shall show, above the AI Use Case Library, the number of use cases by tier, the number active in the Organization or the current Project, the number of custom use cases against the Solution Pack quota, the average completeness of the Prompt Specifications and the number of tasks of the process design supported by AI; it shall filter the library by tier, by active or inactive state and by standard or custom origin, and disable the creation of a custom use case, stating the quota, when the quota is reached.
- **FR-DA-AI-23:** The system shall produce every AI suggestion, in every module — step proposals, questionnaires, training, documents, project drafts and any other — through an AI Use Case linked to the place where it is shown: the effective activation of that use case shall be checked on the server before each generation (FR-DA-AI-05) and the generation logged under FR-DA-AI-07 and FR-DA-AIP-10. A place served by no active AI Use Case shall offer no AI action; a user entitled to manage prompts shall be offered there to suggest one (FR-DA-AIP-11).
- **FR-DA-AI-24:** The system shall resolve the model of every AI generation in a fixed and documented order — the model chosen for the AI Use Case (FR-DA-AI-17), else the Organization-level live model (FR-DA-AI-16), else the user's personal connection (FR-DA-AI-08), else the built-in engine (FR-DA-AI-09) — and apply that order to every AI feature without exception. The settings screens shall state the difference between the Organization model (shared, administered, key stored encrypted on the server) and the personal connection (one user, key kept in the browser only).
- **FR-DA-AI-25:** The system shall create a custom, duplicated or AI-suggested AI Use Case deactivated, with the approval status Draft and a fully populated Prompt Specification, and activate it only after a person reviews it (FR-DA-AIP-06, -11); a duplicate shall record the use case it is based on and copy its Prompt Specification as its own Version 1; a seeded entry may be duplicated and deactivated, never deleted (FR-DA-AI-14).
- **FR-DA-AI-26:** The system shall present an AI proposal of several items with each item editable in place and the actions Accept, Reject, Accept all and Reject all, together with the label "AI-generated — to be reviewed by a person", the tier, the AI Use Case, the engine and, when the live model was not used, the reason (FR-DA-AIP-12), the references retrieved (FR-DA-AI-11), the human checkpoint and a link to the exact prompt (FR-DA-AI-18).

4.5 Standards / Knowledge Base & Retrieval-Augmented Generation

- **FR-DA-KB-01:** The system shall maintain a Standards/Knowledge Base of reference content — regulatory or methodological standards, or per-process/per-stage practitioner guidance — available in every language the platform supports.
- **FR-DA-KB-02:** The system shall index this knowledge base, the tenant's own historical records, and a static application-feature/FAQ corpus using the same retrieval mechanism, so AI Use Case grounding, an AI Assistant's application-help mode, and any chatbot share one retrieval engine rather than several.
- **FR-DA-KB-03:** The system shall scope every retrieval query to the requesting user's own tenant; no cross-tenant record shall appear in a retrieval result or an AI suggestion.

- **FR-DA-KB-04:** The system shall gracefully degrade when a live LLM call is unavailable or fails: the retrieval result (the matched reference content, the computed fact) shall still be returned, rather than the request failing outright.

4.6 Process/Stage Governance (Alerts, Business Rules, Controls, Risks, RACSI, Custom KPIs)

- **FR-DA-GOV-01:** The system shall support full Create, Read, Update, and Delete on Business Rules, each with a condition, an action, a severity, and an owner, gated by a governance-management capability and, where the platform separates GRC from the base module set, by the active Solution Pack's GRC entitlement.
- **FR-DA-GOV-02:** The system shall support full Create, Read, Update, and Delete on Controls, each classified under one of the five COSO Internal Control — Integrated Framework components, with a control type, testing frequency, owner, and effectiveness rating.
- **FR-DA-GOV-03:** The system shall allow a Control to be optionally tagged with the Compliance & Security Standard it was seeded under or otherwise supports, without that tag itself constituting a certification (Section 4.3, FR-DA-CFG-09).
- **FR-DA-GOV-04:** The system shall support full Create, Read, Update, and Delete on Risk/Opportunity items, each with a type, a likelihood (1–5) and impact (1–5) rating, an automatically computed score, an owner, and a status, and shall render a 5×5 likelihood × impact heat-map.
- **FR-DA-GOV-05:** The system shall support full Create, Read, Update, and Delete on Custom KPIs, additive to the platform's built-in KPI set, each with a plain-language formula description, a target, and an owner, available regardless of Solution Pack.
- **FR-DA-GOV-06:** The system shall support a RACSI accountability matrix: full Create, Read, Update, and Delete on RACSI Activities, each optionally linked to a process step or to a specific Business Rule, Control, or Risk/Opportunity record, and on RACSI Assignments, each assigning exactly one of the five RACSI roles to a named person or a role.
- **FR-DA-GOV-07:** The system shall enforce that a RACSI Activity has at most one Accountable assignment at any time, both in the UI and at the data-persistence layer.
- **FR-DA-GOV-08:** The system shall allow a Business Rule, a Control, a Risk/Opportunity, and a RACSI Activity to each be optionally tagged with the owning OBS node and with the process/stage they are most relevant to, and shall surface the items tagged to a given process/stage on that process/stage's own governance view.
- **FR-DA-GOV-09:** The system shall let a RACSI assignment target an OBS Role as well as a named person, and shall resolve a role assignment to the people currently playing that role (Section 3.16) wherever the RACSI is used.
- **FR-DA-GOV-10:** The system shall provide a RACSI editor on every macro process, with one row per task or step and one column per RACSI letter, enforcing exactly one Accountable per row (FR-DA-GOV-07) and defaulting the owner of each step to its Responsible role.

4.7 BPMN 2.0 Process Modeling

- **FR-DA-BPMN-01:** The system shall support full Create, Read, Update, and Delete on BPMN Diagrams, each with a title, description, and a BPMN 2.0-compliant XML payload,

rendered/edited with a standards-compliant BPMN 2.0 modeler/viewer component, not a static image.

- **FR-DA-BPMN-02:** The system shall grant read-only pan/zoom viewing of a BPMN Diagram to any user holding view access, and shall grant the full editing palette and a Save action only to users holding edit access whose Organization's active configuration includes full BPMN editing.
- **FR-DA-BPMN-03:** The system shall render the BPMN editor's tool palette in a dedicated panel beside the diagram canvas, structurally separate from the canvas, rather than floating over it.
- **FR-DA-BPMN-04:** The system shall provide Export (download the current diagram as a schema-valid .bpmn/XML file) and Import (load a .bpmn/XML file, replacing the current diagram) actions, so a diagram is portable to and from any other BPMN 2.0-compliant tool.
- **FR-DA-BPMN-05:** The system shall be seeded, per Organization, with at least one BPMN Diagram modeling the application's own core process end-to-end.
- **FR-DA-BPMN-06:** The system shall offer a shapes palette section grouped by category (tools, events, activities, gateways, data, participants, artifacts), usable by click or drag, and shown as a legend of the main shapes when the diagram is view-only.
- **FR-DA-BPMN-07:** The system shall let the user open a diagram in a full-screen workspace with the palette, zoom controls (slider, zoom in and out, fit), a palette that slides away, and a single action or the Escape key to return to the page.
- **FR-DA-BPMN-08:** The system shall state on the diagram screen whether editing a diagram changes the execution of records; where diagrams only document the process, saving one shall not alter running or future records.

4.8 Reports & Data Export

- **FR-DA-REP-01:** The system shall compute the platform's standard KPIs in real time from current data, scoped to the requesting user's tenant.
- **FR-DA-REP-02:** The system shall provide a set of standard reports, each exportable on demand as a downloadable PDF, Excel (.xlsx), and Word (.docx) file, generated server-side and styled to the common visual identity (Section 5.13).
- **FR-DA-REP-03:** The system shall name every exported report file with the report key and the export date, and shall serve it with the correct MIME type and a Content-Disposition attachment header.
- **FR-DA-REP-04:** The system shall offer a client-side CSV export on its highest-traffic tables, reflecting exactly the rows and columns currently on screen and respecting the same RBAC and scope filtering as the table itself, written with a UTF-8 byte-order mark so accented and Arabic content opens correctly in common spreadsheet tools.

4.9 Notification Center (Alerts)

- **FR-DA-ALT-01:** The system shall define a shared Alert Catalog naming every alert type once, with a human-readable name and a primary process/stage tag, consumed identically by seed data and by live computation, so alert semantics cannot drift between demo data and live behavior.
- **FR-DA-ALT-02:** The system shall surface a persistent, dismissible per-tenant or per-project alert inbox, with a live unread-alert indicator in the primary navigation.

- **FR-DA-ALT-03:** The system shall avoid raising a duplicate alert of the same type against the same triggering entity within the same period.
- **FR-DA-ALT-04:** The system shall allow a user to mark an individual alert as read, and, where dismissal is supported, shall persist the dismissal separately from the alert's own (recomputed) state.
- **FR-DA-ALT-05:** The system shall allow an authorized user to independently enable or disable each alert type, displaying its human-readable name alongside its machine code in both the alert inbox and any governance-settings configuration list.

4.10 AI Assistant / Chatbot

- **FR-DA-AST-01:** The system shall provide a data-query mode that answers natural-language questions about the requesting user's own organization by matching the question to a fixed set of data-query intents.
- **FR-DA-AST-02:** The system shall associate every data-query intent with the single RBAC permission required to answer it, and shall evaluate that permission after matching the intent but strictly before running any query; when the permission is missing, the system shall return an explicit, human-readable refusal naming the missing permission rather than executing the query.
- **FR-DA-AST-03:** The system shall scope every data-query answer to the requesting user's own tenant, consistent with the platform's multi-tenancy requirement (Section 4.1).
- **FR-DA-AST-04:** The system shall provide an application-help mode answering “Can it…?” / “How do I…?” questions by ranking a static FAQ/feature corpus with the same retrieval engine used elsewhere in the platform (Section 4.5).
- **FR-DA-AST-05:** The system shall render the AI Assistant both as a dedicated screen and as a floating widget accessible from every authenticated screen, without navigating away from the user's current work.
- **FR-DA-AST-06:** The system shall gate the AI Assistant behind a single view permission and the active Solution Pack's AI Assistant entitlement, independent of and in addition to any module-specific permission a given data-query intent may separately require.

4.11 Audit Trail & Justification Governance

- **FR-DA-AUD-01:** The system shall write an audit or change-log entry for every state-changing operation on a governed field, capturing the actor, timestamp, entity type/id, and the before/after value.
- **FR-DA-AUD-02:** The system shall follow the stage-then-justify pattern (Section 3.7) for every score or state change to a governed record: staging a new value, requiring or offering a justification note per the platform-wide toggle, and appending both atomically to the record's audit/change log in one update.
- **FR-DA-AUD-03:** The system shall make the audit/change log viewable, read-only, to users holding an audit-view permission, and shall never expose an edit or delete route for it.
- **FR-DA-AUD-04:** The system shall ensure audit-log writes never block or fail the primary operation they accompany.

- **FR-DA-AUD-05:** The system shall allow an authorized user to reopen a completed workflow step before the stage or approval it belongs to is decided, only with a justification recorded per Section 3.7; the reopening shall return the step to its active state and be written to the audit log.

4.12 Help & In-App Guidance

- **FR-DA-HLP-01:** The system shall provide an in-app Help screen presenting topic articles covering every module of the application, accessible to every authenticated user independent of role or permission grants.
- **FR-DA-HLP-02:** The system shall render Help content in the user's currently selected display language, matching the article set one-to-one across every supported language.
- **FR-DA-HLP-03:** The system shall provide a free-text search box on the Help screen filtering topics by a case-insensitive match against the topic title or body.
- **FR-DA-HLP-04:** The system shall provide, next to every scored or classified input (for example complexity scores), a decision matrix describing each level, and shall show a legend for every code or colour vocabulary displayed in a list or matrix (for example the Part letter of a process or the status colours of a matrix).

4.13 Template Libraries

- **FR-DA-TPL-01:** The system shall support a library of pre-filled starting points (templates) for its principal record type, each capturing default field values a new record can be seeded from, while leaving every field editable before submission.
- **FR-DA-TPL-02:** The system shall support full Create, Read, Update, and Delete on templates, versioned via the same version-history mechanism as the AI Use Case Library (Section 4.4), gated by a dedicated template-management capability.
- **FR-DA-TPL-03:** The system shall make templates available regardless of Solution Pack, consistent with the always-standard treatment of Custom KPIs and standard Reports.
- **FR-DA-TPL-04:** Deleting a template shall not affect any record already created from it; a template is a starting point copied at use time, not a live-linked reference.
- **FR-DA-TPL-05:** The system shall manage project templates through the Project Template Catalog of Section 4.33; the rules of FR-DA-TPL-01 to 04 apply to it.
- **FR-DA-TPL-06:** The system shall seed a questionnaire template library covering each focus — All needs, Digital, AI and Custom — and each supported sector, following the rules of FR-DA-TPL-01 to 04.
- **FR-DA-TPL-07:** The system shall allow an Organization to author its own questionnaire templates, visible only within its tenant, alongside the seeded library.
- **FR-DA-TPL-08:** The system shall version every questionnaire template and retain its full history through the generic Version Management service (Section 4.16); a questionnaire shall record the template version it was loaded from.
- **FR-DA-TPL-09:** The system shall allow a questionnaire template to be exported as a portable file and re-imported, in the same or another tenant, as a new template version or a new template, with its focus, sector and language tags preserved.

4.14 Multi-Language & Localization & Currency

- **FR-DA-I18N-01:** The system shall provide complete UI translation in English, French, and Arabic for all navigation, labels, form fields, statuses, and messages, with no hardcoded UI string outside the shared translation dictionary.
- **FR-DA-I18N-02:** The system shall allow a user to change their display language at any time without reloading or losing unsaved work.
- **FR-DA-I18N-03:** The system shall persist a user's language preference and apply it automatically on subsequent logins, resolved per the precedence order in Section 3.6.
- **FR-DA-I18N-04:** The system shall render the entire application right-to-left, including navigation, forms, tables, and charts, when Arabic is selected.
- **FR-DA-I18N-05:** The system shall allow an Organization to configure a tenant-level default language.
- **FR-DA-I18N-06:** The system shall allow adding a new supported language by configuration — populating the shared translation dictionary (Section 3.6) — alone, without a code change or a redeployment of application logic.
- **FR-DA-I18N-07:** The system shall return every server-generated, user-facing message — validation and error messages, notification text, AI Assistant answers, and export labels — in the requesting user's display language, resolved from the same translation dictionary as the UI; a machine code may accompany the translated text but shall not replace it.
- **FR-DA-I18N-08:** The system shall seed its reference and Knowledge Base content in every supported language and shall let the user filter Knowledge Base articles by language, defaulting to the user's display language.
- **FR-DA-I18N-09:** The system shall support several currencies: every amount shall carry its currency, an Organization shall set its default currency and the currencies it enables, a user may enter an amount in any enabled currency, and amounts shall be shown and exported with the formats of NFR-DA-I18N-05; a total shall never add amounts of different currencies without a stated conversion rate and date.
- **FR-DA-I18N-10:** The system shall store a reference to a shared element (a function, a role, a unit, a controlled-list value) by its language-independent identity, or with its names in every configured language, never as the label of the user’s current language, so that a record reads the same and matches the same element in every language; saving a record shall keep the stored translations of every multilingual value the user did not change.

4.15 Work Breakdown Structure & Gantt Scheduling

Any grouping of Actions — regardless of the process, stage, or governed record they originate from — can be organized into a Work Breakdown Structure (WBS) and scheduled as a Gantt chart, so planning is available for any action grouping the user assembles, not only a single predefined project type.

- **FR-DA-WBS-01:** The system shall allow any set of Actions the user selects to be grouped into a Work Breakdown Structure (WBS), organized as a tree of WBS nodes, each node referencing one or more Action IDs as its leaf-level work items.

- **FR-DA-WBS-02:** The system shall allow a WBS node or a referenced Action to carry a planned start date, planned end date or duration, percent-complete, and zero or more predecessor dependencies on other WBS nodes or Actions within the same WBS.
- **FR-DA-WBS-03:** The system shall render a WBS as a Gantt chart, plotting every node and its referenced Actions on a shared timeline, drawing dependency lines between predecessor and successor items, and visually distinguishing planned, in-progress, completed, and overdue items.
- **FR-DA-WBS-04:** The system shall recompute a WBS node's own planned dates and percent-complete from its child nodes/Actions when the node itself is not directly scheduled, so a summary bar always reflects the current state of its children without manual reconciliation.
- **FR-DA-WBS-05:** The system shall support full Create, Read, Update, and Delete on WBS structures, and shall allow the same Action to be referenced from more than one WBS without duplicating the underlying Action record.
- **FR-DA-WBS-06:** The system shall export a WBS/Gantt view to at least one portable format (e.g. PDF or an image), consistent with the platform's standard Reports & Data Export conventions (Section 4.8).

4.16 Version Management

- **FR-DA-VER-01:** The system shall provide a single, generic Version Management service (Section 3.8) that any governed entity type can register with, rather than each entity type implementing its own history mechanism.
- **FR-DA-VER-02:** The system shall create an initial version automatically when a versioned entity is created, mark it Current, and create a new version — never overwrite the prior one — on every subsequent edit.
- **FR-DA-VER-03:** The system shall retain every version of a versioned entity indefinitely, with no hard-delete route for a historical version, and shall allow an authorized user to view the full version history and compare any two versions field-by-field.
- **FR-DA-VER-04:** The system shall allow an authorized user to revert a versioned entity to any historical version; reverting shall duplicate that version's content into a brand-new Current version, never rewrite or delete the entity's history, consistent with the pattern already defined for the AI Use Case Library (FR-DA-AI-03) and Template Libraries (FR-DA-TPL-02).
- **FR-DA-VER-05:** The system shall record, on every version, the acting user, a timestamp, and — where the platform-wide justification toggle (Section 3.7) requires it — the justification note that accompanied the change.
- **FR-DA-VER-06:** The system shall provide one shared version-comparison view, available from the history of every versioned entity, that shows two selected versions side by side and highlights the fields that differ, in support of FR-DA-VER-03.
- **FR-DA-VER-07:** The system shall list, on every versioned entity, all its versions with number, label, author, date, change note and justification, and shall let an authorized user open any version read-only and restore it (FR-DA-VER-04) in one action.
- **FR-DA-VER-08:** The system shall version a composite entity — for example a macro process with its tasks and steps, or a gate with its associated checklists — so that restoring one of its versions restores its children, their order and their links as they were in that version.

- **FR-DA-VER-09:** The system shall support named baselines (Design Releases, Section 3.15) grouping chosen versions of several entities, which can be compared with each other and restored as one unit.

4.17 Multi-Channel Communication

- **FR-DA-COMM-01:** The system shall support delivering a notification or alert (Section 4.9) through one or more channels — at minimum in-app, email, and webhook, with SMS and push as optional channels — behind a single shared dispatch interface (Section 3.9).
- **FR-DA-COMM-02:** The system shall allow a user to independently enable or disable each available channel for each notification category, defaulting every category to the in-app channel at minimum.
- **FR-DA-COMM-03:** The system shall compose a message once, per notification event, and render it in the recipient's own display language (Section 3.6) before dispatch to any channel.
- **FR-DA-COMM-04:** The system shall record a delivery-status record per message per channel (queued, sent, delivered, failed), and shall retry a failed delivery on that channel's own schedule without blocking or duplicating delivery on any other channel.
- **FR-DA-COMM-05:** The system shall never send a tenant's data through an external channel provider (email, SMS, push) beyond the minimum content required for that notification, and shall scope every dispatch to the recipient's own tenant.
- **FR-DA-COMM-06:** The system shall provide a dedicated questionnaire dispatch channel abstraction on top of the Communication Dispatch service (Section 3.9), supporting Face-to-Face (printable or offline packs), Email, Hybrid and Custom channels, so that a new questionnaire channel can be added without changing questionnaire logic.
- **FR-DA-COMM-07:** The system shall let an authorized user set a reminder cadence per channel for each questionnaire, send reminders only to stakeholders who have not responded, and stop them when a response is received or the questionnaire closes.
- **FR-DA-COMM-08:** The system shall log every questionnaire dispatch, reminder and response per channel — recipient, channel, timestamp and delivery status — in an immutable record available for audit (Section 4.11).

4.18 Return on Experience (REX)

A Return on Experience (REX) entry captures what a closed Project (or equivalent principal record) or a closed Action actually taught the organization, so the lesson is queryable and reusable rather than lost in an individual's memory or a narrative report.

- **FR-DA-REX-01:** The system shall prompt for a REX entry when the application's principal record reaches a terminal/closed status, and when an individual Action is marked complete, without blocking the closure itself on the REX entry being filled in.
- **FR-DA-REX-02:** The system shall structure every REX entry with, at minimum: what went well, what did not, a root-cause note, a recommendation/improvement action, a category/tag, and an effectiveness or satisfaction rating on a fixed scale.
- **FR-DA-REX-03:** The system shall support full Create, Read, Update, and Delete on REX entries, each optionally tagged to the owning OBS node and to the process/stage it is most relevant to, consistent with how other governance artifacts are tagged (Section 4.6, FR-DA-GOV-08).

- **FR-DA-REX-04:** The system shall index every saved REX entry into the same retrieval corpus used for AI Use Case grounding, the AI Assistant, and application-help search (Section 4.5), scoped to the requesting user's own tenant, so a relevant prior REX entry can surface in a future suggestion or answer.
- **FR-DA-REX-05:** The system shall provide a REX Register view aggregating entries by category/tag and by rating, so a recurring root cause or a declining rating trend is visible without reading every entry individually.
- **FR-DA-REX-06:** The system shall version every REX entry via the generic Version Management service (Section 3.8/4.16), retaining its full edit history and supporting revert to any prior version.
- **FR-DA-REX-07:** The system shall gate REX entry management behind a dedicated capability, and shall scope every REX read and every retrieval-grounded reference to it to the requesting user's own tenant, consistent with the platform's multi-tenancy requirement (Section 4.1).
- **FR-DA-REX-08:** The system shall present the REX prompt to the user who performed the closing action, immediately after the closure succeeds, with a choice to fill it in now or skip; skipping shall leave the closure intact and the entry available to add later from the REX Register.

4.19 Navigation Shell & Personalization

The navigation shell (Section 3.11) behaves the same way in every application, so a user who has learned one application can find their way around any other.

- **FR-DA-NAV-01:** The system shall generate the primary menu from a single navigation model in which each item declares its module group, label, icon, route, and the view permission and Solution Pack entitlement it requires, and shall show only the items the current user can open.
- **FR-DA-NAV-02:** The system shall group menu items by module, let the user collapse and expand each group, and mark the current screen as the active item.
- **FR-DA-NAV-03:** The system shall let each user pin the menu open or unpin it. Unpinned, the menu shall collapse to an edge handle, slide over the content when the user points at the edge, focuses the handle, or uses the menu button, and slide away again when the pointer leaves or an item is chosen.
- **FR-DA-NAV-04:** The system shall let each user dock the menu on the left, right, top, or bottom edge of the screen, laying it out vertically on the left and right and horizontally, with drop-down groups, on the top and bottom.
- **FR-DA-NAV-05:** The system shall let each user mark menu items as favorites, shown in a Favorites group at the start of the menu.
- **FR-DA-NAV-06:** The system shall store the dock position, pinned state, collapsed groups, and favorites as server-side user preferences, applied on every sign-in and on any device, and shall offer them both from the menu itself and from the user's Settings screen.
- **FR-DA-NAV-07:** The system shall treat left and right as logical start and end under right-to-left languages, mirroring the dock position and the slide direction when Arabic is selected.
- **FR-DA-NAV-08:** The system shall, below a mobile breakpoint, replace the docked menu with a drawer opened from a menu button and closed by Escape, by an outside tap, or by choosing an item, without changing the user's stored desktop preferences.

- **FR-DA-NAV-09:** The system shall make the shell fully keyboard-operable: a skip-to-content link as the first focusable element, menu and group buttons exposing their expanded state, Escape closing any open menu, and the active item marked as the current page for assistive technologies.
- **FR-DA-NAV-10:** The system shall keep the Organization/Project context switcher, language switcher, notification indicator, AI Assistant entry point, and user menu in the header bar of every authenticated screen.
- **FR-DA-NAV-11:** The system shall organize the menu by what users do, in a documented order (home and own work, portfolio of records, process design, governance and risk, AI and knowledge, reports, administration, personal settings), so the same capability is always found in the same place across applications.
- **FR-DA-NAV-12:** The system shall show a Search button at the start of the primary menu, in every dock position, pinned or sliding; unpinned, the button stays visible on the edge handle. It shall open the global search (Section 4.37) and also be reachable by a keyboard shortcut.
- **FR-DA-NAV-13:** The system shall let the user filter the menu itself by typing in the search field of the menu, showing the matching items, favorites first, with their module group.
- **FR-DA-NAV-14:** The system shall show a search bar in the header of every authenticated screen, in addition to the Search button of the menu (FR-DA-NAV-12); typing in it opens the global search (Section 4.37) with the text typed.
- **FR-DA-NAV-15:** The system shall give each catalog that users manage — for example functions, process design, document templates and document layout — a direct menu entry, so that its Create, Read, Update and Delete are reachable in one step from the menu.
- **FR-DA-NAV-16:** The system shall show a top bar 64 pixels high on every authenticated screen, white with a 1-pixel Line bottom border, carrying the horizontal logo (at least 150 pixels wide) at the start, the global search field at the center, and the language switcher, notification indicator, AI Assistant entry point and user menu at the end (FR-DA-NAV-10, FR-DA-NAV-14).
- **FR-DA-NAV-17:** The system shall render menu items 40 pixels high with an 8-pixel radius; the active item shall use the Azure tint background, Navy text and a 3-pixel Azure border on its start edge, and a hovered item the Background color.
- **FR-DA-NAV-18:** The system shall show breadcrumbs on every screen below the first level of the menu: 13-pixel Open Sans in Muted, a 14-pixel chevron as separator (mirrored under right-to-left languages), and the current page in Navy SemiBold, not as a link.
- **FR-DA-NAV-19:** The system shall open a Command Palette with Ctrl+K (Cmd+K on macOS): a 640-pixel overlay with the search input at the top and results below, each with an icon, its label, a type badge and its shortcut, navigated with the arrow keys, chosen with Enter and closed with Escape. It shall run the global search of Section 4.37 and be the keyboard shortcut of FR-DA-NAV-12.
- **FR-DA-NAV-20:** The system shall render tabs with a 2-pixel Azure bottom border on the active tab and Muted labels on the inactive ones, moved between with the arrow keys, and segmented controls inside an 8-pixel-radius Background container with the active segment white.

4.20 Internal Benchmarking

Internal benchmarking (Section 3.12) lets an Organization learn from its own segments and from the other Organizations of its Group. External benchmarking is out of scope.

- **FR-DA-BMK-01:** The system shall define its benchmark metrics in one catalog module, each with a label, a unit, a direction of good, a plain-language explanation, and a computation from the platform's own records, consumed by every benchmarking screen.
- **FR-DA-BMK-02:** The system shall compare, within one Organization, every catalog metric across the segments of at least one dimension the application defines (for example record or project type, process track, or owning OBS node), showing each segment's value, the Organization-wide value, and the median.
- **FR-DA-BMK-03:** The system shall compare, across the Organizations of the requester's Group, per-Organization aggregates of the same metrics, and shall rank each Organization per metric according to the metric's direction of good, giving tied values the same rank.
- **FR-DA-BMK-04:** The system shall include in a Group comparison only per-metric aggregates and each contributing Organization's name and sector — never a record, record title, person's name, or identifier from another Organization.
- **FR-DA-BMK-05:** The system shall flag any segment or Organization with fewer records than a minimum sample size (default 3) as not comparable, and shall exclude it from ranks and medians.
- **FR-DA-BMK-06:** The system shall let an Organization Administrator withdraw the Organization from Group benchmarking and rejoin it; while withdrawn, its figures shall not be shown to the other Organizations of the Group. Every change shall be written to the audit log.
- **FR-DA-BMK-07:** The system shall gate internal benchmarking behind a view capability, Group comparison behind a separate capability, and the sharing setting behind a management capability, each enforced server-side.
- **FR-DA-BMK-08:** The system shall refuse any comparison with an Organization outside the requester's Group and shall not import or display external or third-party benchmark data; an Organization with no Group shall see its internal comparison only, with a notice explaining why.
- **FR-DA-BMK-09:** The system shall render each metric comparison as a chart following the common visual identity, with the requester's own Organization or the selected segment highlighted, and shall let the user filter the comparison by the application's segment dimensions.

4.21 Platform Operations

These requirements keep an application safe to upgrade, recover, and demonstrate, whatever its business domain (Section 3.13).

- **FR-DA-OPS-01:** The system shall take a consistent backup of its data store on demand — from the command line and from the platform administrator's configuration screen — and automatically once a day, while remaining online.
- **FR-DA-OPS-02:** The system shall keep backups for a configurable retention period (default 14 days), delete older ones automatically, and list existing backups with date and size to the platform administrator only.

- **FR-DA-OPS-03:** The system shall document how to restore a backup, and a restored backup from an earlier release shall start without any manual migration step (FR-DA-OPS-04).
- **FR-DA-OPS-04:** The system shall apply additive schema and permission-catalog migrations automatically on startup, never dropping or rewriting existing data or administrator changes, so an existing installation upgrades in place.
- **FR-DA-OPS-05:** The system shall check, when installed and started, the runtime version, port availability, and whether the data store has been initialized, and shall report each problem as a plain-language message naming the corrective action rather than a stack trace.
- **FR-DA-OPS-06:** The system shall expose an unauthenticated health endpoint reporting only liveness, whether the data store is initialized, and the deployment mode — never tenant data.
- **FR-DA-OPS-07:** The system shall ship a deterministic demonstration seed, reset by one command, containing at least one Group of two or more Organizations, at least one Independent Organization, several sectors, a user for every role, and at least ten instances of each principal end-to-end process per Organization, all fictional (Section 2.7).
- **FR-DA-OPS-08:** The system shall include an administrator traceability view mapping each requirement ID of this standard to its implementation status (Met, Partial, Not met, or Deployment responsibility) and a short evidence note.
- **FR-DA-OPS-09:** The system shall seed demonstration data in every module, so that no screen of the delivered application is empty: record teams with roles, templates, attachments in several formats, custom AI Use Cases and project overrides, alert settings and alerts already read, and integration and webhook registrations.

4.22 Portfolio Overview

The portfolio overview answers "where does each record stand in each process?" for any scope the user may see.

- **FR-DA-PFO-01:** The system shall let the user choose a scope: a Group, one or more Organizations, or a set of principal records, each list containing only what the user may see (FR-DA-TEN-15).
- **FR-DA-PFO-02:** The system shall show one row per principal record and one column per process stage, with at each intersection the status of that stage for that record, drawn from one status vocabulary (for example Completed, In progress, At gate, On hold, Stopped, Not started, Not in the record's variant).
- **FR-DA-PFO-03:** The system shall show a legend of the cell statuses, a per-stage total, the number of times a stage ran when more than once, and a hover text naming record, stage and status.
- **FR-DA-PFO-04:** The system shall export the matrix to CSV (FR-DA-REP-04) and shall show records of another Organization read-only.

4.23 Checklist Library

The Checklist Library holds every checklist as an independent entry. A checklist may be related to a gate or not: when it is associated with a gate of the Gate Library (Section 4.35) it becomes part of that gate's exit criteria; when it is not, it can still be attached to a phase, a task or a project and used on its own.

- **FR-DA-CHK-01:** The system shall keep a Checklist Library with full Create, Read, Update, Delete and Duplicate on checklist templates, each with ordered items marked mandatory and evidence-required, classified by scope (universal or one vertical) and optionally by mode and track, and versioned under the generic Version Management service (Section 3.8). A checklist shall not require a gate.
- **FR-DA-CHK-02:** The system shall let a checklist template be related to zero, one or several gates; the association is made from the gate (FR-DA-GTE-02) or from the checklist, and a checklist associated with a gate shall be copied into that gate's checklist when the phase opens. A built-in reference checklist shall apply to a gate that has no associated checklist.
- **FR-DA-CHK-03:** The system shall let an authorized user add items to an open gate or phase checklist from any template or by typing them, never adding the same item twice, and save such a checklist as a new template.
- **FR-DA-CHK-04:** The system shall show for every checklist the gates it is associated with, and let the library be filtered by gate, by "not related to a gate", by vertical, mode and track; it shall seed sector checklists for new Organizations of that sector.
- **FR-DA-CHK-05:** The system shall freeze a checklist, its items and their evidence once the gate it belongs to has been submitted for decision or, for a checklist not related to a gate, once it has been signed off.
- **FR-DA-CHK-06:** The system shall let a checklist that is not related to any gate be attached directly to a phase, a task or a project, with its completion tracked and reported; such a checklist shall not block any progression.
- **FR-DA-CHK-07:** The system shall show, before a checklist is changed or deleted, the gates, templates and open phases that use it; a checklist associated with a published gate shall be retired rather than deleted, and deleting a checklist shall never alter the copies already made in projects.

4.24 Attachments

Actions and checklist items carry the documents that support them.

- **FR-DA-ATT-01:** The system shall let a user attach several files at once to an action or a checklist item, by file picker or drag-and-drop.
- **FR-DA-ATT-02:** The system shall accept a published list of formats — documents, spreadsheets, presentations, images, drawings and CAD, data and models, archives, audio and video, e-mails — with a size and count limit per upload, and shall refuse any other type with a message naming the refused file.
- **FR-DA-ATT-03:** The system shall show for every attachment its type, size, author and date, with download for every viewer and removal for its author or a manager, audited, and never once the stage is submitted.
- **FR-DA-ATT-04:** The system shall store attachments outside every web-served directory, scoped to the tenant, and include them in the backup instructions (NFR-DA-SEC-13, FR-DA-OPS-01).
- **FR-DA-ATT-05:** The system shall version attachments: a user may upload a new version of an attachment with a note; the versions form one chain with number, author and date, the latest is shown by default, and every earlier version stays downloadable.

- **FR-DA-ATT-06:** The system shall let attachments be added to steps, documents and registers as well as to actions and checklist items, under FR-DA-ATT-01 to 05.

4.25 Vertical Framework & Governance

A vertical is a first-class, governed configuration entity. Its content is data, so any application can serve new sectors without code changes.

- **FR-DA-VRT-01:** The system shall support the definition of verticals, each with a unique identifier, a short prefix, a name, a description, its target segments, its value proposition, and its associated macro and E2E processes.
- **FR-DA-VRT-02:** The system shall support a taxonomy of verticals with parent-child and sibling relationships.
- **FR-DA-VRT-03:** The system shall let administrators and partners define new verticals without modifying the platform, and new verticals shall work alongside existing ones.
- **FR-DA-VRT-04:** The system shall manage each vertical through the lifecycle Draft, Review, Approved, Active, Deprecated and Retired, with an approval workflow (configurable approvers, escalation and notifications) for definition, activation, modification, deprecation and retirement.
- **FR-DA-VRT-05:** The system shall version-control vertical definitions under the Version Management service (Section 3.8), so that every change can be tracked, reviewed, approved and rolled back, including the rollback of an activation or configuration change to a previous version.
- **FR-DA-VRT-06:** The system shall let each tenant activate its own set of verticals without affecting other tenants.
- **FR-DA-VRT-07:** The system shall refuse to deactivate a vertical, a macro process or an E2E process while active projects depend on it, shall require approval for a deactivation, and shall preserve all data of a deactivated vertical.
- **FR-DA-VRT-08:** The system shall keep an immutable, searchable and exportable audit trail of every vertical action — definition, activation, modification, deactivation and retirement — retained per policy (Section 4.11).
- **FR-DA-VRT-09:** The system shall record a designated owner for every vertical and shall support change management for it: change request, impact analysis, approval and tracked implementation.
- **FR-DA-VRT-10:** The system shall capture and report governance metrics per vertical: adoption, compliance and performance.

4.26 Vertical Processes

Selecting a vertical activates its own processes, built on the common process model of the application.

- **FR-DA-VPR-01:** The system shall activate the macro processes of a vertical when the vertical is activated, with a configurable range of three to five per vertical.
- **FR-DA-VPR-02:** The system shall describe every vertical macro process with an identifier, name, description, goals, input suppliers, inputs, tasks, outputs, output customers, related E2E processes, related modules and applicable compliance requirements.

- **FR-DA-VPR-03:** The system shall map each vertical macro process to the regulatory and industry standards that apply to it; the mapping shall be configurable and audited.
- **FR-DA-VPR-04:** The system shall let administrators and partners add vertical macro processes without modifying the platform, versioned and reversible like verticals (FR-DA-VRT-05).
- **FR-DA-VPR-05:** The system shall support tasks within each macro process (owner, deadline, dependencies, status) and steps within each task (owner, inputs, outputs, status), and shall enforce the dependencies defined between macro processes, tasks and steps.
- **FR-DA-VPR-06:** The system shall activate the E2E processes of a vertical when the vertical is activated, with a configurable range of two to four per vertical.
- **FR-DA-VPR-07:** The system shall enforce the chain relationships between E2E processes — sequential, branching, parallel, loop-back and terminal — and shall support, per E2E process, triggers (business event, time-based, manual) and terminal states (success, failure, cancellation, loop-back).
- **FR-DA-VPR-08:** The system shall support user-facing tasks in each E2E process, each linked to the macro processes, tasks and steps it relies on.
- **FR-DA-VPR-09:** The system shall capture metrics for each vertical macro and E2E process — cycle time, completion rate, quality and compliance — and produce status, compliance and performance reports from them (Section 4.8).

4.27 Vertical Data, Compliance & Integration

Each vertical may extend the data model, carry its own compliance obligations and connect to sector systems, always without breaking the common core.

- **FR-DA-VDT-01:** The system shall let a vertical extend the common data model with its own objects, attributes, relationships, validation rules and lifecycle states, without altering or breaking the common entities (Section 6).
- **FR-DA-VDT-02:** The system shall exchange vertical data in industry-standard formats, at least JSON, XML, CSV and PDF, and, where the vertical needs it, engineering formats such as STEP, IGES and JT.
- **FR-DA-VDT-03:** The system shall apply per-vertical retention policies and support archiving and retrieval of vertical data, in line with the regulatory retention of the sector.
- **FR-DA-VCP-01:** The system shall support vertical-specific regulations and standards as configurable, audited and reportable compliance requirements, subject to FR-DA-CFG-09.
- **FR-DA-VCP-02:** The system shall monitor compliance per vertical, alert stakeholders of non-compliance through the Notification Center (Section 4.9), and produce compliance reports, audit reports and certification records.
- **FR-DA-VCP-03:** The system shall support internal and external compliance audits — planning, execution, findings and corrective actions — and vertical compliance training with content, delivery, tracking and certification.
- **FR-DA-VIN-01:** The system shall connect vertical-specific external systems through standard interfaces (REST, SOAP, GraphQL, file-based, message-based), under the integration rules of FR-DA-CFG-10 to 14.
- **FR-DA-VIN-02:** The system shall secure, monitor (performance, errors, data quality) and test every vertical integration, and shall handle integration errors by retry, fallback and notification.

4.28 Vertical Configuration

Vertical parameters are configured, validated and versioned before activation.

- **FR-DA-VCF-01:** The system shall let an administrator configure, per vertical: macro and E2E process activation, checklist templates, gate criteria and data model extensions.
- **FR-DA-VCF-02:** The system shall provide a pre-built configuration template per vertical for rapid deployment.
- **FR-DA-VCF-03:** The system shall validate a vertical configuration for completeness, consistency and compliance before activation, and shall refuse an activation that fails validation, naming each failed rule.
- **FR-DA-VCF-04:** The system shall version every vertical configuration change and audit who changed what, when and why.

4.29 SME Mode & Tracks

The SME mode is a cross-sector enabler: an SME in any vertical can run it alongside that vertical.

- **FR-DA-SME-01:** The system shall define an SME by configurable employee thresholds per region (default 1–250 in the EU, up to 500 under the US SBA definition) and shall segment SMEs into Micro (1–10), Small (10–50) and Medium (50–250).
- **FR-DA-SME-02:** The system shall let an Organization activate the SME mode together with any vertical, and each tenant shall activate its own tracks and configuration without affecting other tenants.
- **FR-DA-SME-03:** The system shall support two to four configurable SME tracks, each defining its number of macro processes, E2E processes, gates, checklist items per gate, and typical duration.
- **FR-DA-SME-04:** The system shall activate three to five SME macro processes and two to six SME E2E processes when a track is selected, with the same taxonomy, task, step, chain and metric rules as vertical processes (FR-DA-VPR-02, 05, 07, 08 and 09) plus track applicability and cost metrics.
- **FR-DA-SME-05:** The system shall let administrators define new SME tracks, macro processes and configurations without modifying the platform.
- **FR-DA-SME-06:** The system shall assign a project in SME mode to the track recommended by its complexity score (Section 4.30); a manual override shall require a written justification and an approval, and shall be audited and reportable.
- **FR-DA-SME-07:** The system shall capture and report, per track, its usage, effectiveness and outcomes.
- **FR-DA-SME-08:** The system shall govern the SME framework like verticals: a designated owner, an approval workflow for track definition, activation, modification, deprecation and retirement, change management, an immutable audit trail and adoption, compliance and performance metrics.

4.30 Complexity Scoring

A complexity score gives every project the right amount of rigor, consistently and traceably.

- **FR-DA-SCO-01:** The system shall compute a complexity score for every project from weighted criteria and shall use it to recommend the mode and track.
- **FR-DA-SCO-02:** The system shall let an administrator configure the criteria — at least strategic impact, technical novelty, regulatory burden, investment size, cross-functional scope, market uncertainty and time-to-market pressure — and their weights, which shall sum to 100 %; a set of weights that does not sum to 100 % shall be refused.
- **FR-DA-SCO-03:** The system shall let each vertical add its own complexity drivers (for example regulatory burden, industry standards, sector-specific risks) with their weights, versioned, so that the score and the recommended track reflect the vertical of the project.
- **FR-DA-SCO-04:** The system shall keep an audit trail of every score — criteria, weights, values, result and track assignment — and shall report the distribution of scores, track assignments and outcomes.

4.31 SME Onboarding & Experience

SMEs need to go live quickly, often with little prior experience of the application's domain.

- **FR-DA-ONB-01:** The system shall run an onboarding plan for an SME within a configurable period (default 30 days): tenant provisioning (FR-DA-TEN-14), track configuration, data import, validation, user training and go-live.
- **FR-DA-ONB-02:** The system shall import data from spreadsheets, CSV files, legacy systems and cloud storage, preserving data integrity and relationships, and shall validate the imported data for completeness, consistency and accuracy before go-live, listing every rejected record with its reason.
- **FR-DA-ONB-03:** The system shall provide self-service training and guided onboarding: interactive tutorials, guided workflows, sample data and contextual help (Section 4.12).
- **FR-DA-ONB-04:** The system shall offer SME users simplified navigation, a mobile-responsive or progressive web app experience, and self-service support through the knowledge base, the AI Assistant (Section 4.10) and a community space.
- **FR-DA-ONB-05:** The system shall measure onboarding: time-to-value, data quality, user adoption and satisfaction.

4.32 Packaging & Pricing

SME packs, vertical packs and their bundles extend the Solution Pack model of Section 4.3.

- **FR-DA-PKG-01:** The system shall let an administrator define three to five SME packs, each with configurable contents, target segment and price, and bundles of a vertical pack with an SME pack, each recorded as a Solution Pack variant so that entitlement stays enforced by FR-DA-CFG-06.
- **FR-DA-PKG-02:** The system shall enforce configurable pricing rules — minimum price per user per month, minimum user count per pack, volume discounts and bundle discounts — and shall show how each price was obtained.
- **FR-DA-PKG-03:** The system shall support upgrading and downgrading between packs and bundles with prorated billing and without loss of data.
- **FR-DA-PKG-04:** The system shall report revenue, discounts, adoption, upgrades and downgrades per pack and bundle, exportable to PDF, XML and CSV.

4.33 Project Template Catalog

Every application offers a catalog of project templates. A template is a starting point copied at creation time (FR-DA-TPL-04).

- **FR-DA-PTC-01:** The system shall keep a Project Template Catalog in which every template is classified by scope — Universal (common to all sectors) or one Vertical — and by mode — Full, or SME with its track.
- **FR-DA-PTC-02:** The system shall support full Create, Read, Update, Delete and Duplicate on project templates, versioned under Section 3.8; a template shall carry its default fields, its phases, the gates (Section 4.35) and checklists (Section 4.23) attached to each phase, its OBS roles and its milestones.
- **FR-DA-PTC-03:** The system shall present the catalog as a browsable list with search and filters on scope, vertical, mode, track and status, a preview of each template's phases, gates and checklists, and its use count.
- **FR-DA-PTC-04:** The system shall manage each template through Draft, Published and Retired; only published templates shall be offered at project creation, and retiring a template shall not change any project created from it.
- **FR-DA-PTC-05:** The system shall provide universal templates for Full and SME mode on every installation, seed vertical templates when a vertical is activated, and let an authorized user save an existing project as a new template.
- **FR-DA-PTC-06:** The system shall restrict catalog changes to a dedicated permission, audit every change, and let platform templates be shared read-only with all Organizations while Organization templates stay private to their tenant.
- **FR-DA-PTC-07:** The system shall, from the project creation screen, let the user view the selected template's blueprint and open it for customization (FR-DA-PTB-01 to 10).
- **FR-DA-PTC-08:** The system shall offer, at project creation, the library templates and the organization's own published templates, the organization's first, marked as such.

Project template blueprints. A project template is a complete blueprint of the projects created from it, with full Create, Read, Update and Delete at every level. The (New) SRS revision 1.7 specifies this family in its Section 4.49; the IDs below are the same.

- **FR-DA-PTB-01:** The system shall hold, for every project template, a Project Template Blueprint: the end-to-end processes, macro processes, tasks and steps it includes, its gates, business rules, controls, risks and opportunities, alerts, KPIs and its Reporting Plan.
- **FR-DA-PTB-02:** The system shall offer a project template editor with one view per part of the blueprint — overview, processes and steps, rules, controls, risks, alerts, KPIs and reporting — each with full Create, Read, Update and Delete.
- **FR-DA-PTB-03:** The system shall let a template include or exclude each phase, macro process, task and step of the reference process design, and set for each included element its name, owner and roles.
- **FR-DA-PTB-04:** The system shall let a template add Custom Elements — phases, macro processes or steps — to the reference process design; a Custom Element exists only in projects created from that template and never changes the reference process design.

- **FR-DA-PTB-05:** The system shall manage the template status (draft, published, retired), increase its version on every change, refuse to publish a template with no included step, and delete a template only when no project uses it (otherwise retire it).
- **FR-DA-PTB-06:** The system shall, when a project is created from a template, apply its blueprint exactly: included phases, macro processes and steps with their names, owners and roles, gates, KPIs and risks; add its rules and controls to the organization when missing; apply its alert settings to the project; and record the template code and version in the project.
- **FR-DA-PTB-07:** The system shall let a template define its Reporting Plan — for each report its name, audience, frequency, format and owner — and plan those reports in every project created from it.
- **FR-DA-PTB-08:** The system shall let the rules, controls, risks and opportunities, alerts and KPIs of a blueprint be taken from the Organization's libraries, entered manually or proposed by the AI with human validation, and record the origin of each row (FR-DA-LNK-02).
- **FR-DA-PTB-09:** The system shall keep, in every project created from a template, a trace of the blueprint applied — template code and version and the blueprint element each project element came from — so that a later version of the template never changes an existing project.
- **FR-DA-PTB-10:** The system shall let an authorized user duplicate a template, compare two versions of its blueprint, and export the blueprint in Word, PDF and Excel.

4.34 Project Creation Modes

A project can be created in three ways; all three end in the same, fully editable project.

- **FR-DA-PCM-01:** The system shall offer three creation modes on one entry point: From the catalog, Manual, and With AI.
- **FR-DA-PCM-02:** From the catalog: the system shall propose the templates that match the Organization's verticals and the project's mode and track, and copy the chosen template's fields, phases, gates, checklists and roles into the new project.
- **FR-DA-PCM-03:** Manual: the system shall let the user enter the project and choose its vertical (or none), mode and track, and build its phases by attaching gates from the Gate Library and checklists from the Checklist Library, or by creating new ones.
- **FR-DA-PCM-04:** With AI: the system shall draft the project — fields, complexity scores, recommended mode and track, closest template, phases, gates and checklists — from a short description, as an Assistive AI suggestion that the user accepts, modifies or rejects item by item before anything is saved (FR-DA-AI-07); the draft shall work with the built-in engine and use the live model when configured.
- **FR-DA-PCM-05:** The system shall compute the complexity score (Section 4.30) in every creation mode, apply the quota checks of NFR-DA-SEC-08, and record in the audit trail the creation mode, the template and version used, and the AI suggestions accepted.

4.35 Gate Library

The Gate Library holds every gate definition. Checklists from the Checklist Library (Section 4.23) are associated with a gate; gates are then attached to any phase of any project or project template.

- **FR-DA-GTE-01:** The system shall keep a Gate Library with full Create, Read, Update, Delete and Duplicate on gate definitions, versioned: name, purpose, entry and exit criteria, approvers (roles or people), decision options Go, No-Go, Hold and Recycle, and applicability by vertical, mode and track.
- **FR-DA-GTE-02:** The system shall let an authorized user associate zero, one or several checklists of the Checklist Library with a gate, optionally per track, each marked mandatory or optional, in a chosen order; a checklist created from the gate screen shall be saved in the Checklist Library, and the same checklist may be associated with several gates.
- **FR-DA-GTE-03:** The system shall let an authorized user attach one or more gates to any phase of a project or a project template, reorder them, and detach them while the phase has not started; attaching a gate brings its associated checklists with it.
- **FR-DA-GTE-04:** The system shall configure the gates of a project from its vertical and track: the number of gates depends on the track and the criteria on the vertical, and a gate may be marked optional for a track.
- **FR-DA-GTE-05:** The system shall block the progression of a phase while a mandatory item of a mandatory checklist associated with its gate is incomplete; enforcement shall be configurable per gate and a waiver shall require justification and approval.
- **FR-DA-GTE-06:** The system shall capture every gate decision with its approvers, criteria results and comments, and keep an immutable audit trail of gate actions — creation, modification, checklist association, attachment to a phase, review, decision and waiver.
- **FR-DA-GTE-07:** Changing a gate, or a checklist associated with it, shall not alter the copies already attached to started phases; the user shall be offered to update not-yet-started phases to the new version.

4.36 Questionnaire & Survey Management

An application that collects evidence from people — employees, managers, clients or partners — designs questionnaires from a governed template library, distributes them through the channels respondents actually use, and turns consented, complete responses into a dataset its own processes consume.

- **FR-DA-QLT-01:** The system shall support full Create, Read, Update, and Delete on questionnaire templates, each carrying a focus tag (All needs, Digital, AI or a named Custom focus), a sector tag, a language, and a version (FR-DA-TPL-06 to 09).
- **FR-DA-QLT-02:** The system shall let an authorized user elaborate a questionnaire by Load (copying a template), by AI (generating questions from the approved scope through a governed AI use case, Section 4.4), or Manually; the modes may be combined, and the mode of each question shall be recorded.
- **FR-DA-QLT-03:** The system shall label every AI-generated or AI-tailored question and shall block distribution until an authorized user has validated each such question; the validation shall record the validating user and time.
- **FR-DA-QLT-04:** The system shall allocate RACSI roles to each questionnaire section, reusing the RACSI model of Section 4.6, and shall block distribution while any section has no Accountable; a section shall have exactly one Accountable.

- **FR-DA-QLT-05:** The system shall distribute a questionnaire through Face-to-Face, Email, Hybrid and Custom channels, selectable per stakeholder or stakeholder segment, through the questionnaire dispatch abstraction of FR-DA-COMM-06.
- **FR-DA-QLT-06:** The system shall capture the respondent's consent before storing any response; a response submitted without consent shall not be stored, and the rejection shall be logged.
- **FR-DA-QLT-07:** The system shall send reminders on the cadence configured per channel for each questionnaire (FR-DA-COMM-07).
- **FR-DA-QLT-08:** The system shall support offline response collection for the Face-to-Face channel and synchronize captured responses when connectivity returns, without duplicates and with the original capture time preserved.
- **FR-DA-QLT-09:** The system shall compute a completeness percentage for every response and validate it against a configurable threshold before consolidation; a response below the threshold shall be flagged and excluded unless an authorized user includes it with a justification (Section 3.7).
- **FR-DA-QLT-10:** The system shall consolidate validated responses into a structured dataset scoped to the requesting user's tenant, never mixing responses from another Organization.
- **FR-DA-QLT-11:** The system shall make the consolidated dataset available as an input to the application's downstream processes — for example a needs analysis, a diagnostic or a report — without export or re-keying.
- **FR-DA-QLT-12:** The system shall report response rate, engagement rate and completeness per channel, per focus and per stakeholder segment, exportable under the conventions of Section 4.8.

4.37 Global Search

A user finds anything they may open from one place, whatever the screen they are on.

- **FR-DA-SRCH-01:** The system shall provide a global search, opened from the Search button of the primary menu (FR-DA-NAV-12) or its keyboard shortcut, from every authenticated screen.
- **FR-DA-SRCH-02:** The system shall search, in the current Organization and within the scope of the user's rights: principal records, projects, process elements (functions, processes, phases, tasks, steps, gates, checklists, templates), governance items (rules, controls, risks, KPIs, RACSI), actions, registers, documents, attachment names, AI Use Cases, people and OBS units and roles, and Help topics.
- **FR-DA-SRCH-03:** The system shall match codes and identifiers exactly (for example a step code) and text by words, ignoring case and accents, in every supported language, and shall rank exact code matches first.
- **FR-DA-SRCH-04:** The system shall show results grouped by type, each with its code, title, location (project, process or register) and status, and open the chosen result on its own screen.
- **FR-DA-SRCH-05:** The system shall let the user filter results by type, project, process, status and date, and keep the user's recent searches as a personal preference.
- **FR-DA-SRCH-06:** The system shall apply to every result the RBAC, Solution Pack and tenant rules of the screen it opens (NFR-DA-SEC-18).

- **FR-DA-SRCH-07:** The system shall update the search index when a record is created, changed or deleted, so a change is findable within one minute.

4.38 Process Design Management

The process design is managed as data, with full history, by the people entitled to change it.

- **FR-DA-PDM-01:** The system shall support full Create, Read, Update, Delete and Duplicate on every Process Element: functions, end-to-end processes, phases, macro processes, tasks, steps, gates, checklists and their items, step forms and their fields, templates (project, checklist, document, questionnaire) and AI Use Cases, gated by a process-design management capability.
- **FR-DA-PDM-02:** The system shall version every Process Element under the generic Version Management service (Section 3.8): an edit creates a new version with its author, date, change note and justification (Section 3.7), and no version is rewritten or deleted.
- **FR-DA-PDM-03:** The system shall let an authorized user open any version of a Process Element read-only, compare any two versions field by field (FR-DA-VER-06), and restore any version as the new Current Version.
- **FR-DA-PDM-04:** The system shall version composite elements with their children (FR-DA-VER-08): restoring a version of a macro process restores its tasks and steps, their order, forms, links to gates, checklists, RACSI and AI Use Cases as they were.
- **FR-DA-PDM-05:** The system shall manage Design Releases through Draft, In review, Published and Retired; changes made in a draft release shall not affect running work until the release is published with the approval of an authorized user, and a published release shall be restorable.
- **FR-DA-PDM-06:** The system shall bind every running record to the element versions current when it started, or when its phase opened. A new version applies to new records; an authorized user may migrate running records to it after an impact preview, with a justification, and completed steps shall never change.
- **FR-DA-PDM-07:** The system shall show, before an element is changed, retired or deleted, what uses it: child elements, templates, releases and running records. An element used by a published release or a running record shall be retired rather than deleted; a retired or deleted element shall be restorable from its history.
- **FR-DA-PDM-08:** The system shall let an authorized user reorder elements, move a task or step to another parent and link or unlink elements, with the referential checks of FR-DA-TEN-05; identifiers shall be stable and never reused.
- **FR-DA-PDM-09:** The system shall let the platform share a reference process design read-only with all Organizations, and let an Organization customize it by copy; the copy shall keep the source and version it came from and be offered the changes of a newer source version, element by element.
- **FR-DA-PDM-10:** The system shall export a process design, or part of it, to a portable file (JSON and spreadsheet, with BPMN 2.0 for the diagrams) and import it back as new versions, never overwriting history.
- **FR-DA-PDM-11:** The system shall write every Create, Update, Delete, restore, migration and publication of a Process Element to the audit trail (Section 4.11).

4.39 OBS: Functions, Roles and People

The OBS is made of units, roles and people. A role is linked to one or more functions, and a person can play one or more roles.

- **FR-DA-OBS-01:** The system shall keep a catalog of functions per Organization, with full Create, Read, Update and Delete, versioned, each with a code, a name, a description and an optional parent function, and seed a default set on provisioning (FR-DA-TEN-14).
- **FR-DA-OBS-02:** The system shall support full Create, Read, Update and Delete on OBS Roles, each defined in an OBS unit, with a name, a mission, responsibilities, required competences, an optional reporting role, and the default access roles it proposes to its holders.
- **FR-DA-OBS-03:** The system shall let an OBS Role be linked to one or more functions, and a function to several roles; a role shall be linked to at least one function.
- **FR-DA-OBS-04:** The system shall let a person play one or more OBS Roles, in the same or different units, through Role Assignments with a start date, an optional end date, an allocation percentage and a holder type (holder, deputy, acting), and shall let a role be played by several people.
- **FR-DA-OBS-05:** The system shall show the OBS by unit (units, roles and their holders), by function (the roles and people linked to each function) and by person (the roles a person plays), and as an organization chart.
- **FR-DA-OBS-06:** The system shall let step owners, RACSI letters, approvers, gate approvers, reviewers and notification targets be assigned to OBS Roles, and resolve them to the people currently playing the role; a role with no current holder shall raise an alert (Section 4.9).
- **FR-DA-OBS-07:** The system shall, when a Role Assignment ends, move the open work assigned through that role to its remaining or next holder, and list the work assigned to the person by name for re-assignment.
- **FR-DA-OBS-08:** The system shall offer, wherever a person or a unit is chosen, pickers drawn from the OBS — people with their roles, roles with their holders, units with their type — rather than free text.
- **FR-DA-OBS-09:** The system shall keep OBS Roles distinct from RBAC access roles (Section 3.2); holding an OBS Role shall grant no permission by itself.
- **FR-DA-OBS-10:** The system shall version units, roles, role-function links and Role Assignments under Section 3.8, so the OBS as it was on any past date can be shown, and enforce the quotas of FR-DA-TEN-07.

4.40 AI Use Case Mapping & Prompt Specification

Every AI Use Case is attached to the right task or step and carries a complete, structured and versioned prompt.

- **FR-DA-AIP-01:** The system shall link every AI Use Case to the tasks or steps it supports, the link naming the input fields it reads and the output fields it proposes, and shall show the use case only on those tasks and steps (FR-DA-AI-12, FR-DA-AI-13).
- **FR-DA-AIP-02:** The system shall check each link for fit — the use case's expected output matches the output or a field of the step, and its inputs exist at that point of the process — and flag a link that fails the check for review.

- **FR-DA-AIP-03:** The system shall hold, for every AI Use Case, a Prompt Specification made of separate fields: Role (persona); Context (the process, task or step, the standards and clauses applicable, the Organization profile); Task (the instruction); Inputs (named variables bound to record and step data); Knowledge sources (the corpora and references to retrieve from); Constraints (what the answer must and must not do, including the grounding rule of FR-DA-AI-11); Examples (one or more input and output pairs); Output format (structure, fields, length and a schema where the output fills a form); Tone and language; Quality criteria (how a good answer is judged); Human checkpoint (who reviews and what they check); and Model parameters (model, temperature, maximum length).
- **FR-DA-AIP-04:** The system shall let an authorized user edit each field of a Prompt Specification on its own, and shall version each change under Section 3.8 with its author, date and note.
- **FR-DA-AIP-05:** The system shall let the user see the history of the whole specification and of each field, compare any two versions, and restore any version of one field or of the whole specification as the new Current Version.
- **FR-DA-AIP-06:** The system shall require, before an AI Use Case can be activated, that Role, Context, Task, Constraints, Output format and Human checkpoint are filled, and shall show the completeness of every specification.
- **FR-DA-AIP-07:** The system shall populate every seeded or AI-suggested use case fully for the task or step it is linked to: the Context shall name that task or step, its inputs, outputs and applicable clauses; the Inputs shall bind to its actual fields; the Examples and the Output format shall match its form; every seeded specification shall be complete (100 %), its Examples holding at least one input and output pair built from the form of the linked step.
- **FR-DA-AIP-08:** The system shall assemble the prompt from the fields in a fixed, documented order, resolve every variable, and refuse to run when a required variable has no value, naming it.
- **FR-DA-AIP-09:** The system shall let an authorized user test a Prompt Specification on a chosen record before publishing a new version, without writing the result to the record.
- **FR-DA-AIP-10:** The system shall record in the AI Usage Log (FR-DA-AI-07) the use case, the specification version, the engine and model used, and the outcome of every run.
- **FR-DA-AIP-11:** The system shall let an AI Use Case be suggested for a task or step that has none — by rule or with AI — as a draft with a fully populated Prompt Specification, activated only after a person reviews it.
- **FR-DA-AIP-12:** The system shall show with every AI answer the engine that produced it (Engine Transparency) and, when the Organization's live model was not used, a visible message giving the reason — no model enabled, or the error the model returned — with a link to the model settings for users entitled to change them.
- **FR-DA-AIP-13:** The system shall make the built-in engine produce, when it answers, content of the type the use case expects (for example a draft text built from the records already captured), never an echo of the input.
- **FR-DA-AIP-14:** The system shall send as prompt inputs only the values the user typed or the records named in the specification, never the state of interface controls (for example the choice made in a list of outcomes).
- **FR-DA-AIP-15:** The system shall insert an accepted or edited suggestion into the output field the use case names (for example the final validated text), never into an input field.

- **FR-DA-AIP-16:** The system shall let an AI suggestion fill several rows of a record table, each row linked to the records it concerns and marked with the Entry Origin AI, for the user to review, edit or delete before completion.

4.41 Process Element Naming & Descriptions

Names say what is done; descriptions say how.

- **FR-DA-NAM-01:** The system shall require an explicit name for every task and step, beginning with an action verb followed by its object (for example “Identify the interested parties”), unique within its parent, in every supported language.
- **FR-DA-NAM-02:** The system shall check names on save — missing verb, generic wording, abbreviation, duplicate — and show each finding as a warning that the user may accept or correct.
- **FR-DA-NAM-03:** The system shall give every step a brief (one line, shown by default) and a detailed description (purpose, inputs, how to fill each field, expected result), expandable and collapsible, in every supported language.
- **FR-DA-NAM-04:** The system shall show on every step its process, inputs and outputs, the standards and clauses of the record that it covers, and the rules and controls that apply to it.

4.42 Typed Step Forms & Process Readiness

Each step collects the information it needs in a form suited to it, and its completion feeds the registers that follow.

- **FR-DA-SFM-01:** The system shall give every step a Step Form Kind from a catalog — at least record table, assessment matrix, decision, objectives, plan, communication, training, monitoring, review, assignment, configuration, document and free record — each with its own fields, checks and completion effects.
- **FR-DA-SFM-02:** The system shall capture lists as tables of rows, one row per item, with add, edit and delete, each row carrying its source (a named record, study, interview or date) rather than free text.
- **FR-DA-SFM-03:** The system shall provide a decision or assessment matrix with criteria, weights, scores on a stated scale and the facts justifying each score, computing the weighted score and showing the scale next to it (FR-DA-HLP-04).
- **FR-DA-SFM-04:** The system shall, when a step is completed, create or update the records it implies — rows of a plan become actions, objectives become entries of the objectives register, KPI rows become KPI values with their alerts — and link each created record back to its row.
- **FR-DA-SFM-05:** The system shall let a KPI be chosen from the KPI catalog, or created in place, with the reason it is monitored, and let people and units be chosen from the OBS (FR-DA-OBS-08).
- **FR-DA-SFM-06:** The system shall validate the mandatory fields of a form before completion, naming each missing field, and lock the step once the gate of its phase is passed (FR-DA-AUD-05 governs reopening).
- **FR-DA-SFM-07:** The system shall show, before a macro process starts, an optional Readiness Checklist: required inputs, completed predecessor processes, owner and RACSI assigned,

templates available and KPIs defined. Items the system can verify shall be checked automatically; the checklist shall inform and never block.

- **FR-DA-SFM-08:** The system shall choose the Step Form Kind from the meaning of the step, not only from its object: in particular, a step that maps functions to decision levels shall be a Function × Level Matrix — one row per function of the Organization, one tick column per decision level, an action adding every function not yet listed and an AI proposal derived from the OBS — never a generic list of items with a category and a description.
- **FR-DA-SFM-09:** The system shall, when a step is reopened (FR-DA-AUD-05), clear every completion mark of that step — its step record and the step list of its task — so that the step reads as open everywhere, and record the reopening with its justification.

4.43 Documented Information Templates & Generated Documents

Controlled documents and records are produced from the application's own data, not retyped.

- **FR-DA-DOC-01:** The system shall keep a library of documented-information templates with full Create, Read, Update, Delete and Duplicate, versioned, each with a code, a category, the formats it produces (Word, PDF, Excel), and ordered sections typed as text, table or approval block.
- **FR-DA-DOC-02:** The system shall let each section be bound to a named data source of the application (for example context issues, interested parties, objectives, KPIs, risks, RACSI, audits, nonconformities, actions), with free text that may use variables such as the Organization name.
- **FR-DA-DOC-03:** The system shall list, per standard in scope, the documented information it requires — to maintain (a document) or to retain (a record) — with the template and the document that answer each requirement, and create a missing one in one action.
- **FR-DA-DOC-04:** The system shall generate a document from a template and the current data of a project, as a draft version, and let it be regenerated from newer data or edited section by section.
- **FR-DA-DOC-05:** The system shall manage document versions through Draft, In review and Published, with approval by a person other than the author (two-person rule) and a change note; one version is published at a time and earlier versions stay downloadable.
- **FR-DA-DOC-06:** The system shall download every document version in each of its formats, with a cover page, a table of contents where relevant, headers, footers and page numbers.
- **FR-DA-DOC-07:** The system shall let an Organization set its document layout — logo, colors, fonts, header and footer — applied to every generated document.
- **FR-DA-DOC-08:** The system shall let an Organization copy a library template to adapt it; its copy replaces the library template within the tenant and keeps the version it came from.
- **FR-DA-DOC-09:** The system shall let a document be retired, with a justification, rather than deleted once published; a draft may be deleted.
- **FR-DA-DOC-10:** The system shall link generated documents to the step, process and project they come from, and show them on that step.
- **FR-DA-DOC-11:** The system shall seed the template library and, in the demonstration data (FR-DA-OPS-09), documents generated for every seeded project, ready to download.

Document formatting and pictures.

Note: These requirements were previously covered inside FR-DA-DOC-06 (cover page, table of contents, headers, footers and page numbers) and FR-DA-DOC-07 (Organization document layout); they are given their own IDs here to align with the (New) SRS, which specifies them in its Section 4.50.

- **FR-DA-DFP-01:** The system shall let a document template set its Document Formatting: body and heading fonts and sizes, alignment, page orientation, margins, cover page, table of contents, heading numbering, page numbers, colors, header and footer.
- **FR-DA-DFP-02:** The system shall let a document template choose its logo (the Organization logo, another uploaded logo or none), the position of the logo on the cover page, and whether the logo appears in the page header.
- **FR-DA-DFP-03:** The system shall let each section of a template or draft document carry its Section Formatting: alignment, text size and color, background (callout), bold, italic, page break before and a hidden title.
- **FR-DA-DFP-04:** The system shall support Picture Sections showing an uploaded PNG or JPEG picture or logo with a caption, a width as a percentage of the page and an alignment; the picture shall keep its proportions and never exceed the page.
- **FR-DA-DFP-05:** The system shall apply the Document Formatting, Section Formatting and Picture Sections in every output format (Word, PDF, Excel), using the organization layout (FR-DA-DOC-07) for any value the template leaves empty.
- **FR-DA-DFP-06:** The system shall preview the Document Formatting while it is edited.
- **FR-DA-DFP-07:** The system shall let a draft document edit its own sections, pictures and formatting without changing the template it was generated from.
- **FR-DA-DFP-08:** The system shall let the user add, move, duplicate and delete sections of every kind — text, table, approval block and picture — in a template and in a draft document.
- **FR-DA-DOC-12:** The system shall produce Word, Excel and PowerPoint files that open in their reference applications without a repair prompt: each package shall pass Office Open XML schema validation, every shape and drawing object of a slide shall have a unique identifier, every chart axis referenced by a series shall be declared, and the package shall hold no directory entries; a release shall check every generated file type automatically.

4.44 Generated Documents: Consistency & Fine-Grained Content

A generated document is detailed, complete for its type and consistent with the application's data and with every other document.

- **FR-DA-DGC-01:** The system shall draw every value of a generated document from the application's records through a Data Source Binding; the same item shall appear with the same identifier, name and values in every screen and every document that shows it.
- **FR-DA-DGC-02:** The system shall trace every data-bound table row and value of a document to its source record (identifier and version), list the data sources used at the end of the document, and show the date and time of the data ("data as of").
- **FR-DA-DGC-03:** The system shall, when the register a section is bound to has no entry yet, use the records typed in the steps that feed it; when no data exists at all, it shall print a visible notice naming the step or register that provides the data, never an empty section or an empty table.

- **FR-DA-DGC-04:** The system shall define a Document Profile per document type, and each generated document shall carry at least: an identification block (reference, title, version, status, owner, approver, dates, classification, distribution), purpose and scope, references and definitions, method and rating scales where scores are used, the data sections of its type, analysis and conclusions, actions with owner and due date, a revision history and an approval block.
- **FR-DA-DGC-05:** The system shall produce data sections at the finest grain of the data: one row per item, each with its own category, description, relevance or score, owner, dates, status and source or evidence, rather than grouped or summarized text.
- **FR-DA-DGC-06:** The system shall, for a procedure or process description, include the process flow as a diagram (BPMN 2.0 with lanes per role, gateways and exception flows), the inputs and outputs of each activity (SIPOC per step), the decision points with their Go / No-Go criteria, evidence and decider, the records produced, the indicators and the risks and controls.
- **FR-DA-DGC-07:** The system shall run Consistency Checks before a document is sent for review and before it is published: referenced items exist and are current; totals, counts and scores agree with their rows; identifiers and names match the records; every mandatory section of the Document Profile is filled; dates are coherent (next review after last review); and the document agrees with the other published documents of the project that show the same items.
- **FR-DA-DGC-08:** The system shall list the findings of the Consistency Checks with their location in the document and a link to the record to correct; findings marked blocking shall prevent publication, the others shall be accepted with a justification (Section 3.7).
- **FR-DA-DGC-09:** The system shall show, for every published document, whether its source data changed since it was generated, list the changed items, and propose a new version; a data change that affects a published document shall raise an alert to its owner (Section 4.9).
- **FR-DA-DGC-10:** The system shall keep the documents of a project mutually consistent: a change of a shared item (for example a party, a risk, an objective, a process owner) shall list every document that shows it and let the owner regenerate them together.
- **FR-DA-DGC-11:** The system shall lay out each format for reading: a cover page and a table of contents where relevant; landscape pages for wide tables; table headers repeated on each page; no heading or table header left alone at the bottom of a page; diagrams rendered as images in Word and PDF; one sheet per table after a cover sheet in Excel; status values shown with the semantic color scale.
- **FR-DA-DGC-12:** The system shall generate a document in any supported language, with the data in that language, and keep the language of each version.
- **FR-DA-DGC-13:** The system shall keep a master list of the project's documented information up to date automatically: reference, title, version, status, owner, approval date, review frequency, next review, retention for records, and the requirement each document answers.

4.45 Generated Documents: Full CRUD at Every Level

Every level of a generated document can be created, read, changed and removed by the people entitled to, with full history.

- **FR-DA-DCR-01:** The system shall support Create, Read, Update, Duplicate and Delete on documents: create from a template or blank, read on screen and in every format, update the metadata (title, owner, approver, review frequency, classification, distribution), duplicate to another project, and delete a document never published; a published document shall be retired with a justification, not deleted (FR-DA-DOC-09).
- **FR-DA-DCR-02:** The system shall support Create, Read, Compare, Restore and Delete on document versions: create a minor or major version by regeneration from newer data or by copy of the current content; read and download any version; compare any two versions section by section and row by row; restore any version as a new draft; delete a draft version only.
- **FR-DA-DCR-03:** The system shall support, on a draft version, Create, Read, Update, Reorder and Delete on sections: add a section (text, data-bound table, diagram, key-value block, approval block), rename it, move it, delete it, edit its text, and change its Data Source Binding, filters and columns.
- **FR-DA-DCR-04:** The system shall let an authorized user edit a row of a data-bound table from the document through Edit-through — add, change or delete the source record under its own rights, checks and audit — and regenerate the section, so the document and the application stay the same.
- **FR-DA-DCR-05:** The system shall let an authorized user set a Section Override for text or a value that belongs to the document only; the override shall be marked, kept when the document is regenerated, listed in the Consistency Checks and removable to return to the source data.
- **FR-DA-DCR-06:** The system shall keep, when a document is regenerated, the manual text and overrides of its sections, and show the user what the regeneration changed before the new version is saved.
- **FR-DA-DCR-07:** The system shall apply the same CRUD to templates (FR-DA-DOC-01): sections, their order, text, data sources and Document Profile; a template change shall apply to documents generated after it and be offered to existing drafts.
- **FR-DA-DCR-08:** The system shall support Create, Read, Update and Delete on the document layout of an Organization (FR-DA-DOC-07) and on its per-type variants, with a preview, versioned under Section 3.8.
- **FR-DA-DCR-09:** The system shall gate every document, version, section, row and layout action by its own permission, enforce the two-person rule for approval (FR-DA-DOC-05) and write every action to the audit trail (Section 4.11).
- **FR-DA-DCR-10:** The system shall show where to customize documents from the documents screen and from every document: templates (all documents of a type), layout (all documents of the Organization) and structure (this document only).

4.46 Data Entry Usability

Every form names what it expects, proposes values and gives room to type.

- **FR-DA-DEU-01:** The system shall qualify every field and column title with the object it holds in its context (for example "External issue", "Interested party", "Macro process 1"), never a

generic title such as "Item"; a reusable form shall take its titles from the step or record type that uses it.

- **FR-DA-DEU-02:** The system shall present every choice as a Controlled List with a Custom value that opens a free text field, and shall offer again, in the same list, the values already typed in the same table.
- **FR-DA-DEU-03:** The system shall let the user open any row of a record table in a Row Editor — one large field per column, scrolling, previous and next row, and a return to the table — in edit mode and in read-only mode; a new row shall open in the Row Editor.
- **FR-DA-DEU-04:** The system shall let reference items (processes, phases, functions, people, units, records) be chosen from pickers of the current project or Organization, shown as "ID (name)", with a Custom value where an item outside the application is allowed.
- **FR-DA-DEU-05:** The system shall explain every computed or scored field under the field — what it measures, how it is computed and where it is used — and every free-text field that feeds a document.
- **FR-DA-DEU-06:** The system shall keep the user's unsaved input when a dialog or Row Editor is closed with Back, and ask for confirmation only when the input would be lost.

4.47 Linked Records, Choice Libraries & Register Propagation

Records are captured once, linked to each other and reused everywhere.

- **FR-DA-LNK-01:** The system shall support Record Links many to many between records (for example a need or expectation and the interested parties it concerns, a risk and the processes it affects), each end chosen from the records already identified in earlier steps or registers, or typed as a Custom value.
- **FR-DA-LNK-02:** The system shall let multi-valued records be entered in three ways — typed (Manual), taken from a Choice Library (Library) or suggested by the AI (AI) — and record the Entry Origin of each.
- **FR-DA-LNK-03:** The system shall build each Choice Library from the values entered before in the Organization, in any project, together with a reference list managed as configuration data; library values are not linked to any record until the user links them.
- **FR-DA-LNK-04:** The system shall apply Register Propagation when a step is completed: each row of an identification step creates or updates the matching register entry, matched by its identifier or its name, and keeps the identifier of that entry; linked records update the entries they concern.
- **FR-DA-LNK-05:** The system shall keep steps, registers, pickers and documents consistent: an item typed in a step is offered in the pickers of later steps, shown in its register, and printed in the documents bound to that register (FR-DA-DGC-03).

4.48 Audits, Inspections & Graded Findings

Where an application plans audits or inspections, it plans them at a stated frequency and grades each finding.

- **FR-DA-AFP-01:** The system shall let the frequency of an audit or inspection be chosen from a list (for example monthly, quarterly, semi-annual, annual, every two years, every three years) or set as Custom with its description, and derive the next planned date from it.

- **FR-DA-AFP-02:** The system shall record, for each planned audit, its objectives, criteria, scope, processes, duration, lead and team with their qualification, and justify its frequency from the risks and the results of earlier audits.
- **FR-DA-AFP-03:** The system shall grade every finding on a configurable scale — at least major nonconformity, minor nonconformity, observation and opportunity for improvement — with the grading rules shown to the user.
- **FR-DA-AFP-04:** The system shall hold a detailed record for each nonconformity: requirement and clause, statement, objective evidence, process and auditee, root cause, correction, corrective action with owner and due date, verification of effectiveness and status, linked to the corrective action process of the application.
- **FR-DA-AFP-05:** The system shall produce the audit report from these records: summary, strengths, findings by grade and by requirement, one detailed sheet per nonconformity, conclusion and follow-up (Section 4.44).

4.49 Multi-Record Data Entry

Note: the (New) SRS revision 1.7 uses section 4.49 for Project Template Blueprints. In this document, project template blueprints are specified in Section 4.33 (Project Template Catalog), and the FR-DA-PTB family of Section 4.49 of the New file corresponds to FR-DA-PTB-01 to 10 there, under the same IDs. Section 4.49 here remains Multi-Record Data Entry.

Applies to every screen where users create, edit, duplicate or delete several records of the same kind, in particular the record tables of typed step forms (Section 4.42) and registers. It details, for multi-record entry, the data entry usability requirements of Section 4.46. FR-DA-MRE-15 to 23, added in revision 1.8, apply to every Multi-Entry as defined in Section 1.4, whatever its pattern.

Note: The (New) SRS revision 1.7 carries FR-DA-MRE-01 to FR-DA-MRE-08. Those eight requirements are a subset of FR-DA-MRE-01 to FR-DA-MRE-14 below; the longer set in this document supersedes and extends them, and no requirement of the shorter set is lost.

- **FR-DA-MRE-01:** The system shall choose the entry pattern of each record type from its number of fields, and document the choice: an Inline Editable Table for several records of two to eight fields; Master–Detail for several records of more than eight fields; a sectioned form with progressive disclosure for one record of many fields; a simple form for one record of few fields. The Row Editor (FR-DA-DEU-03) shall remain available from every row of an Inline Editable Table.
- **FR-DA-MRE-02:** The system shall lay out an Inline Editable Table with one row per record and one column per field, a sticky header row, a sticky first column (row number or identifier) when the table scrolls horizontally, and an Add row button always visible below the table.
- **FR-DA-MRE-03:** The system shall show the status of each row in its first column: no mark when clean, an Azure dot when changed and not saved, a spinner while saving, a Green check when saved, and an error marker in the Critical status color when a cell is invalid.
- **FR-DA-MRE-04:** The system shall support the keyboard map of Table 4.49-1 in every Inline Editable Table, keep every cell focusable, and keep focus on the active cell across keystrokes (Section 5.14).
- **FR-DA-MRE-05:** The system shall offer, in a column pinned at the end of each row and revealed on hover and on keyboard focus, the Duplicate row, Insert row below and Delete row actions

as 24-pixel line icons, with any other action in an overflow menu. A duplicated row shall copy every value and receive focus on its first editable cell; a deleted row shall be restorable for at least 5 seconds — 10 seconds for a deletion of several rows — from an Undo notification and with Ctrl/Cmd + Z. The deletion of several selected rows shall be confirmed in an accessible dialog stating the number of rows and the Undo period, never stating that the deletion cannot be undone.

- **FR-DA-MRE-06:** The system shall reduce repeated typing: a Duplicate previous row action above the table; carry-over of the categorizing value (for example the PESTLE category) from the row above when the user leaves it unset; and a Controlled List with a Custom value (FR-DA-DEU-02), never free text, wherever a known list of values exists.
- **FR-DA-MRE-07:** The system shall accept the paste of several rows of tab-separated content from a spreadsheet, expanding it into the table (tabs as columns, line breaks as rows), keeping every valid cell and flagging only the invalid ones. FR-DA-MRE-19 to 23 extend this requirement to complete records copied from any external file.
- **FR-DA-MRE-08:** The system shall validate a cell when the user leaves it after changing it — a value typed and then cleared counts as a change — never on every keystroke, never when the user only crosses an untouched cell, never on a new empty row or a new row opened in the Row Editor, and never only on submission; re-validate it on change after a first error; show the error under the cell in 12-pixel Open Sans SemiBold with a 1-pixel border in the Critical status color on its start edge; never reset the value typed; and reflect the error in the row status.
- **FR-DA-MRE-09:** The system shall save each row automatically about 500 milliseconds after the user leaves it, never lose a row on navigation, validation failure, panel change, re-render or refresh, warn before the user leaves a page with unsaved rows, and offer a Save all action whenever a row has pending changes.
- **FR-DA-MRE-10:** The system shall confirm a saved row with a 400-millisecond Green tint highlight and a Green check, and show a running count of the records in Muted text in the form "7 external issues identified."
- **FR-DA-MRE-11:** The system shall let the user resize columns from the edge of their header (minimum 80 pixels), reorder them by dragging the header, sort by a header in three steps (none, ascending, descending), choose the visible columns (the pinned first column cannot be hidden) and pin the first column; these choices shall be kept per user and per table. While a sort is applied, rows not yet saved shall stay at the end, in the order they were added, so that a row never moves while it is typed.
- **FR-DA-MRE-12:** The system shall, where the volume of records warrants it, offer row selection by checkbox (Shift-click selecting a range and the header checkbox selecting every filtered row), deletion of the selected rows after confirmation, and setting one field to the same value on every selected row.
- **FR-DA-MRE-13:** The system shall give every record table a toolbar (search, filter chips, density toggle, column chooser, and a bulk-action bar when rows are selected) and a footer (pagination with "Page X of Y", the record count and the page size). A filter shall be added as field, operator and value, removed on its own or with Clear all, and the number of active filters shall be shown on the filter button.
- **FR-DA-MRE-14:** The system shall not use, for multi-record entry, a dialog opened and closed for each row, repeated form cards for tabular data, a wizard for typing rows, or horizontal

scrolling without a sticky first column. A file import (FR-DA-MRE-30) is not the typing of rows and may be organized in stages.

Full-screen entry workspace, zoom and return. Every Multi-Entry can be opened full screen, where the user types, saves, changes and deletes several entries with room to work, and returns to the page in one step.

- **FR-DA-MRE-15:** The system shall let the user open every Multi-Entry in a Full-Screen Entry Workspace, from an Expand action (a 24-pixel line icon with the accessible name “Open full screen”) always visible in the toolbar of the Multi-Entry and from the keyboard shortcut of Table 4.49-1. The workspace shall fill the whole viewport above the navigation shell and the side panels; keep in a fixed header the title of the Multi-Entry, its context (for example the project and the step), the record count, the save status, the Zoom Controls and the return action; and show the same records, columns, filters, sort and selection as the page.
- **FR-DA-MRE-16:** The system shall offer Zoom Controls in the workspace: zoom with Ctrl/Cmd and the scroll wheel, or with a pinch on a touch screen or trackpad; a zoom slider; Zoom in and Zoom out buttons; a Fit action that scales the table so that all its visible columns fit the width of the workspace; and Reset to 100 %. The zoom shall range from 50 % to 200 % in steps of 10 %, show its current level as a percentage, keep the active cell in view when the level changes, scale the records area only (never the header, the controls or the dialogs), and be kept per user and per Multi-Entry (TableViewPreference). Without Ctrl/Cmd, the scroll wheel shall scroll the table as it does on the page.
- **FR-DA-MRE-17:** The system shall let the user do in the workspace everything the page allows on the Multi-Entry, so that several entries can be typed, saved, changed and deleted without leaving it: add a row; edit any cell in place; open the Row Editor; duplicate and delete rows with Undo; select rows and apply bulk actions; Save and Save all; filter, sort, resize, reorder and choose columns; and use the keyboard map of Table 4.49-1. Row status, validation, autosave, permissions, quotas and the audit trail shall be those of the page (FR-DA-MRE-03 to 13); the workspace edits the same records as the page, never a copy.
- **FR-DA-MRE-18:** The system shall return the user from the workspace to the page with a single action — one return button always visible at the end of the workspace header, named after the place it returns to (for example “Back to <task name>”, its accessible name including “Return to page”), the identifier and status of that place being shown in the header — or with the Escape key. When a cell is being edited, Escape shall first cancel that edit (Table 4.49-1) and a second Escape shall return to the page; when a menu or dialog is open, Escape shall first close it (NFR-DA-FOC-03). On return, the page shall show every change made in the workspace, with pending rows saved or kept with their pending status (FR-DA-MRE-09), the same row scrolled into view, and keyboard focus restored to the active cell or, when none, to the Expand action; no input shall be lost, and a confirmation shall be asked only when a change would otherwise be discarded. The browser Back action shall also return to the page.

Paste of complete records from external files. A user who already holds records in a document or a spreadsheet copies them and pastes them in one step, as a Word table is pasted into Excel.

- **FR-DA-MRE-19:** The system shall let the user copy one or several complete records from a word-processing document (for example a Word table, or lines of tab-separated values), a spreadsheet (for example Excel), a CSV or text file, a web page or another application, and

paste them into any Multi-Entry, on the page or in the workspace, with Ctrl/Cmd + V or a Paste action: each copied row shall become one record and each copied column one field, starting at the active cell, in the order copied. The system shall read the clipboard as an HTML table, as tab-separated text, or as comma- or semicolon-separated text, and shall add rows below the last one when the pasted rows outnumber the rows below the active cell.

- **FR-DA-MRE-20:** The system shall apply a pasted block directly, with an Undo notification, when its columns match the fields from the active cell and it overwrites no value; otherwise — when it holds a header row, more or fewer columns than the target, columns in a different order, or would overwrite existing values — it shall first show a Paste Preview: the mapping of each pasted column to a field, proposed by header name and then by position and changeable by the user; the columns that will be ignored; the rows that will be added and those that will be overwritten; and the invalid cells, marked. The user shall Apply, adjust the mapping or Cancel. A mapping confirmed by the user shall be proposed again for the next paste into the same Multi-Entry.
- **FR-DA-MRE-21:** The system shall convert each pasted value to its field: dates and numbers read in the user's locale (NFR-DA-I18N-05); a value of a Controlled List matched to its label ignoring case and accents, and otherwise kept as a Custom value (FR-DA-DEU-02); a reference given as an identifier or as “ID (name)” resolved to its record (FR-DA-DEU-04); line breaks inside a cell kept and character formatting removed. Each cell shall be validated like a typed cell (FR-DA-MRE-08): valid cells are kept, invalid cells are flagged with their message and the row status, and no row is ever discarded silently. A paste shall be undone in one action, recorded with the Entry Origin Manual, and written to the audit trail as one operation listing the records created and changed.
- **FR-DA-MRE-22:** The system shall accept up to a configurable number of rows per paste (default 1,000) and apply to every pasted row the user's rights, the Solution Pack quotas (NFR-DA-SEC-08) and the state of the Multi-Entry; it shall refuse a paste into a read-only Multi-Entry or a locked step with a message naming the reason, and skip computed columns, which are recomputed. The server shall check every row of a paste or an import with the same rules as a typed row, including the required values of a new row; an overwrite is checked on the cells it changes.
- **FR-DA-MRE-23:** The system shall let the user copy selected cells, or whole rows with their column titles, from any Multi-Entry, as tab-separated text and as an HTML table, so that they paste with their columns into Word, Excel or another application, and back into another Multi-Entry under FR-DA-MRE-19.

Finding errors and never losing work. Errors are found and fixed wherever they are, and work typed is never lost — not by a refresh, a crash, a lost connection, another user’s change or leaving the page.

- **FR-DA-MRE-24:** The system shall make every error findable: an Issues list in the toolbar giving the number of errors and, for each one, its row, its field and its message, each opening its cell (or the Row Editor for a hidden field); a “Rows with errors” filter; an error marker in the status column of every invalid row, so that an error scrolled out of view stays visible; and a Retry action per failed row and “Retry all failed”.

- **FR-DA-MRE-25:** The system shall show the save state of the Multi-Entry at all times in its footer — “All changes saved”, the number of unsaved rows or the number of rows with errors — offer a Save all action only while changes are pending, and offer Discard, confirmed in an accessible dialog, which returns changed rows to their saved values and removes new unsaved rows; no confirmation or alert of the browser itself (alert, confirm) shall be used.
- **FR-DA-MRE-26:** The system shall keep a Draft Backup of the pending rows of every Multi-Entry in the user’s browser, versioned and timestamped, written on every change and removed once nothing is pending; on return to the Multi-Entry it shall restore the draft with the notice “Draft restored from <date and time>” and a Discard the draft action. Every saved row shall carry its Row Stamp; a save based on an older stamp shall be refused as an Edit Conflict naming the user who changed the row, with the choice to reload the row or overwrite it with the user’s version, and a restored draft row whose record changed or was deleted since shall be listed with, for each one, the choice to keep the user’s version or take the server version.
- **FR-DA-MRE-27:** The system shall queue, not fail, a save that cannot reach the server: the row shall show “Waiting for the connection”, a banner shall state that the changes will be saved when the connection returns, the save shall be retried with exponential backoff (for example 1, 2, 4 seconds up to 30 seconds) and at once when the device reports the connection again; a save refused by the server (validation, rights, lock, conflict) shall not be retried by itself.

Large volumes, import and export. A Multi-Entry stays fast with thousands of rows, accepts a file as well as a paste, and gives its rows back as a file.

- **FR-DA-MRE-28:** The system shall render, in a Multi-Entry of more than a few dozen rows, only the visible rows and a small buffer (virtual rows), inside a scroll area of fixed height, with uniform row heights while long text is not wrapped and a “Wrap long text” choice; continuous scrolling shall be the default for entry, and pages (“Page X of Y”, FR-DA-MRE-13) shall remain available as a view choice for browsing; the ARIA row and column indexes shall stay consistent with the rows rendered.
- **FR-DA-MRE-29:** The system shall run a bulk paste, import, update or delete as a Batch Operation: batches of 200 to 500 rows (configurable), each saved before the next, a failed batch not stopping the others, a progress indicator announced to assistive technologies with a Cancel action between batches, and a Resume action continuing from the first batch not done; a bulk change made on screen (for example a field set on every selected row) shall be sent about two seconds after the last change.
- **FR-DA-MRE-30:** The system shall import rows from a file in an Import Session of five stages shown in a stepper: (1) template — the formats accepted (spreadsheet workbook, delimited text, paste), the limits, and a downloadable template with the column titles, required ones marked, and an example row; (2) upload — drag and drop, file picker or pasted text, with the type, size, row count and text encoding checked and the first ten rows previewed; (3) column mapping — titles recognized, columns matched by position otherwise, every match changeable with sample values shown, and the next stage refused while a required field has no column; (4) error repair — the rows needing attention listed first with their count, each cell corrected in place and checked again at once, duplicates within the file and against the existing rows flagged, guided corrections for the common errors (fill an empty required column, replace a value not in a list, leave the duplicates out) and any row left out on request; (5)

summary — the rows ready, with a warning, with errors and left out, the choice to leave rows with errors out or import them as rows to correct, an optional value given to every imported row, the commit as a Batch Operation (FR-DA-MRE-29) and its result with an Undo action for at least 60 seconds. Cancel shall be offered at every stage, confirmed once data was loaded, and moving between stages shall keep everything entered.

- **FR-DA-MRE-31:** The system shall export the current view of a Multi-Entry — its filters, sort and visible columns — or only the selected rows, as delimited text (UTF-8 with byte-order mark) and as a spreadsheet workbook, with the grid's column titles and display values, the file named after the Multi-Entry's parent and the date (<parent>-<yyyy-mm-dd>), and spreadsheet formulas neutralized (NFR-DA-SEC-21).

Readable on every screen and safe to leave. A Multi-Entry reads without horizontal scrolling on a common screen and on a phone, and nothing is lost when the user leaves the page.

- **FR-DA-MRE-32:** The system shall ask, when the user leaves a page that holds rows not saved — a link, the browser Back action, closing or switching a panel, closing the browser tab — with three choices: Save and leave, Discard and leave, or Stay on the page, stating the number of rows and of rows with errors and keeping the user on the page when rows could not be saved; returning from the Full-Screen Entry Workspace to the page shall never ask, nothing being lost there (FR-DA-MRE-18).
- **FR-DA-MRE-33:** The system shall keep a Multi-Entry readable without horizontal scrolling: when it has more than six columns, optional long-text columns shall be shown in the Row Editor rather than the table until the user shows them (the column chooser stating it); the footer shall show the totals of the number columns, the number of rows shown and selected; the Row Editor shall offer Save and new; and an empty Multi-Entry shall offer Add the first row, Import, Paste and, where an AI Use Case serves it, Propose with AI.
- **FR-DA-MRE-34:** The system shall show, below a phone breakpoint (768 pixels wide by default), no grid but a read-only list of the rows with their status and first fields, an "Edit one by one" action opening the Row Editor with previous, next and Save and new, Add row and Import; errors shall be shown on each listed row.
- **FR-DA-MRE-35:** The system shall undo and redo, with Ctrl/Cmd + Z and Ctrl/Cmd + Shift + Z and with toolbar actions, every change of a Multi-Entry as one step each — a cell edit, an added, inserted or duplicated row, a deletion, a value set on the selected rows, a fill down and a paste or an import — the server following the rows (a row brought back by Undo is restored or created again; a paste or an import is undone as its audited operation).

Table 4.49-1 — Keyboard map of an Inline Editable Table

Key	Action
Tab / Shift+Tab	Move to the next or previous cell
Enter	Commit the cell and move down
Enter in the last cell of the last row	Commit the row, add a new row and focus its first cell
Up / Down arrows	Move between rows in the same column
Escape	Cancel the edit of the cell and restore its value; with no cell being edited and no menu or dialog open, leave the Full-Screen Entry Workspace and return to the page (FR-DA-MRE-18)

Key	Action
Ctrl/Cmd + Enter	Commit the current row
Ctrl/Cmd + D	Duplicate the current row
Ctrl/Cmd + S	Save every pending row
Ctrl/Cmd + Shift + F	Open the Full-Screen Entry Workspace of the current Multi-Entry (FR-DA-MRE-15)
Ctrl/Cmd + scroll wheel, or pinch	Zoom in or out in the workspace (FR-DA-MRE-16)
Ctrl/Cmd + plus / Ctrl/Cmd + minus	Zoom in or out by 10 % in the workspace
Ctrl/Cmd + 0	Reset the zoom of the workspace to 100 %
Ctrl/Cmd + Shift + 1	Fit the table to the width of the workspace
Ctrl/Cmd + C	Copy the selected cells, or the selected rows with their column titles (FR-DA-MRE-23)
Ctrl/Cmd + V	Paste complete records copied from Word, Excel or another file, starting at the active cell (FR-DA-MRE-19)
Ctrl/Cmd + Z	Undo the last change — a cell edit, a paste, a duplication, a deletion, a value set on several rows, a fill down (FR-DA-MRE-05, 21, 35); inside a cell being typed, the field's own undo applies first
Ctrl/Cmd + Shift + Z, or Ctrl + Y	Redo the change undone last (FR-DA-MRE-35)
Ctrl/Cmd + ' (apostrophe), or Ctrl/Cmd + Shift + D	Fill down: copy into the cell the value of the cell above
Left / Right arrows at the start or end of the text	Move to the previous or next cell
Page Up / Page Down	Move up or down ten rows in the same column
Ctrl/Cmd + Home / Ctrl/Cmd + End	Move to the first or last row in the same column
Ctrl/Cmd + A	Select every row shown, after the text of the cell when it holds text
Shift + click on a row selection box	Select the range of rows from the last selected row (FR-DA-MRE-12)

4.50 Resizable Layout Panels

Note: the (New) SRS revision 1.7 uses section 4.50 for Document Formatting & Pictures. In this document, document formatting and pictures are specified in Section 4.43 (FR-DA-DOC-06 and 07), and the FR-DA-DFP family is added there as a new family (FR-DA-DFP-01 to 08). Section 4.50 here remains Resizable Layout Panels.

The application shell is made of three horizontal Layout Panels: navigation at the start, content at the center and context at the end. This section applies when the menu is docked on the start or end edge (FR-DA-NAV-04); when it is docked on the top or bottom edge, only the context panel is resizable.

Note: The (New) SRS revision 1.7 carries FR-DA-PNL-01 to FR-DA-PNL-05. Those five requirements are a subset of FR-DA-PNL-01 to FR-DA-PNL-08 below; the longer set in this document supersedes and extends them.

- **FR-DA-PNL-01:** The system shall let the user resize the navigation panel and the context panel by dragging their edge or with the keyboard (arrow keys by 16 px, Home and End to the limits), within limits (navigation 180–480 px, context 220–560 px, content at least 400 px), and reset a panel by double-clicking its edge; the content panel shall take the remaining width and never have a fixed width.

- **FR-DA-PNL-02:** The system shall apply the default, minimum and maximum widths of Table 4.50-1 and, when the window is resized, clamp the side panels so that the content panel never drops below its minimum width.
- **FR-DA-PNL-03:** The system shall place a drag handle on the inner edge of each side panel, show a resize cursor and a 2-pixel Azure highlight on hover and while dragging, and suspend text selection and pointer interaction in the panels during the drag.
- **FR-DA-PNL-04:** The system shall make each handle keyboard-operable: focusable with a visible focus ring, the left and right arrows changing the width by 16 pixels, Home and End setting the minimum and maximum width; the handle shall expose the separator role, its vertical orientation, its current, minimum and maximum values and a label such as "Resize navigation panel".
- **FR-DA-PNL-05:** The system shall reset a panel to its default width when its handle is double-clicked, and may offer a button that collapses a side panel.
- **FR-DA-PNL-06:** The system shall store the panel widths as a user preference with the other shell preferences (FR-DA-NAV-06), cached in the browser for an immediate layout, and clamp them to their limits when they are applied.
- **FR-DA-PNL-07:** The system shall disable resizing below 768 pixels of viewport width and fall back to a single-pane layout (FR-DA-NAV-08).
- **FR-DA-PNL-08:** The system shall treat the start and end panels as logical sides, so that under a right-to-left language the panels, their handles and the directional icons are mirrored (FR-DA-NAV-07).

Table 4.50-1 — Panel widths

Panel	Default	Minimum	Maximum
Navigation (start)	260 px	180 px	480 px
Context (end)	320 px	220 px	560 px
Content (center)	Fills the remaining width	400 px	—

4.51 Charts & Data Visualization

Note: the (New) SRS revision 1.7 uses section 4.51 for Data Tables & Pagination. In this document, pagination is specified inside FR-DA-MRE-11 to MRE-13 (Section 4.49), and the FR-DA-TBL family is added as a new family at the end of this Section 4.51. The chart rules FR-DA-VIZ-01 to 08 are unchanged.

Every chart, KPI tile and heatmap shown on a screen, in a report or in a generated document follows one set of rules, so that a reader recognizes the same meaning in every application.

- **FR-DA-VIZ-01:** The system shall choose the chart type from the purpose of the figure, as in Table 4.51-1, and shall never use a pie chart or a three-dimensional chart.
- **FR-DA-VIZ-02:** The system shall draw the primary or current series in solid Azure, a comparison or prior period in dashed Muted, a growth or positive trend in solid Green and an overlay or secondary metric in solid Teal, with gridlines in Line; it shall never use the default palette of a chart library, a rainbow palette, gradients, shadows or 3D effects on data.

- **FR-DA-VIZ-03:** The system shall label axes in 12-pixel regular Muted text and values in 12-pixel SemiBold Navy text, place the legend below the chart, horizontally, with dot markers, and give every chart a short italic caption in Muted below it.
- **FR-DA-VIZ-04:** The system shall resize charts with their container, with a minimum height of 240 pixels, show a skeleton of the chart's size while loading and an icon with "No data for this period." when there is nothing to show.
- **FR-DA-VIZ-05:** The system shall show a single indicator as a KPI Tile: a card with 24-pixel padding, the label in 13-pixel Open Sans SemiBold uppercase Muted, the value in 36-pixel Montserrat Bold Navy, the change in 12 pixels with an arrow icon (Green when favorable, the Critical status color when unfavorable) and an optional 24-pixel trend line.
- **FR-DA-VIZ-06:** The system shall draw maturity heatmaps with the five Status Scale colors in their fixed order, label every row and column, never color a cell outside the scale, and show the level as text or a number in each cell so that the meaning does not rest on color alone.
- **FR-DA-VIZ-07:** The system shall let the user, within the export rights of the user's role, copy a chart as an image, download it as PNG at twice its resolution or as SVG, download the CSV of its underlying data, and print it in a print-friendly rendering.
- **FR-DA-VIZ-08:** The system shall give every chart a text alternative and a data-table view reachable by keyboard, and under a right-to-left language mirror the axis origin without reversing the data.

Table 4.51-1 — Chart type by purpose

Purpose	Chart
Trend over time	Line
Compare categories	Bar or column
Composition	Stacked or 100 % stacked bar
Part of a whole	Donut (never pie, never 3D)
Distribution	Histogram or box plot
Correlation	Scatter
Contribution to a change	Waterfall
Maturity score at a glance	Radial gauge
Maturity on several axes	Radar
Single indicator	KPI Tile (FR-DA-VIZ-05)
Trend inside a table cell	Sparkline

Data tables and pagination.

Note: Previously covered inside FR-DA-MRE-11 to MRE-13 and Section 4.49. The (New) SRS specifies this family in its Section 4.51.

- **FR-DA-TBL-01:** The system shall show a list of more than 20 records by pages, with a Page Size of 20 (default), 50, 100 or all, the range shown (“1–20 of 345”), previous and next and “Page X of Y”; the chosen Page Size shall be kept for the user.
- **FR-DA-TBL-02:** The system shall scroll a table inside its own frame, horizontally and vertically, with a sticky header row and a sticky first column; the page itself shall never scroll sideways.

- **FR-DA-TBL-03:** The system shall sort a table by a column header in three states — none, ascending, descending — with a visible indicator of the current state.
- **FR-DA-TBL-04:** The system shall let each user choose the Density of tables, comfortable (default) or compact, and keep the choice for that user.
- **FR-DA-TBL-05:** The system shall highlight the row under the pointer, alternate the row backgrounds and draw thin, non-black borders.
- **FR-DA-TBL-06:** The system shall give every table a loading state, an empty state with its create action, a filtered-empty state with Clear filters and an error state with Retry (FR-DA-STA-01, 02).
- **FR-DA-TBL-07:** The system shall let the user resize the columns of an editable table (minimum 80 pixels) and keep the widths for that user.

4.52 Feedback & System States

Note: the (New) SRS revision 1.7 uses section 4.52 for Inline Multi-Record Entry. In this document, inline multi-record entry is specified in Section 4.49 (FR-DA-MRE-01 to 14). Section 4.52 here remains Feedback & System States (FR-DA-STA-01 to 07).

Every situation a user can meet has a designed state that says what happened and what to do next, in the voice of Section 5.13 (evidence first, no apology, no exclamation).

- **FR-DA-STA-01:** The system shall show a skeleton of the main content region while a page loads, a spinner inside the button that started an in-page action (the button disabled, the page not blocked), and a progress bar with an estimated time for a long operation.
- **FR-DA-STA-02:** The system shall show an empty state centered, at most 400 pixels wide, with a 48-pixel Muted icon, a title, one sentence and the primary action, leading with the action rather than an apology (for example "No surveys yet. Create your first assessment."); a filtered list with no result shall offer Clear filters, and a failed load shall offer Retry.
- **FR-DA-STA-03:** The system shall report an error at the level where it occurred: under the field for invalid input, in a notification for a transient failure (network, saving), on a full page with a headline and a primary action for a missing page or a server error, and in a dialog offering "Sign in again" when the session has expired.
- **FR-DA-STA-04:** The system shall confirm success with a notification for a background operation, an inline check for a row of a table, and a full-page summary at the end of a multi-step process.
- **FR-DA-STA-05:** The system shall show notifications at the bottom end of the screen on desktop and at the top center on mobile, with an 8-pixel radius and the card shadow, in four variants (information in Azure tint, success in Green tint, warning in the Developing status color, error in the Critical status color, all with Navy text), for 5 seconds by default and until dismissed for an error, with an optional action such as Undo.
- **FR-DA-STA-06:** The system shall show a persistent banner while the device is offline ("You're offline. Changes will sync when you're back."), queue the user's changes locally and synchronize them on reconnection, where the screen supports offline work (for example Face-to-Face questionnaire capture, Section 4.36).
- **FR-DA-STA-07:** The system shall present a capability the user may not use in one of three ways, chosen per capability: hidden; visible but disabled with a tooltip explaining why; or, for a whole screen, the access notice of FR-DA-RBAC-08 with a Request access action.

4.53 Layout Panels & Navigation Robustness (Alias)

The (New) SRS revision 1.7 specifies layout panels and navigation robustness in its Section 4.53. In this document that content is specified in Section 4.50 (FR-DA-PNL-01 to 08) and FR-DA-NAV-16 to 20 (Section 4.19). This section holds no requirement; it is a pointer only.

5. Common Non-Functional Requirements

5.1 Performance

- **NFR-DA-PERF-01:** Dashboard and list views shall render within 2 seconds under nominal per-tenant data volumes.
- **NFR-DA-PERF-02:** An AI agent suggestion shall return within 5 seconds under its deterministic path.
- **NFR-DA-PERF-03:** A knowledge-base or FAQ retrieval search shall return ranked results within 1 second for a tenant's full corpus.
- **NFR-DA-PERF-04:** A single standard-report export (PDF, Excel, or Word) shall complete within 3 seconds for a tenant's current data volume.
- **NFR-DA-PERF-05:** With 100 concurrent users on one tenant, the 95th-percentile response time of read endpoints shall stay at or below 1 second, measured by a repeatable load test. A per-user read cache, where used, shall hold an entry for no more than 30 seconds and shall be invalidated by any write in the same Organization.
- **NFR-DA-PERF-06:** Activating a vertical, with its processes and configuration, shall complete within 5 minutes (configurable target).
- **NFR-DA-PERF-07:** A bulk data import shall load 10,000 records within 30 minutes, and an SME shall be able to complete onboarding within 30 days (configurable targets).
- **NFR-DA-PERF-08:** A production deployment offering the Vertical and SME layers shall be sized for configurable targets, by default 1,000 API calls per second sustained and 10,000 concurrent users, with reports generated within 60 seconds.
- **NFR-DA-PERF-09:** The global search shall show its first suggestions within 300 milliseconds of typing and full results within 1 second at the 95th percentile, for a tenant's full data volume.
- **NFR-DA-PERF-10:** Restoring a version of a process element with up to 1,000 descendants shall complete within 5 seconds; generating a document of up to 50 pages from a template shall complete within 10 seconds.
- **NFR-DA-PERF-11:** Generating a document of up to 50 pages with diagrams (for example BPMN flows) shall complete within 15 seconds, and the consistency checks of a project's documents within 30 seconds, at the 95th percentile.
- **NFR-DA-PERF-12:** A long list shall render at most the selected Page Size of rows at a time (FR-DA-TBL-01); opening a project template blueprint of up to 1,500 steps shall take less than 1 second on the server and less than 2 seconds to display.
- **NFR-DA-PERF-13:** The Full-Screen Entry Workspace shall open within 1 second for a Multi-Entry of up to 1,000 records; a zoom change shall be rendered within 100 milliseconds; the Paste Preview of 1,000 rows shall appear within 2 seconds and applying it shall complete within 5 seconds, at the 95th percentile.
- **NFR-DA-PERF-14:** A Multi-Entry of 10,000 rows shall open within 2 seconds and scroll and accept typing without visible delay (a keystroke shown within 50 milliseconds), and an import of 1,000 rows shall be committed within 10 seconds, at the 95th percentile.

5.2 Security

- **NFR-DA-SEC-01:** All API endpoints except login shall require a valid, signed, time-boxed authentication token.
- **NFR-DA-SEC-02:** Passwords shall be stored only as salted cryptographic hashes; plaintext passwords shall never be logged or persisted.
- **NFR-DA-SEC-03:** Every authorization decision shall be enforced server-side; client-side permission and Pack-entitlement checks are a UX convenience only, never the security boundary.
- **NFR-DA-SEC-04:** No data query shall be constructed by concatenating untrusted input; all data access shall use parameterized statements or an equivalent safe-query mechanism.
- **NFR-DA-SEC-05:** Owner/Evaluator segregation of duties, and any equivalent server-enforced business rule, shall be enforced server-side, not merely hidden in the UI.
- **NFR-DA-SEC-06:** A one-Accountable-per-RACSI-activity rule, and any equivalent uniqueness rule, shall be enforced by a data-persistence-layer constraint, not merely by disabling an option in the UI.
- **NFR-DA-SEC-07:** A Real LLM Provider Connection's API key shall be captured and stored only client-side; the backend shall never persist it to any table, write it to any log line, or return it in any API response, and any server-side proxying of a forwarded key shall use it for exactly one outbound call and never write, log, cache, or hold it beyond that request.
- **NFR-DA-SEC-08:** Every quota check (Projects, OBS nodes, AI Use Cases, and any other Pack-bound dimension) shall be enforced at the point of creation/activation on the server, never solely by disabling a client-side control; an over-quota request shall be rejected with HTTP 409, never silently truncated or partially applied.
- **NFR-DA-SEC-09:** Solution Pack entitlement shall be treated as an authorization axis independent of, and enforced in addition to, RBAC permissions; a route gated by both shall require both checks to pass, each independently testable and independently capable of producing a 403.
- **NFR-DA-SEC-10:** Individually identifiable records requiring restricted visibility (e.g. individual-level readiness or sentiment data) shall be restricted, by RBAC default, to an explicitly named set of roles; other roles shall see only aggregated, de-identified views.
- **NFR-DA-SEC-11:** Tenant data isolation shall be verified by an automated cross-tenant access test suite exercised in the build/release pipeline, not solely by code review.
- **NFR-DA-SEC-12:** The login endpoint shall be rate-limited per client, and every response shall carry standard security headers (content-type sniffing protection, frame protection, and a restrictive cross-origin resource policy).
- **NFR-DA-SEC-13:** Caches, backups, and logs shall never make one tenant's data available to another; backups shall be stored outside every web-served directory and be readable only by the service account and the platform administrator.
- **NFR-DA-SEC-14:** The system shall support standard federated authentication (SAML 2.0, OAuth 2.0, OpenID Connect) and multi-factor authentication.
- **NFR-DA-SEC-15:** Role-based access control shall support scopes by vertical and by SME track in addition to the tenancy scopes.

- **NFR-DA-SEC-16:** Data shall be encrypted at rest (AES-256 or equivalent) and in transit (TLS 1.2 or later), with configurable data residency regions and optional customer-managed keys (bring your own key).
- **NFR-DA-SEC-17:** An Organization-level live-model key (FR-DA-AI-16) and any integration secret stored on the server shall be encrypted at rest with authenticated encryption (for example AES-256-GCM) under a key held outside the database; it shall never be returned in full, logged, cached beyond its request, exported or included in a backup restore to another tenant. NFR-DA-SEC-07 continues to govern a personal connection.
- **NFR-DA-SEC-18:** A search result, count, suggestion or snippet shall never reveal a record, a title or a value that the user could not open on its own screen.
- **NFR-DA-SEC-19:** A generated document shall contain only data the requesting user may read; a document shared or downloaded shall carry its classification and shall not include hidden data (comments, hidden sheets, metadata) beyond what it shows.
- **NFR-DA-SEC-20:** Picture uploads (FR-DA-DFP-04) shall accept PNG and JPEG only, up to 5 MB, stored under generated, per-tenant names on validated paths; every formatting value (font, size, color, alignment, position) shall be checked on the server against its list of allowed values.
- **NFR-DA-SEC-21:** Pasted content shall be treated as untrusted input: markup, scripts, styles, embedded objects and images on the clipboard shall be discarded, never rendered or stored; a text value beginning with =, +, -, @, a tab or a carriage return shall be stored as text, never evaluated, and neutralized when exported to a spreadsheet or CSV (formula injection); and every pasted row shall pass the same server-side validation, authorization and tenant checks as a typed row.
- **NFR-DA-SEC-22:** An imported file shall be treated as untrusted input: its type checked by extension and by content, its size and row count bounded by configurable limits, its text read as UTF-8 or, when it is not valid UTF-8, in a stated fallback encoding that the user can change; a spreadsheet workbook shall be read on the server keeping only cell text (the last value of a formula); the rights, quotas and checks of a paste (FR-DA-MRE-22) shall apply to every imported row.

5.3 Reliability & Data Integrity

- **NFR-DA-REL-01:** The system shall degrade gracefully if an AI agent or live-LLM call fails: the underlying CRUD operation shall still succeed, with the AI suggestion simply unavailable or served from its deterministic fallback.
- **NFR-DA-REL-02:** The system shall target 99.9% uptime in a production deployment.
- **NFR-DA-REL-03:** A report-export failure shall return a structured error without corrupting or partially writing the underlying report data view.
- **NFR-DA-REL-04:** The persistence layer shall support scheduled backup with a defined retention policy; a production deployment shall not rely on a single unreplicated data file as its sole durability boundary.
- **NFR-DA-REL-05:** The backup schedule shall bound data loss to at most 24 hours, and restoring a backup shall be tested at least once per release.
- **NFR-DA-REL-06:** A production deployment offering the Vertical and SME layers shall support disaster recovery with an RTO under 4 hours and an RPO under 1 hour, automatic failover and automated backups, tightening NFR-DA-REL-05 for that deployment.

- **NFR-DA-REL-07:** Every version of a versioned entity shall be stored complete and checksummed, so that viewing or restoring it reproduces its exact content; a failed restore shall leave the Current Version unchanged.
- **NFR-DA-REL-08:** Generating a document twice from the same data and template version shall produce the same content; every published document version shall keep the data snapshot it was generated from, so it can be reproduced.
- **NFR-DA-REL-09:** A page shall never stay blank: a failure to load or render shall show the full-page error state with a headline and a primary action (FR-DA-STA-03).
- **NFR-DA-REL-10:** Inline data entry shall never lose a row: every row typed shall be saved, or kept with its pending status, across navigation, validation failure, panel change, re-render, refresh, crash of the browser and loss of connection (FR-DA-MRE-09), by the Draft Backup (FR-DA-MRE-26), the offline queue (FR-DA-MRE-27) and the Leave Guard (FR-DA-MRE-32).
- **NFR-DA-REL-11:** A save shall never overwrite a newer version of a record without the user's explicit choice: every save of a row shall be checked against its Row Stamp (FR-DA-MRE-26).

5.4 Scalability

- **NFR-DA-SCALE-01:** The architecture shall support horizontal scaling of the web tier independently of the API tier.
- **NFR-DA-SCALE-02:** The system shall support at least 100 concurrent users per tenant without functional degradation.
- **NFR-DA-SCALE-03:** The persistence schema and the retrieval index shall each be swappable for a distributed equivalent (e.g. a managed relational database, a vector database) without changing the API contract.
- **NFR-DA-SCALE-04:** The system shall scale to configurable limits, by default 1,000 tenants per deployment including 10,000 SME tenants in shared mode, 100 million records, 500 macro processes, 200 E2E processes and 100 checklist items per gate.
- **NFR-DA-SCALE-05:** Adding a vertical, an SME track, a template, a gate or a checklist shall require no downtime.

5.5 Usability & Accessibility

- **NFR-DA-UX-01:** The UI, and every generated document, shall follow the common visual identity (Section 5.13) (color palette, typography, iconography) consistently across all modules and output formats.
- **NFR-DA-UX-02:** The UI shall be responsive and usable on desktop, tablet, and mobile viewport widths, with touch-target sizing meeting WCAG 2.1 AA on mobile.
- **NFR-DA-UX-03:** All interactive controls shall be keyboard-navigable and carry accessible labels (WCAG 2.1 AA target).
- **NFR-DA-UX-04:** Status, severity, and classification fields shall use one consistent color/badge vocabulary across every module, so a user who has learned a meaning in one module recognizes it in every other.
- **NFR-DA-UX-05:** Every interactive target shall measure at least 24 × 24 CSS pixels (WCAG 2.2 SC 2.5.8), and at least 44 × 44 CSS pixels on touch screens.

- **NFR-DA-UX-06:** Every release shall pass an automated accessibility scan (e.g. axe-core) of its main screens with no critical or serious violation. Text shall meet WCAG 2.1 AA contrast (4.5:1, or 3:1 for large text); where a brand color falls short, its darker tone shall be used for text, and any remaining brand conflict shall be recorded in Appendix A.
- **NFR-DA-UX-07:** Every confirmation, warning, and error message shown as a transient notification shall be translated (FR-DA-I18N-07), announced to assistive technologies through a polite live region, and remain readable long enough to act on.
- **NFR-DA-UX-08:** Wide matrices and tables shall scroll inside their own container, keeping the first column visible, so the page never scrolls horizontally on a 390-pixel viewport.
- **NFR-DA-UX-09:** SME users shall complete onboarding within 1 day and need less than 4 hours of training, helped by guided workflows (configurable targets).
- **NFR-DA-UX-10:** User satisfaction shall be measured, with a configurable target (default NPS of 30 or more).
- **NFR-DA-UX-11:** Search shall be fully keyboard-operable (shortcut, arrow keys, Enter, Escape), group its results by type, highlight the matched words, and mirror under right-to-left languages.
- **NFR-DA-UX-12:** Every step shall show a one-line brief with an expandable detailed description, and every form field shall show how to fill it, in the user's display language.
- **NFR-DA-UX-13:** Every dialog shall keep its title and actions in view and scroll its content with a visible scroll bar; read-only text shall be shown in full rather than inside a small scrolling box.
- **NFR-DA-UX-14:** Every identifier shown to a user shall be followed by the name of the item it identifies, in the form "ID (name)" — for example a phase, a process, a step, a function or a record — in lists, pickers and headers.
- **NFR-DA-UX-15:** A computed field shall be read-only, show how it is computed and why it exists, and never be typed.
- **NFR-DA-UX-16:** Text fields shall be sized for their content; a field meant for more than one sentence shall offer at least five lines, in the form or in the Row Editor.
- **NFR-DA-UX-17:** Every icon-only button shall carry an accessible name; every form field shall have a label bound to it; errors shall be announced through a polite live region; table headers shall declare their scope; dialogs shall trap focus and return it on closing.
- **NFR-DA-UX-18:** Meaning shall never be carried by color alone: a status shall show its label or level, a comparison series shall be dashed as well as grey, and an error shall show a message as well as a red border.
- **NFR-DA-UX-19:** Brand colors shall be applied with the contrast table of Appendix J: Navy, Ink and Muted for any text; Azure only for large bold text and interface components (3.9 : 1); Teal and Green for fills only (2.9 : 1 and 2.2 : 1).
- **NFR-DA-UX-20:** The Full-Screen Entry Workspace shall be accessible: announced to assistive technologies as a labelled dialog, it shall move focus to the active cell on opening, keep focus within it and return focus on leaving (NFR-DA-UX-17); every Zoom Control shall be a labelled button or slider operable by keyboard, with the zoom level announced; the workspace zoom shall complement, never replace or block, the zoom of the browser (WCAG 2.1 SC 1.4.4 and 1.4.10); its opening and closing transitions shall last 200 to 300 milliseconds and be removed when the user prefers reduced motion (NFR-DA-VDS-16); below 768 pixels it shall show the

records as the mobile page does, with pinch zoom; and under a right-to-left language the workspace, its controls and its return action shall be mirrored (FR-DA-PNL-08).

- **NFR-DA-UX-21:** Every Multi-Entry shall follow the W3C ARIA grid pattern: grid, row, column header and grid cell roles, row and column indexes kept consistent with virtual rows and pages, the sort state of each column announced, a live region announcing the rows added, deleted, saved and failed, batch progress and a restored draft; errors shown with an icon and text, never by color alone; focus moved to a sensible cell after an addition or a deletion; focus trapped only in dialogs; and animations removed when the user prefers reduced motion.

5.6 Localization

- **NFR-DA-I18N-01:** Every layout — navigation, tables, and charts — shall render correctly mirrored under Arabic RTL, not just with mirrored text direction.
- **NFR-DA-I18N-02:** The system shall support configurable languages (default English, French, Arabic, Spanish, German), currencies (default USD, EUR, MAD, ZAR) and regional date, time and number formats.
- **NFR-DA-I18N-03:** Labels, buttons and headings shall accommodate text 30 % longer in French and 20 % longer in Arabic than in English without clipping, and headlines shall be kept short so that they survive translation.
- **NFR-DA-I18N-04:** Under Arabic the layout shall be mirrored in full: the navigation panel moves to the right, chevrons and arrows and every icon with a direction are reversed, icons without a direction (search, settings) are not, and numbers, code and Latin brand names stay left to right.
- **NFR-DA-I18N-05:** Dates, numbers and currencies shall be formatted by the locale of the user (for example fr-FR, ar-MA) through the platform's internationalization services; Eastern Arabic numerals shall be used only where the locale requires them; amounts shall show the currency's ISO code with the symbol of the locale.
- **NFR-DA-I18N-06:** No text shall be part of an image; idioms and culture-specific references shall be avoided in interface text.
- **NFR-DA-I18N-07:** Files exported or downloaded shall keep non-Latin text intact: delimited text in UTF-8 with a byte-order mark, and a file name with non-Latin characters sent in its encoded form (RFC 6266, filename*=UTF-8''...) with a Latin fallback, so that a download never fails because of the language of its name.

5.7 Portability / Installability

- **NFR-DA-PORT-01:** Installation shall succeed on a fresh target machine with only a supported Node.js runtime installed — no native C/C++ build toolchain, no required external database server, and no mandatory external API key.
- **NFR-DA-PORT-02:** The same source shall run unmodified across supported operating systems via standard package-manager install/build/start commands.
- **NFR-DA-PORT-03:** Every change shall be built by a continuous-integration pipeline on Linux, Windows, and macOS that installs dependencies, seeds the demonstration data, runs the automated tests — including the cross-tenant suite of NFR-DA-SEC-11 — and builds the web client.

- **NFR-DA-PORT-04:** Each tier shall install, seed, and start with its package manager's standard commands (install, seed, dev or start), documented in a README at the root of the delivery.
- **NFR-DA-PORT-05:** The continuous-integration build shall run automated tests of the multi-record entry logic (validation, undo and redo, batching with failure, cancellation and resumption, draft restore and conflicts, column mapping and error repair, export, retries) and an end-to-end import from file upload to commit and undo; test suites that share one database shall run one after another, so that no test fails for a lock.

5.8 Maintainability / Extensibility

- **NFR-DA-MAINT-01:** Cross-cutting engines — the Action/Task Register, the Alert & Notification Engine, the RBAC layer, the KPI/Reporting engine, and the Audit Trail Service — shall be implemented as composable modules reusable by sibling applications via the same API contract.
- **NFR-DA-MAINT-02:** AI agent implementations shall be isolated behind a stable internal interface so a rule-based implementation can be replaced by an LLM/vector-DB-backed one without changing calling code.
- **NFR-DA-MAINT-03:** The Solution Pack feature and quota matrix shall be defined in exactly one place and consumed by every enforcing route, so entitlements can never drift between routes.
- **NFR-DA-MAINT-04:** The AI Use Case Library's generation service shall be the single designated server-side integration point for a live LLM call and its provider-specific credential/endpoint logic; no other module shall embed its own LLM provider client; and no module shall produce an AI suggestion, even from the built-in engine, outside an AI Use Case (FR-DA-AI-23).
- **NFR-DA-MAINT-05:** The alert type-to-name and type-to-stage/process mapping shall be defined in exactly one shared module, consumed by both seed data and live computation.
- **NFR-DA-MAINT-06:** The Installation Guide and User Guide shall be generated from source in Word and PDF, each with an automatic table of contents, and the User Guide's step-by-step walkthroughs shall be replayed automatically against a freshly seeded instance, so the documentation cannot drift from the application.
- **NFR-DA-MAINT-07:** Every User Guide walkthrough shall start from the tenancy — create or choose the Group (Yes/No), create the Organization, then the principal record(s) — and the guide shall include the decision matrix of FR-DA-HLP-04.
- **NFR-DA-MAINT-08:** Verticals, SME tracks, complexity criteria, packs, project templates, gates and checklists shall be configuration data loaded without code changes, and the platform shall be modular with zero-downtime upgrades, monitoring, alerting and comprehensive logging.
- **NFR-DA-MAINT-09:** The process design, functions, OBS roles, Prompt Specifications, step form kinds and documented-information templates shall be configuration data, changed without code changes or downtime.
- **NFR-DA-MAINT-10:** Controlled Lists, Document Profiles, step-specific field titles and data source bindings shall be configuration data, changed without code changes or downtime.
- **NFR-DA-MAINT-11:** An upgrade shall migrate the data of an existing installation in place at its first start — changed entity definitions, completed catalogs, new fields with their derived values — each migration idempotent, versioned, recorded with what it changed, and run once; an upgrade shall never require the data to be reseeded, and its release notes shall state that the server is restarted after the files are replaced.

5.9 Compatibility

- **NFR-DA-COMPAT-01:** The application shall target evergreen desktop, tablet, and mobile browsers; no legacy-browser support is in scope.
- **NFR-DA-COMPAT-02:** BPMN diagrams shall be stored and exchanged as schema-valid BPMN 2.0 XML, verified by a round-trip import/export, so a diagram authored in one application opens without error in any other BPMN 2.0-compliant tool, and vice versa.
- **NFR-DA-COMPAT-03:** The application's core workflows (record CRUD, approvals, alerts) shall remain fully usable on a mobile viewport without requiring a separate native application; a native or installable Progressive Web App wrapper, where offered, shall consume the same REST API (Section 7.2) with no mobile-specific endpoint variant.

5.10 Auditability & Compliance

- **NFR-DA-COMP-01:** Every AI-generated suggestion shall be logged with a confidence indicator and be attributable to a specific agent/use case and record, or, for the AI Assistant, to the requesting user and question.
- **NFR-DA-COMP-02:** The Controls module shall support demonstrating COSO Internal Control — Integrated Framework coverage across all five components.
- **NFR-DA-COMP-03:** The system shall retain the full version history of every AI Use Case and every template indefinitely (no hard deletion of prior versions on edit or revert).
- **NFR-DA-COMP-04:** The system shall never represent, in any UI text, Help content, FAQ answer, or generated document, that activating a Compliance & Security Standard constitutes a certification, an external audit, or a legal attestation of compliance.
- **NFR-DA-COMP-05:** The system shall let verticals map to sector standards — for example IATF 16949 (automotive), AS9100 (aerospace), EU MDR and ISO 13485 (medical devices), FDA 21 CFR Part 11 (pharmaceutical), HACCP (food safety) — as scaffolds subject to FR-DA-CFG-09; certification (ISO/IEC 27001, SOC 2, GDPR) is an attribute of a deployment, not of the software.
- **NFR-DA-COMP-06:** Every requirement of an application shall be traceable to its business requirement, user story, test case, release and, where applicable, customer commitment, in line with ISO/IEC/IEEE 29148; requirement changes shall be classified Minor, Moderate, Major or Breaking with the matching approval and regression scope.

5.11 Data Retention & Privacy

- **NFR-DA-DATA-01:** By default, and whenever no Real LLM Provider Connection is configured, the system shall not transmit tenant data to any third-party service; every AI capability shall run entirely within the deployment's own infrastructure using rule-based/statistical methods and RAG grounding. Only an explicitly user-configured, client-stored Real LLM Provider Connection may cause a single generation request to be sent to that user-selected third-party provider — opt-in per user, never enabled by default, and never applied to a permission-scoped data-query mode.
- **NFR-DA-DATA-02:** The system shall support both shared-infrastructure (SaaS) and dedicated/sovereign deployment options per Organization, without changing the tenant-isolation guarantee.

5.12 Questionnaire Data

- **NFR-DA-QLT-01:** A questionnaire response shall be stored within 2 seconds of submission, at the reference load of Section 5.1.
- **NFR-DA-QLT-02:** Questionnaire response data shall be retained according to the tenant's retention policy, by default 5 years, and then deleted or anonymized.
- **NFR-DA-QLT-03:** For a questionnaire marked anonymized, the system shall strip the respondent's identity before storage, so that no stored response can be linked back to a stakeholder.

5.13 Visual Design System

This section details NFR-DA-UX-01. It makes the AI Value Graphical Chart the visual and interaction reference of every application and of every document it generates. Where this section and the Graphical Chart differ, Appendix A records the difference and the rule that applies.

Tokens, color and type

- **NFR-DA-VDS-01:** Every color, font, type size, spacing, radius, shadow, focus ring, duration, easing, breakpoint, panel width and layer of the user interface shall come from the design tokens of Appendix J, defined once as CSS custom properties prefixed --aiv- and never written as a raw value in a component. An automated check in continuous integration shall reject a raw color value outside the token file.
- **NFR-DA-VDS-02:** The interface shall be light: page background in Background (#F5F8FB), surfaces in White, borders and dividers 1 pixel in Line (#E1E8F0), never black and never heavy. Dark mode shall not be the default; a dark background shall appear only as a deliberate section divider. The Azure–Teal–Green gradient shall appear only in the logo swoosh and in at most one hero element per page, never on data, charts or text; purple, pink and rainbow gradients shall not be used.
- **NFR-DA-VDS-03:** Body text shall be in Ink (#2C3E50) or Navy (#123A5F), metadata and captions in Muted (#5A6B7B). Azure, Teal and Green shall be used for fills, borders, buttons, badges, icons, charts and large bold text (at least 18.66 pixels bold or 24 pixels) only, never for smaller text on a light surface (Appendix J, color contrast).
- **NFR-DA-VDS-04:** Status, health, maturity, audit and severity values shall use the five Status Scale colors in their fixed order, always together with a text label or level number, and no other color; this applies NFR-DA-UX-04 across every module.
- **NFR-DA-VDS-05:** Headings and KPI values shall be set in Montserrat (Bold, ExtraBold, Black), body and interface text in Open Sans (Regular, SemiBold, Bold) and data and code in JetBrains Mono, with the fallbacks of Appendix J. Inter, Roboto, Arial, Lato and system interface fonts shall not be used as primary fonts. Fonts shall be served with the application, not from a third-party service, so that an on-premises installation renders identically.
- **NFR-DA-VDS-06:** Type sizes shall be taken from the scale 12, 14, 16, 18, 20, 24, 30, 36 and 48 pixels (13 pixels for form labels and breadcrumbs), with the hierarchy of Appendix J; hierarchy shall be carried by weight and color, not size alone. Eyebrow labels shall be bold, uppercase and letter-spaced.

Layout, shape, elevation and icons

- **NFR-DA-VDS-07:** Spacing shall be taken from the scale 0, 4, 8, 12, 16, 24, 32, 48, 64 and 96 pixels, on a vertical rhythm of multiples of 8 pixels, in a container at most 1,200 pixels wide with 24-pixel gutters and page margins of 24 pixels on mobile and 32 pixels on desktop.
- **NFR-DA-VDS-08:** Radii shall be 8 pixels for buttons, inputs and menus, 12 pixels for cards, 16 pixels for dialogs and 6 pixels or a circle for badges and avatars; no fully rounded (pill) button shall be used.
- **NFR-DA-VDS-09:** Elevation shall use only the card, hover and dialog shadows of Appendix J, tinted with Navy; a card shall have either the card shadow or a Line border, never both; no colored shadow or glow shall be used except on primary buttons.
- **NFR-DA-VDS-10:** Icons shall come from the Lucide set only, as line icons with a 2-pixel stroke, at 16, 20, 24, 32 or 48 pixels; filled icons, other icon families and emoji used as icons shall not appear. Icon badges shall be solid circles (Azure or Green for emphasis, Navy or Muted for neutral) with a white icon at 50 to 55 % of the diameter, and one screen shall not color its icons in several accent colors.

Components

- **NFR-DA-VDS-11:** Every interactive component shall define its default, hover, focus-visible, active, disabled (40 % opacity, not-allowed cursor), loading (spinner replacing the label, width kept), error and success states where they apply, and never rely on browser defaults. Buttons shall come in the Primary (Azure, Navy on hover), Success (Green), Secondary (Azure outline), Ghost and Danger variants, in sizes of 32, 40 and 48 pixels.
- **NFR-DA-VDS-12:** Form fields shall be 48 pixels high with 12-by-14-pixel padding, a Line border and an 8-pixel radius; on focus, a 2-pixel Azure border with a soft ring; on error, a 2-pixel border and a 12-pixel SemiBold message in the Critical status color; read-only fields shall have the Background color and no border. Labels shall be 13-pixel SemiBold Navy above the field, help text 12-pixel Muted below it; a required field shall be marked with an asterisk before its label and an optional one suffixed "(optional)". Forms shall use one column up to 720 pixels, two up to 1,024 and three above, with 24 pixels between fields and 32 between groups, validate on leaving a field, list the errors at the top when there are more than three, and never disable the submit button to prevent errors.
- **NFR-DA-VDS-13:** Dropdowns shall share the field style, open a white menu with an 8-pixel radius, the card shadow and a maximum height of 320 pixels, with 40-pixel options, the hover in Azure tint, a check on the selected option, chips for multiple selection and a filter at the top for long lists. Checkboxes and radio buttons shall be 16 pixels and filled with Azure when selected; toggles shall be 36 by 20 pixels, Muted when off and Green when on, with the label after the control.
- **NFR-DA-VDS-14:** Dialogs shall have an overlay of Navy at 45 % opacity, a white panel with a 16-pixel radius and the dialog shadow, widths of 560, 720 or 960 pixels, 32-pixel padding, a Montserrat title with a close icon, and actions at the end of the footer with the primary action last; Escape shall close a dialog unless the action is destructive, focus shall be trapped inside and returned on closing. Side panels shall be 320, 400 or 480 pixels wide with a fixed header and footer. Tooltips shall be Navy with white 12-pixel text, at most 240 pixels wide, shown after 300 milliseconds.

- **NFR-DA-VDS-15:** Tables shall have a header row in solid Navy or Azure with bold white text, body rows alternating White and Background, Line borders, the hovered row in Azure tint and a selected row in Azure tint with a 2-pixel Azure border on its start edge. They shall offer a Comfortable density (56-pixel rows, 16-pixel cell padding) and a Compact density (40-pixel rows, 8-pixel padding), kept per user, and show 6 skeleton rows while loading.

Motion, responsiveness and voice

- **NFR-DA-VDS-16:** Motion shall clarify a change, never decorate: 150 milliseconds for hover and focus, 200 for menus and tabs, 300 for dialogs, drawers and page transitions, with the standard and emphasis easing tokens, animating transform and opacity rather than layout. Non-essential motion shall stop when the user prefers reduced motion, and no video shall play automatically.
- **NFR-DA-VDS-17:** The layout shall adapt at the breakpoints of Appendix J: below 768 pixels a single pane in which the menu becomes a drawer and tables become cards; from 768 to 1,024 pixels two panes with the menu reduced to icons; from 1,024 pixels three panes (menu, content and an optional context panel); from 1,280 pixels the full layout with wider table columns.
- **NFR-DA-VDS-18:** Interface text shall lead with the number or the finding, in plain professional language, active voice and short sentences, without superlatives that are not evidenced, without exclamations or emoji (for example "3 of 4 findings resolved. 1 open." rather than "Great job! Almost there!"). Brand and product names shall be written exactly as their owners define them (for example AI Value: two words, both capitalized), and an application's own product name shall be written the same way on every screen and in every document.

Logo, documents and governance of the design system

- **NFR-DA-VDS-19:** The product logo of the application, or the AI Value logo where the application has none, shall appear as the horizontal lockup in the top bar and in document headers and as the icon alone for the favicon and application icon; it shall respect its clear space and minimum sizes (150 pixels or 35 millimeters wide for the full logo, 32 pixels or 8 millimeters for the icon) and shall never be recolored, stretched, rotated, cropped, outlined or given a shadow or a glow.
- **NFR-DA-VDS-20:** Generated documents and exports shall follow the Graphical Chart: Open Sans 11-point Ink body, Montserrat Navy headings, tables with a Navy header row and alternating White and Background rows, 2-centimeter margins, A4 by default and Letter where the region requires it, fonts embedded, hyperlinks and color profile preserved (Navy CMYK 100, 80, 30, 20) and the vector logo never flattened to a bitmap. Spreadsheet exports shall use the Status Scale alone for conditional formatting and the chart series colors of FR-DA-VIZ-02; slide exports shall use a Navy title slide and white content slides.
- **NFR-DA-VDS-21:** The tokens and components shall be maintained as one shared, semantically versioned design system (a code package and the AI Value Design System library), from which every application takes them; each release of an application shall state the version of the Graphical Chart it conforms to, and no visual asset shall be considered final unless it references a token.

5.14 Focus & Input Integrity

While a user types in or operates a field, the application never takes the field away.

- **NFR-DA-FOC-01:** Keyboard focus shall stay in the field the user is typing in or operating through every keystroke, dropdown change, validation, saving, state update, re-render and panel change.
- **NFR-DA-FOC-02:** No input shall be lost on a validation failure, a panel change or a refresh; a field shall never be reset because its value is invalid.
- **NFR-DA-FOC-03:** Every interactive element shall show a visible focus ring (3 pixels of Azure at 35 % opacity, 2 pixels from the element); the tab order shall follow the reading order; every action shall be reachable with the keyboard alone; Escape shall close the topmost overlay.
- **NFR-DA-FOC-04:** Code review shall reject the known causes of focus loss: a component defined inside a render function; an unstable key (the row position for reorderable rows, a random or time-based value, or a value derived from changing state) instead of the record identifier and column name; automatic focus that runs again after the first display; an effect that moves focus on every change; form fields re-created on every render; a context value re-created on every render; a field that swaps element type (single-line and multi-line) while in use.
- **NFR-DA-FOC-05:** Before a table or form is released, automated tests shall confirm that typing in every free-text cell and changing every dropdown keeps focus, that Tab, Shift+Tab, Enter, Escape and the arrow keys behave as in Table 4.49-1, that validation and saving do not move focus, that changing record or panel does not move focus unexpectedly, and that no new console warning appears.

Input-integrity defect catalog. The Universal Field-Typing Fix identifies twenty-three causes of Input-Integrity Defects, C1 to C23, listed in Table L-1. NFR-DA-FOC-04 covers C1, C2, C3, C7, C8 and C10; the requirements below cover the others and set the audit an application undergoes to remove them.

- **NFR-DA-FOC-06:** A field shall keep the same component identity, element type, name and identifier for as long as it is in use: no list, row or field key derived from an edited value (C2, C12); no name or identifier generated from the current value (C13); no dialog, drawer, panel, wizard step or form region re-created through a changing key, a component defined during rendering or injected HTML (C4, C14); no field removed from the page because its own value, another value being typed or its validation state changes (C18).
- **NFR-DA-FOC-07:** A controlled field shall receive a value of the primitive type it edits, never an object or an array (C6), and shall stay controlled for its whole life, a missing value being normalized to the empty value the component expects rather than switching between undefined, null and a value (C22); the public data shape shall not change to achieve this.
- **NFR-DA-FOC-08:** Validation and normalization shall never rewrite the value a user is typing nor move focus (C11, C16); validation shall run when the user leaves the field or at the lifecycle point the screen already uses. Autosave, save, checkpoint and synchronization shall never clear, reset or reinitialize an active form or replace its Active Draft; a form is reset only on an explicit close, cancel, discard or save-and-close (C17).
- **NFR-DA-FOC-09:** An asynchronous result — search, autocomplete, remote lookup, server validation, autosave, query result or debounced request — shall update a field only when it

answers the current request; a stale result shall be discarded through cancellation, request identifiers or sequence checks (C19). A debounce or throttle shall be a single instance for the lifetime of its component, its pending work cancelled when the component is intentionally removed (C20).

- **NFR-DA-FOC-10:** The Active Draft shall take precedence over parent, derived, subscribed and server state while the user edits; external state is merged at a lifecycle boundary only (C21). Switching route, tab, panel or wizard step shall not destroy a form whose draft the user expects to keep (C15). Form libraries shall use their stable field identifiers and shall not rebuild field arrays, re-register fields or reset default values while a user edits (C5). A memoized child shall not receive a callback re-created on each render where that identity disturbs typing (C9).
- **NFR-DA-FOC-11:** The pointer, focus, blur, click and change events of dropdowns, comboboxes, date and time pickers, popovers, inline editors and Row Editors shall be ordered so that an interaction completes before its field is blurred, committed, reset, closed or removed (C23).
- **NFR-DA-FOC-12:** An Input-Integrity Defect shall be corrected at its cause. The following shall not be accepted as a fix: re-focusing a field after rendering or after a delay; focusing a field after every state update; disabling validation or autosave globally; disabling strict mode, lint rules, tests or type checks; suppressing blur events globally; silencing an error instead of correcting its cause.
- **NFR-DA-FOC-13:** A remediation pass for input integrity shall follow the audit of Appendix L: a Typeable Field Inventory covering every screen and shared field component, including custom and third-party date pickers, comboboxes, autocompletes, masked and numeric inputs and rich-text editors; a check of every cause C1 to C23, each confirmed instance recorded and fixed or deferred with its reason, and each absent cause reported as “Checked — no instance found”; the smallest correct change, preserving public component interfaces, routes, API contracts, data shapes, business logic and visual design; a final re-scan of the touched areas; and the report of Table L-3.

6. Common Data Model

The following core entities recur, under equivalent names, across applications built to this standard. Each entity carries a tenant (organization_id) foreign key except Group and any global, platform-wide catalog.

Entity	Description
Group	Optional holding entity above Organization; groups sibling Organizations.
Organization	The tenant boundary; holds sector/country/profile attributes and all tenant-isolated data.
Project	Optional sub-entity of an Organization; the principal record type may reference one; count enforced against the active Pack's quota.
ObsNode	A node in the OBS tree; referenced by users and by every governance entity below; count enforced against the active Pack's quota.
User / Role / Permission	Person, named role, and a global permission catalog entry (module.action) granted to roles.
License / Configuration	Configuration Management record: Solution Pack, Add-On flags, Compliance Standard flags, deployment option, support tier, seats, billing.
GovernanceSettings	KPI thresholds, alert toggles, and process-default/justification-requirement settings per Organization.
Standard / StandardRequirement	A reference standard and its clause-level requirement breakdown, each illustrated with an example.
Principal Record (e.g. a case, a project, an assessment)	The central aggregate the application manages through its defined multi-stage lifecycle.
Action / ActionEvaluation / ActionEvidence	Abstract action base, its Owner/Evaluator-segregated effectiveness verdict, and supporting evidence attachments.
NotificationAlert	An instance of a cataloged alert type, tagged with its process/stage.
AIAgentLog / AIUsageLog	Immutable logs of every AI agent invocation and every AI suggestion's accepted/edited/rejected outcome.
AuditLog	Immutable log of every create/update/delete on a governed entity.
AIUseCase / AIUseCaseVersion / AIUseCaseProjectOverride	Governed AI Use Case metadata (including the stored owning module, trigger and expected output, the use case it is based on and its origin: standard, manual, duplicate or AI-suggested), its versioned content, and its per-Project tri-state activation override.
BusinessRule / Control / RiskOpportunity	Typed governance records, each optionally tagged to an OBS node, a process/stage, and (Control) a Compliance Standard.
RacsiActivity / RacsiAssignment	A governed activity and its R/A/C/S/I assignments, at most one Accountable per activity.
BpmnDiagram	A stored BPMN 2.0 process diagram (XML payload), owned by an OBS node.

Entity	Description
CustomKpi	An organization-defined KPI additive to the platform's built-in KPI set.
Template (Sheet/Charter/Communication/Training, etc.)	A versioned, pre-filled starting point for the application's principal record type or a related artifact.
WbsNode	A node in a Work Breakdown Structure, referencing one or more Action IDs as leaf-level work items, with planned dates, percent-complete, and dependency links to other WBS nodes/Actions.
EntityVersion	A generic, immutable version record produced by the Version Management service for any entity type registered with it; carries the entity's full state at that version, the acting user, and a timestamp.
CommunicationDispatch / DeliveryStatus	A composed message routed through one or more channels (in-app, email, SMS, push, webhook) and its per-channel delivery status (queued/sent/delivered/failed).
RexEntry	A Return on Experience record captured at the closure of a principal record or an Action: what went well, what didn't, root cause, recommendation, category/tag, and an effectiveness rating; versioned and indexed into the retrieval corpus.
ExternalIntegration	A registered connection to an external application (CRM, ERP, calendar, storage, identity provider, etc.), its own enable/disable state, Solution-Pack/Add-On gate, credentials reference, and connection-health/audit log.
UserPreference	Per-user settings stored server-side: display language, menu dock position, pinned state, collapsed menu groups, favorites, and notification channel choices.
BenchmarkMetric / BenchmarkSharing	A catalog entry of a comparable metric (unit, direction of good, computation), and an Organization's Group-sharing setting, audited on every change.
Backup	A dated, consistent copy of the data store, with its size and retention expiry; listed to the platform administrator only.
RequirementTrace	A requirement ID of this standard with its implementation status and evidence note, shown in the traceability view.
ObsMember	A person placed in an OBS node, with the standard role and the role played there.
ChecklistTemplate	A versioned list of checklist items in the Checklist Library, universal or specific to a vertical, mode or track, associated with zero or more gates (GateChecklist).
Attachment	A file attached to an action or a checklist item: name, type, size, author, date, tenant-scoped storage reference.
Vertical	A sector entity: prefix, name, taxonomy parent, lifecycle state, owner, version; links to its macro and E2E processes, data extensions, compliance mappings and configuration.
VerticalActivation	The activation of a vertical by an Organization, with its version, validation result and activation order.
SmeTrack	An SME rigor level: processes, gates, checklist items per gate, typical duration, lifecycle state.
ComplexityCriterion	A scoring criterion and its weight, universal or specific to one vertical, versioned.

Entity	Description
ComplexityScore	The scored criteria of a project, the result, the recommended and chosen track, and any override justification and approval.
ProjectTemplate	A catalog entry: scope (universal or vertical), mode and track, status, version, fields, phases with their attached gates and checklists, roles and milestones.
GateDefinition	A Gate Library entry: criteria, approvers, decision options, applicability, version.
GateChecklist	The association of a checklist template with a gate, optionally per track, with its order and whether it is mandatory.
PhaseAttachment	The attachment of a gate, or of a checklist not related to a gate, at a given version, to a phase of a project or a template.
Pack	An SME pack, vertical pack or bundle: contents, target segment, pricing rules.
OnboardingPlan	The onboarding of an SME: steps, imports, validation results, dates, metrics.
QuestionnaireTemplate	A versioned questionnaire template — sections and questions — tagged by focus, sector and language; seeded or organization-authored.
Questionnaire	A questionnaire instance loaded, generated or written for an engagement, with its elaboration mode per question, AI-validation state, RACSI allocation per section, status and due date.
QuestionnaireDistribution	The dispatch of a questionnaire to one stakeholder through one channel, with its reminders and per-channel delivery status.
QuestionnaireResponse	One respondent's answers, with channel used, consent, capture and submission times, completeness percentage and validation; stripped of identity when the questionnaire is anonymized.
Stakeholder	A registered respondent: role, function, decision level, site, preferred channel and consent status.
Function	A business function of an Organization; versioned; linked to OBS Roles and process elements.
OrgRole	An OBS Role defined in an OBS unit, with mission, responsibilities and proposed default access roles.
RoleFunction	The link between an OBS Role and a Function (many to many).
RoleAssignment	A person playing an OBS Role: start and end dates, allocation, holder type.
ProcessElement / ProcessElementVersion	Any element of the process design and its immutable versions.
DesignRelease	A named snapshot of chosen element versions, with its lifecycle status.
RecordDesignBinding	The element versions a running record, or one of its phases, is bound to.

Entity	Description
PromptSpecification / PromptFieldVersion	The structured prompt of an AI Use Case and the versions of each field.
AIUseCaseLink	The link of an AI Use Case to a process element it serves — its primary step, or a further step, task, macro process or end-to-end process — with the input fields it reads and the output fields it proposes.
StepForm / StepFormField	The form kind of a step and its fields, checks and completion effects.
ReadinessItem	An item of the readiness checklist of a macro process, with its automatic check.
DocumentTemplate / TemplateSection	A documented-information template and its typed, data-bound sections.
Document / DocumentVersion	A controlled document of a project and its versions, with status, approval and content.
DocumentLayout	The logo, colors, fonts, header and footer of an Organization's documents.
AttachmentVersion	A version of an attachment in its chain, with number, author, date and note.
SearchIndexEntry	A tenant-scoped, rights-aware index entry of a searchable item.
DocumentProfile	The minimum content and checks of a document type.
DocumentSection	A section of a document version: kind, order, text, Data Source Binding, filters and columns.
DataSourceBinding	The link of a section to a named data source, with filters and sort order.
SectionOverride	Text or a value typed in a document section that replaces the source data for that document.
DocumentSourceTrace	The source record (identifier and version) of each data-bound value of a document version.
ConsistencyFinding	A finding of a Consistency Check: location, rule, severity, status, justification.
ChoiceList / ChoiceValue	A Controlled List and its values, reference or typed by users, with language labels.
RecordLink	A many-to-many link between two records, with its type.
EntryOrigin	The origin of a record entered in a table: Manual, Library or AI (attribute of the record).
ModelProfile	The parameters a language model accepts, its reasoning behavior and its output limits.
AuditPlan	A planned audit or inspection with frequency (list or custom), objectives, criteria, scope and team.
Finding	A graded audit or inspection finding and, for a nonconformity, its detailed record.
UiPreference	A user's layout preferences: panel widths, collapsed panels, density mode and the version of the design system last applied; stored with the shell preferences of FR-DA-NAV-06.

Entity	Description
TableViewPreference	A user's settings for one table: column widths, order, visibility and pinning, sort, page size, density, the zoom level of its Full-Screen Entry Workspace and the last confirmed paste mapping. Also called TablePreference.
PanelWidths	A user's navigation and context panel widths, stored with the shell preferences (FR-DA-NAV-06); held within UiPreference.
DesignTokenSet	The Design Tokens of the brand graphical chart, with their version (Appendix J).
ProjectTemplateBlueprint	The content a project template applies to every project created from it: processes, macro processes, tasks and steps, gates, rules, controls, risks and opportunities, alerts, KPIs and Reporting Plan (FR-DA-PTB-01).
CustomElement	A phase, macro process or step a template adds to the reference process design (FR-DA-PTB-04).
TemplateListRow	One row of a blueprint list — rule, control, risk or opportunity, alert, KPI or report — with its inclusion and origin (FR-DA-PTB-07, 08).
ProjectBlueprintTrace	The template code and version applied to a project, and the blueprint element each project element came from (FR-DA-PTB-06, 09).
DocumentFormatting	The Document Formatting of a document template, including its logo choice and position (FR-DA-DFP-01, 02).
SectionFormatting	The Section Formatting of a template or document section (FR-DA-DFP-03).
DocumentPicture	The uploaded picture of a Picture Section, with its caption, width and alignment (FR-DA-DFP-04).
PasteOperation	One Structured Paste into a Multi-Entry: user, time, clipboard format, column-to-field mapping, records created, changed and rejected, and undo state; written to the audit trail (FR-DA-MRE-19 to 22).
ImportSession	One file import into a Multi-Entry: the source (file name, type, size, encoding, sheet), the column-to-field mapping, the rows with their repair state and the PasteOperations that committed them.
BatchOperation	A bulk paste, import, update or delete sent in batches: target, batch size, batches done and failed with their errors, cancelled or finished, and the batch to resume from.
DraftBackup	Kept in the user’s browser, not on the server: the pending rows of one Multi-Entry for one user, with its version, the time it was taken and the Row Stamp each row was based on.

Table 6-1 maps the entity names of the (New) SRS revision 1.7 to the entities above, so that each concept has one canonical entity.

Table 6-1 — Entity names of the (New) SRS revision 1.7

Entity in the (New) SRS	Entity in this document
TableViewPreference	TableViewPreference
TablePreference	TableViewPreference
ProjectTemplateBlueprint	ProjectTemplateBlueprint
CustomElement	CustomElement
TemplateListRow	TemplateListRow
ProjectBlueprintTrace	ProjectBlueprintTrace
DocumentFormatting	DocumentFormatting
SectionFormatting	SectionFormatting
DocumentPicture	DocumentPicture
PanelWidths	PanelWidths
DesignTokenSet	DesignTokenSet

7. Common External Interface Conventions

7.1 User Interface Conventions

Every screen is accessible only when the authenticated user holds the corresponding view permission and, where applicable, the required Solution Pack entitlement. A Dashboard, a Configuration Management screen, a Permission Matrix screen, a Help screen, an Alerts inbox, a Reports screen with per-format download actions, and an AI Assistant (dedicated screen plus floating widget) are present in every application.

Every screen sits inside the common navigation shell (Section 4.19). A Benchmarking screen, with Within the organization and Across the group views, is present in every application that holds comparable records (Section 4.20), and a Backups panel is present on the platform administrator's configuration screen (Section 4.21). A screen the user may not open shows an access notice (FR-DA-RBAC-08) rather than an empty page.

Revision 1.2 adds a tenancy screen (Group Yes/No > Organization > records), a portfolio overview, a checklist template library in the process design, an attachments panel on every action, AI badges on the process design, and a live AI model card in personal settings and in administration.

Revision 1.3 adds screens for verticals and their activation, SME tracks and complexity criteria, packs, the project template catalog, a project creation entry with three tabs (From the catalog, Manual, With AI), a Checklist Library (checklists related to a gate or not) and a Gate Library (gates with their associated checklists), both attachable to phases.

Revision 1.5 adds a Search button at the start of the primary menu with a global search screen; a process design workspace with version history, comparison and restore on every element; OBS views by unit, by function and by person with an organization chart; a Prompt Specification editor with one field per aspect; typed step forms with record tables and decision matrices; a readiness checklist on every macro process; and a Documents screen with required documents, templates and layout.

Revision 1.6 adds a search bar in the header of every screen; direct menu entries for managed catalogs (functions, document templates and layout); a Row Editor on every record table; Controlled Lists with a Custom value; "ID (name)" labels; scrollable dialogs with fixed title and actions; a Consistency Check panel, a version comparison and section editing on documents; and an engine notice on every AI answer.

Revision 1.7 applies the AI Value Graphical Chart to every screen: a 64-pixel top bar with the horizontal logo, the global search and Command Palette (Ctrl/Cmd+K) and the language, notification, AI Assistant and user controls; breadcrumbs; a resizable navigation panel and context panel around a fluid content area; Inline Editable Tables with sticky headers, row status, row actions, filters and density; designed loading, empty, error, success, offline and permission states; KPI tiles and charts in the AI Value series colors with export; and every component designed for its hover, focus, active, disabled, loading, error and success states.

The (New) SRS revision 1.7 introduces the same screens under different section numbers: a project template editor (overview, processes and steps, rules, controls, risks, alerts, KPIs, reporting); structure, formatting and picture editors with a preview for document templates and

draft documents; a pager with page size and density under long lists; Inline Editable Tables; resize handles on the navigation and context panels; a scrollable BPMN palette; toasts with Undo; and a full-page error state. In this document those screens are specified in Sections 4.33, 4.43 and 4.49 to 4.52 and Appendix K.

Revision 1.8 adds to every Multi-Entry an Expand action that opens a Full-Screen Entry Workspace with Zoom Controls (Ctrl/Cmd + scroll, slider, Zoom in, Zoom out, Fit and Reset) and a “Return to page” action, also reached with Escape; a Paste Preview with column mapping for records pasted from Word, Excel and other files; and the copy of rows with their column titles.

Revision 1.9 adds an AI Use Case page and form, a coverage view of the process design and, on the steps, the tier, AI Use Case, engine and sources of every AI proposal; in every Multi-Entry, an Issues list, a “Rows with errors” filter, Undo and Redo, Import and Export in the toolbar, a footer with totals and the save state, banners for a restored draft and for an offline queue, a batch progress bar, dialogs for edit conflicts, deletion and leaving with unsaved rows, a five-stage import with a stepper and, on phones, a list with one-by-one editing.

7.2 Software Interfaces (REST API)

All interfaces are exposed as a JSON REST API, secured by a bearer token (except the login endpoint), and enforced per-endpoint by the RBAC permission catalog and, where applicable, Solution Pack entitlement middleware. At minimum, every application exposes equivalents of the following endpoint families:

- Auth: login, current-user profile, language preference update.
- Hierarchy & OBS: Group/Organization/Project CRUD, OBS CRUD, quota-checked creation.
- Users, Roles & Permissions: user CRUD, role assignment, permission-matrix read/write.
- Configuration Management: read effective configuration and static catalog (open to any authenticated user); write Pack/Add-On/Compliance selection (permission-gated); read live usage against quota.
- Governance: Business Rules, Controls, Risks & Opportunities, RACSI, Custom KPIs — each with process/stage filtering.
- AI Use Cases: catalog CRUD with duplicate, links and linkable process elements, library figures, coverage of the process design, activation, per-Project override, version history and revert, append-only usage log; step proposals through the AI Use Case linked to the step (FR-DA-AI-21 to 26).
- BPMN: diagram CRUD, view requiring at least view-mode entitlement, write requiring full-edit entitlement.
- AI Assistant: a data-query endpoint and an application-help endpoint, each Pack-gated.
- Reports: per-report data endpoints and a per-report export endpoint accepting a target format (pdf/xlsx/docx).
- Alerts: list, run/recompute, mark-read.
- User preferences: read and update the current user's navigation and display preferences.
- Document formatting and pictures: read and update the Document Formatting of a template, the Section Formatting of a section and its Picture Sections (upload, caption, width, alignment); formatting preview (FR-DA-DFP-01 to 08).

- Data tables and pagination preferences: read and update a user's Page Size, Density, column widths and sort state per table (FR-DA-TBL-01 to 07).
- Workflow steps: complete, and reopen with justification (FR-DA-AUD-05).
- Benchmarking: metric catalog, within-organization comparison, Group comparison, and read/update of the Organization's sharing setting.
- Operations: health (unauthenticated, no tenant data); backup list and on-demand backup (platform administrator).
- Tenancy and portfolio: tenancy tree, scope lists (groups, organizations, records the user may see), records across the scope, portfolio matrix.
- Checklist templates: library CRUD with version history; add to a gate or phase checklist from a template or by hand; save it as a template.
- Attachments: accepted types, multi-file upload, download, delete.
- AI model: model and provider catalog; connection test.
- Verticals: CRUD, lifecycle transitions and approvals, versions and rollback, activation with validation, deactivation with dependency check.
- SME and scoring: tracks CRUD, complexity criteria and weights, score computation, track override with approval.
- Project templates: catalog list with filters, CRUD, duplicate, publish and retire, version history, save a project as a template.
- Project creation: create from a template, manually, or from an AI draft (draft, then accept with the retained items).
- Checklist Library: CRUD, duplicate and versions; filter by gate or "not related to a gate"; gates using a checklist; attach a stand-alone checklist to a phase, task or project.
- Gate Library: CRUD, duplicate and versions; associate, order and detach checklists; attach, reorder and detach gates on a phase; decisions and waivers.
- Packs and onboarding: pack and bundle CRUD with pricing rules, upgrade and downgrade; onboarding plan, import and validation.
- Bulk entry: validate a batch of pasted rows without saving (preview with per-cell results), apply it as one audited operation and undo it; read and update a user's workspace zoom level and paste mapping per Multi-Entry (FR-DA-MRE-15 to 23).
- Bulk import and export: read a spreadsheet workbook (sheets and cell text), export the given view as a workbook, delete a batch of rows with a result per row, read the import limits and batch size; a row update carries its Row Stamp and answers a conflict when the row changed (FR-DA-MRE-26 to 31, NFR-DA-SEC-22).

7.3 Hardware Interfaces

None. An application is a standard web application with no direct hardware interface requirement.

7.4 Communications Interfaces

HTTP/HTTPS using JSON request/response bodies for all client-server communication, except report-export responses (application/pdf, .xlsx, .docx binary bodies with a Content-Disposition attachment header). Authentication is via a signed bearer token issued at login.

Appendix A — Conformance Record & Resolution Rules

This standard states the target platform behavior every application should converge on. Applications do not reach it at the same pace, so this appendix makes each gap explicit rather than silently absorbed. Section A.1 is the conformance record each application keeps against this standard; Section A.2 records where this standard and the AI Value Graphical Chart differ, and the rule that applies.

A.1 Conformance record

Each application copies Table A-1 into its own SRS or release documentation, sets the status of every area — Met, Partial, Not met, To be assessed, or Deployment responsibility — and cites the evidence (automated test, walkthrough replayed on a fresh installation, load-test result, accessibility scan, remediation report) or describes the gap. The administrator traceability view of FR-DA-OPS-08 shows the same status requirement by requirement. A new application starts with every area To be assessed. The assessment notes record lessons from earlier revisions that apply to any application.

Table A-1 — Conformance record (one copy per application)

Area	Requirements	Evidence expected	Assessment notes
Server-side authorization	NFR-DA-SEC-01, 03, 09; FR-DA-RBAC-03; FR-DA-CFG-06	Automated tests calling each endpoint without a token, without the permission and without the entitlement.	RBAC and Pack gating enforced only in the client is Not met, whatever the screens show. Closing this gap is the highest-priority item of any production-grade build.
Tenant isolation and identifiers	FR-DA-TEN-08, 11; NFR-DA-SEC-11, 13	Cross-tenant test suite run in the release pipeline; 404 for another tenant's record.	Record identifiers drawn from one global sequence answer 404 correctly but can hint at overall platform activity: Partial for FR-DA-TEN-11.
Persistence model	Sections 2.5, 3.13; NFR-DA-REL-04; NFR-DA-SCALE-03	Schema description; concurrent-write test.	Any relational or document store that preserves per-entity querying conforms. A store that rewrites a whole tenant document on every save, without optimistic locking, suits a single-machine demonstration only and is Partial for production.

Area	Requirements	Evidence expected	Assessment notes
Solution Pack model	Sections 3.3, 4.3	Configuration screen; entitlement tests.	The three independent axes — Pack, Add-Ons, Compliance Standards — are the requirement. The names of the Packs and the modules each unlocks are product-specific and expected to differ.
Compliance & Security Standards catalog	FR-DA-CFG-04, 05, 09	Catalog listing; idempotent seeding test.	GDPR, ISO/IEC 27001 and SOC 2 are the minimum; an application may offer more (for example ISO/IEC 27701, HIPAA, FIPS 140-2 or ISO 9001).
AI Assistant	Section 4.10	Intent-permission tests; help-mode retrieval test.	A further retrieval source (for example a process knowledge base) extends FR-DA-AST-04 and is not a divergence.
Governance-artifact scoping	FR-DA-GOV-08	Governance view of one process or stage.	Items may be tagged to a lifecycle stage or to a macro-process catalog; either conforms at the granularity of the application's own process model.
Shared platform services	Sections 4.15 – 4.18; FR-DA-CFG-10 – 14	Demonstration and automated tests of WBS and Gantt, version management, multi-channel dispatch, REX and the integration registry.	These are shared services. An application that versions each entity type with its own mechanism is Partial for Section 4.16; in-app-only notification is Partial for Section 4.17.
Mobile compatibility	NFR-DA-UX-02, 05, 08; NFR-DA-COMPAT-03	Screens checked at a 390-pixel viewport.	No horizontal page scroll and 44-pixel touch targets are the test; a native or installable wrapper is optional.
Navigation shell, benchmarking and operations	Sections 4.19 – 4.21	Preferences followed across devices; minimum-sample rule; backup and restore test; startup guards.	A runtime version checked at install time but not by the server at startup is Partial for FR-DA-OPS-05. Off-site backup copies are a Deployment responsibility (Section 2.7).
Tenancy, portfolio, checklists and attachments	Sections 4.1, 4.22 – 4.24	Provisioning test, including the refusal of a duplicate e-mail domain; User Guide walkthroughs replayed.	—
AI placement, model choice and BPMN workspace	FR-DA-AI-12 – 20; FR-DA-BPMN-06 – 08	Tier badges runnable in place; connection-test outcomes.	A live call can be verified only with a user's own key; record which outcomes were verified.

Area	Requirements	Evidence expected	Assessment notes
Verticals, SME, templates, creation modes and gates	Sections 4.25 – 4.35	Configuration without code change; activation, rollback and blueprint tests.	Gates or tracks fixed in code are Not met for the Gate Library; a sector attribute with seeded templates alone is Partial for Section 4.25.
Questionnaires	Section 4.36; NFR-DA-QLT-01 – 03	Consent, offline synchronization and completeness tests.	—
Search, process design, OBS and prompts	Sections 4.37 – 4.40	Rights-aware search test; restore of a composite version; prompt completeness check.	Named Design Releases and design import and export are often delivered after the core; record them separately.
Naming, step forms and documents	Sections 4.41 – 4.45	Generated documents compared with their records; Consistency Check findings.	Consistency Checks, change detection on published documents, Edit-through and Section Overrides are frequent gaps; record each one.
Data entry, linked records and audits	Sections 4.46 – 4.48	Walkthrough of a record table, a many-to-many link and a register propagation.	—
AI transparency and model profiles	FR-DA-AIP-12 – 16; FR-DA-AI-19, 20	Engine notice shown with and without a live model.	Capability profiles built in for listed models rather than configurable are Partial for FR-DA-AI-19.
Visual design system	NFR-DA-VDS-01 – 21; FR-DA-VIZ; FR-DA-STA	Token check in continuous integration; screen review against Appendix K.	An application styled against an earlier brand guide is To be assessed against the Graphical Chart 1.2.
Contrast of brand colors	NFR-DA-UX-06, 19	Automated accessibility scan.	A brand-mandated color pair below WCAG AA is recorded here with its contrast ratio rather than silently changed.
Multi-record entry and focus	FR-DA-MRE-01 – 14; NFR-DA-FOC-01 – 05	Focus tests of NFR-DA-FOC-05 on every record table.	Loss of focus while typing in a record table is the defect that led to Section 5.14 and Appendix L; test every table, not a sample.
Full-screen entry workspace and paste	FR-DA-MRE-15 – 23; NFR-DA-PERF-13; NFR-DA-SEC-21; NFR-DA-UX-20	Workspace opened from every Multi-Entry; zoom, entry and return tested by mouse and by keyboard; paste from Word, Excel, a CSV file and a web page.	New in revision 1.8: To be assessed for every application.
Input integrity	NFR-DA-FOC-06 – 13	Remediation report of Table L-3.	The first audit records its confirmed, fixed and deferred instances for each cause C1 to C23.

Area	Requirements	Evidence expected	Assessment notes
Fourth-round and realigned families	FR-DA-PTC-07, 08; FR-DA-PTB; FR-DA-DFP; FR-DA-TBL; FR-DA-PNL	Template editor, formatting preview, pagination and panel tests.	Conformance records written against FR-DA-DOC-06, 07 and FR-DA-MRE-11 – 13 remain valid for the new families.
AI Use Case Library and governed generation	FR-DA-AI-01, 07, 13, 21 – 26; FR-DA-AIP-07; NFR-DA-MAINT-04	Use case created, edited, duplicated and deleted in its form and shown on its page; coverage view; every AI action traced to an active use case in the usage log; model precedence tested with and without each model.	New or amended in revision 1.9: to be assessed for every application.
Multi-Entry at scale and without loss	FR-DA-MRE-05, 08, 11, 14, 18, 22, 24 – 35; NFR-DA-PERF-14; NFR-DA-REL-10, 11; NFR-DA-UX-21; NFR-DA-SEC-22	Automated tests of NFR-DA-PORT-05; a 10,000-row table timed; a draft restored after a refresh; an edit conflict and an offline queue reproduced; an import of a file with errors repaired and undone; the leave guard and the phone list checked.	New or amended in revision 1.9: to be assessed for every application.
Data rules learned in acceptance	FR-DA-SFM-08, 09; FR-DA-I18N-10; FR-DA-DOC-12; NFR-DA-I18N-07; NFR-DA-MAINT-11	A matrix step typed in each language; a reopened step read as open; every generated Office file validated; a download with a non-Latin name; an upgrade over an existing database without reseeding.	New in revision 1.9.

A.2 Resolution rules

Where this standard and the AI Value Graphical Chart, or this standard and an earlier edition, say different things, Table A-2 gives the rule that applies to every application.

Table A-2 — Resolution rules

Area	Standard requirement	Difference	Rule
Preference storage (FR-DA-MRE-11, FR-DA-PNL-06, NFR-DA-VDS-15)	Panel widths, column settings, density and workspace zoom are user preferences kept on the server.	The Graphical Chart (§14.9, §16.5, §22.3) stores them in the browser only. This standard keeps them server-side, with a browser cache, so that they follow the user on every device as FR-DA-NAV-06 requires.	This standard applies.

Area	Standard requirement	Difference	Rule
Row Editor and inline entry (FR-DA-DEU-03, FR-DA-MRE-01)	Both patterns available.	The Graphical Chart makes inline entry the default for records of two to eight fields; revision 1.6 requires a Row Editor on every record table. Inline editing is the default and the Row Editor stays available from every row, on the page and in the workspace.	Both apply.
Menu docking and panels (FR-DA-NAV-04, FR-DA-PNL)	Menu docked on any edge; resizable panels.	The Graphical Chart describes a menu on the start edge only. Resizing applies to the navigation panel when the menu is docked on the start or end edge; on the top or bottom edge only the context panel is resizable.	This standard applies.
Small colored text (NFR-DA-VDS-03, NFR-DA-UX-06)	Azure, Teal and Green never used for text below 18.66 pixels bold on a light surface.	The Graphical Chart sets eyebrow labels in 12-pixel Azure and the tagline in 13-pixel Green, which its own §3.6 rates below WCAG AA. On light surfaces these labels use Navy (or are enlarged to 18.66 pixels bold); the brand colors keep their place on dark or tinted fills.	This standard applies.
Languages and currency (NFR-DA-I18N-02, 05; FR-DA-I18N-09)	Configurable languages and currencies.	The Graphical Chart requires English, French and Arabic and makes MAD the default currency; this standard keeps its wider configurable list (adding Spanish and German; USD, EUR, MAD, ZAR). The Graphical Chart set is the minimum; the default currency is set per Organization.	Both apply.
Role names (Graphical Chart §24.3)	Role names come from each application's RBAC catalog.	The Graphical Chart lists example role names drawn from one product (Administrator, Manager, Auditor, Contributor, Viewer). They are not imposed; each application displays its own role names consistently on every screen.	This standard applies.

Area	Standard requirement	Difference	Rule
Revision 1.7 ID aliases (FR-DA-MRE, FR-DA-PNL)	FR-DA-MRE-01 – 23 and FR-DA-PNL-01 – 08 govern.	The (New) SRS revision 1.7 carries the shorter sets FR-DA-MRE-01 – 08 and FR-DA-PNL-01 – 05. A conformance record written against them is covered by the longer sets of this standard; no capability is lost.	This standard applies.
New ID families (FR-DA-DFP, FR-DA-TBL)	FR-DA-DFP-01 – 08; FR-DA-TBL-01 – 07.	New identifiers over content previously carried in FR-DA-DOC-06, 07 and FR-DA-MRE-11 – 13; conformance records written against those requirements remain valid.	Both apply.
Workspace zoom and browser zoom (FR-DA-MRE-16, NFR-DA-UX-20)	Zoom Controls scale the records area of the workspace.	The Graphical Chart describes no zoom inside an application. The workspace zoom scales the records area only and never replaces or blocks the zoom of the browser, which WCAG 2.1 (SC 1.4.4, 1.4.10) relies on.	Both apply.
Escape key (FR-DA-MRE-18, NFR-DA-FOC-03, Table 4.49-1)	Escape cancels a cell edit, closes the topmost overlay and leaves the workspace.	Three requirements give Escape a meaning. The innermost one applies first: the cell being edited, then an open menu or dialog, then the Full-Screen Entry Workspace.	This standard applies.
Paste (FR-DA-MRE-07, FR-DA-MRE-19 – 23)	Paste of rows from a spreadsheet; paste of complete records from any external file.	FR-DA-MRE-07 remains the base; revision 1.8 extends it to word-processing, CSV and web content, with a Paste Preview, conversion and a single Undo.	Both apply.

An application that closes one of its gaps updates its conformance record in the same release, so that the standard and the observed state of the application never drift silently apart.

Appendix B — Revision 1.1 Change Log

Revision 1.1 is additive. Every 1.0 requirement keeps its ID and wording, except the editorial changes marked Clarified below. New requirements take the next free number in their section, so a conformance record written against 1.0 remains valid against 1.1.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.1; reference implementation	Version raised to 1.1. A reference implementation added as a third source; purpose and overview explain the revision.
1.2 Scope	Added	Navigation shell, internal benchmarking, and platform operations brought into scope; external benchmarking stated out of scope.
1.4 Definitions	6 terms added	Navigation Shell, Pinned / Sliding Menu, Dock Position, Internal Benchmarking, Minimum Sample Size, Additive Migration.
2.2, 2.3	3 principles, 3 functions	Personal navigation shell; compare within the Group only; upgrade and recover in place.
2.5 Operating Environment	Clarified	Recommended Node.js raised from 22.5+ to 22.13+ (built-in SQLite without an experimental flag); required version declared in the package manifest.
2.6, 2.7	1 constraint, 1 assumption	Benchmark data leaving a tenant is aggregated only; off-site backup copies are the operator's responsibility.
3.1 Tenancy	Clarified	Group membership is the benchmarking boundary; Group Yes/No and Independent shown on hierarchy screens.
3.11 – 3.13	New sections	Navigation Shell & Personalization, Internal Benchmarking, and Platform Operations architectures.
4.1 Tenant & Hierarchy	FR-DA-TEN-09 – 11	Group membership display; same-tree OBS parents; opaque identifiers and 404 for foreign records.
4.2 Identity & RBAC	FR-DA-RBAC-08 – 10	Access notice instead of a blank page; restricted evaluation-content permission; additive permission migration at startup.
4.11 Audit Trail	FR-DA-AUD-05	Reopen a completed step, with justification, before its stage is decided.

Section	Requirement IDs / Type	Change
4.14 Localization	FR-DA-I18N-07 – 08	Server messages and assistant answers translated; multilingual Knowledge Base seed with language filter.
4.16 Version Management	FR-DA-VER-06	One shared side-by-side version comparison with changed fields highlighted.
4.18 REX	FR-DA-REX-08	REX prompt shown to whoever closes the record, with Skip that keeps the closure.
4.19 Navigation Shell	FR-DA-NAV-01 – 10 (new)	Menu from one navigation model; collapsible groups; pin / slide away; dock left, right, top, or bottom; favorites; server-side preferences; RTL mirroring; mobile drawer; keyboard operation; standard header bar.
4.20 Internal Benchmarking	FR-DA-BMK-01 – 09 (new)	Metric catalog; within-organization segments; Group ranks with ties; aggregates only; minimum sample; opt-out with audit; permissions; no external benchmarking; charts with filters.
4.21 Platform Operations	FR-DA-OPS-01 – 08 (new)	On-demand and daily backup; retention; restore; additive migrations; startup guards; health endpoint; deterministic seed; traceability view.
5.1 Performance	NFR-DA-PERF-05	95th-percentile read time of 1 s or less at 100 concurrent users; rules for a per-user read cache.
5.2 Security	NFR-DA-SEC-12 – 13	Login rate limiting and security headers; no cross-tenant exposure through caches, backups, or logs.
5.3 Reliability	NFR-DA-REL-05	Data loss bounded to 24 hours; restore tested each release.
5.5 Usability & Accessibility	NFR-DA-UX-05 – 07	Target sizes 24 px / 44 px; automated accessibility scan and contrast rule; accessible, translated notifications.
5.7 Portability	NFR-DA-PORT-03 – 04	CI on Linux, Windows, and macOS with seed, tests, and build; standard install / seed / start commands.
5.8 Maintainability	NFR-DA-MAINT-06	Guides generated in Word and PDF with a TOC; walkthroughs replayed against a fresh seed.
6 Data Model	4 entities	UserPreference, BenchmarkMetric / BenchmarkSharing, Backup, RequirementTrace.
7.1, 7.2 Interfaces	Clarified	Shell, Benchmarking screen, Backups panel, and access notice; endpoint families for preferences, workflow steps, benchmarking, and operations.

Section	Requirement IDs / Type	Change
Appendix A	Updated	Reference-implementation status added to six 1.0 target rows and server-side authorization; six rows added for 1.1 capabilities; closing note widened to any application.

Nothing was removed. Product-specific material of the reference implementation — its phase-gate tracks, gate criteria, industry content, and commercial packaging — was deliberately left out, because it does not apply to every application.

Appendix C — Revision 1.2 Change Log

Revision 1.2 is additive, like 1.1: every earlier requirement keeps its ID and wording, and new requirements take the next free number in their section.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.6	Version 1.2	Version raised to 1.2; purpose and overview describe the revision.
1.4 Definitions	4 terms added	Tenancy Tree, Provisioning, Portfolio Overview, Stage Checklist Template.
4.1 Tenant & Hierarchy	FR-DA-TEN-12 – 16	Tenancy tree and table; organization chosen before creating a record; provisioning of new organizations with an optional starting team; read-only visibility within the group; OBS members with roles.
4.4 AI Use Case Library	FR-DA-AI-12 – 15	Tier badges shown and runnable on the process design; links to step, process, task or E2E process; delete for custom use cases; model list with custom choice, custom endpoint and connection test.
4.7 BPMN	FR-DA-BPMN-06 – 08	Shapes palette section (legend when view-only); full-screen workspace with zoom and return; statement of whether diagrams drive execution.
4.12 Help	FR-DA-HLP-04	Decision matrix next to scored inputs; legend for every code or colour vocabulary.
4.19 Navigation	FR-DA-NAV-11	Menu organized by what users do, in a documented order.
4.21 Operations	FR-DA-OPS-09	Demonstration seed covers every module.
4.22 Portfolio Overview	FR-DA-PFO-01 – 04 (new)	Scope, record × stage matrix, legend and totals, CSV export, read-only foreign records.
4.23 Stage Checklist Library	FR-DA-CHK-01 – 05 (new)	Versioned templates per stage and variant; linked templates copied at stage opening; add from template or by hand without duplicates; sector templates; frozen after submission.
4.24 Attachments	FR-DA-ATT-01 – 04 (new)	Multi-file upload; published accepted formats with limits; metadata, download, audited removal; storage and backup.
5.5 Usability	NFR-DA-UX-08	Wide matrices scroll in their own container; no page-level horizontal scroll on phones.
5.8 Maintainability	NFR-DA-MAINT-07	User Guide walkthroughs start from the tenancy and include the decision matrix.

Section	Requirement IDs / Type	Change
6 Data Model	3 entities	ObsMember, ChecklistTemplate, Attachment.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for tenancy and portfolio, checklist templates, attachments and the AI model.
Appendix A	3 rows	Conformance of the 1.2 additions in the reference implementation.

Product-specific material of this round — its sectors, its phase-gate vocabulary and its demonstration content — stays in the documentation of the reference implementation.

Appendix D — Revision 1.3 Change Log

Revision 1.3 keeps every earlier requirement ID. It is additive except in Section 4.23, where five requirements are reworded so that checklists and gates are separate libraries; new requirements take the next free number in their section or a new family.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.5, 1.6	Version 1.3	Version raised to 1.3; purpose and overview describe the revision; the Verticals and SME source added to the references.
1.4 Definitions	11 terms added, 1 reworded	Stage Checklist Template renamed Checklist Template. Added: Vertical, Vertical Instance, SME, Mode (Full / SME), SME Track, Complexity Score, Project Template Catalog, Creation Mode, Phase, Gate Definition, Pack / Bundle.
2.3 Product Functions	1 function added	Verticals, SME mode, complexity scoring, template catalog, creation modes, gates and checklists on any phase.
3.14 (new)	Architecture	Five requirement layers; verticals and tracks as configuration data; ordered, additive activation.
4.13 Template Libraries	FR-DA-TPL-05	Project templates are managed through the catalog of Section 4.33.
4.23 Checklist Library (renamed)	FR-DA-CHK-01 – 05 reworded; CHK-06 – 07 (new)	Renamed from Stage Checklist Library. A checklist is an independent library entry, related to zero, one or several gates; stand-alone checklists attach to a phase, task or project without blocking progression; impact shown before change or deletion.
4.25 Vertical Framework & Governance	FR-DA-VRT-01 – 10 (new)	Definition, taxonomy, extensibility, lifecycle and approval, versioning and rollback, tenant isolation, guarded deactivation, audit, ownership and change management, metrics.
4.26 Vertical Processes	FR-DA-VPR-01 – 09 (new)	Macro processes (3–5) and E2E processes (2–4) per vertical; taxonomy, compliance mapping, tasks and steps, chains, triggers, terminal states, metrics and reports.
4.27 Vertical Data, Compliance & Integration	FR-DA-VDT-01 – 03, VCP-01 – 03, VIN-01 – 02 (new)	Data model extension, exchange formats, retention and archiving; compliance requirements, monitoring, audits and training; connectors, security, monitoring and error handling.
4.28 Vertical Configuration	FR-DA-VCF-01 – 04 (new)	Configurable parameters, pre-built templates, validation before activation, versioning and audit.

Section	Requirement IDs / Type	Change
4.29 SME Mode & Tracks	FR-DA-SME-01 – 08 (new)	SME definition and segments, cross-sector activation, 2–4 tracks, SME processes, extensibility, assignment and override, metrics, governance.
4.30 Complexity Scoring	FR-DA-SCO-01 – 04 (new)	Weighted configurable criteria summing to 100 %, vertical drivers, audit and reporting.
4.31 SME Onboarding & Experience	FR-DA-ONB-01 – 05 (new)	30-day onboarding plan, import and validation, guided training, simplified mobile experience and self-service support, metrics.
4.32 Packaging & Pricing	FR-DA-PKG-01 – 04 (new)	SME packs and vertical bundles as Solution Pack variants, pricing rules, upgrade and downgrade, reports.
4.33 Project Template Catalog	FR-DA-PTC-01 – 06 (new)	Universal or vertical, Full or SME templates; CRUD, duplicate and versions; catalog browsing; Draft, Published, Retired; seeding and save-as-template; permissions.
4.34 Project Creation Modes	FR-DA-PCM-01 – 05 (new)	From the catalog, Manual, With AI (reviewed item by item); scoring, quotas and audit in every mode.
4.35 Gate Library	FR-DA-GTE-01 – 07 (new)	Gate definitions with CRUD and versions; association of checklists from the Checklist Library; attachment of gates to any phase; vertical- and track-aware gates; enforcement and waivers; decisions and audit; library changes never alter started phases.
5.1, 5.2, 5.3, 5.4	NFR-DA-PERF-06 – 08, SEC-14 – 16, REL-06, SCALE-04 – 05	Activation, import and production sizing targets; federated authentication and MFA, vertical and track scopes, encryption, residency and BYOK; disaster recovery; scale limits and zero-downtime additions.
5.5, 5.6, 5.8, 5.10	NFR-DA-UX-09 – 10, I18N-02, MAINT-08, COMP-05 – 06	SME onboarding and training targets, satisfaction; languages, currencies and formats; configuration-driven layers; sector standards scaffolds; requirement traceability and change classes.
6 Data Model	11 entities, 1 reworded	Vertical, VerticalActivation, SmeTrack, ComplexityCriterion, ComplexityScore, ProjectTemplate, GateDefinition, GateChecklist, PhaseAttachment, Pack, OnboardingPlan; ChecklistTemplate no longer tied to one stage.
7.1, 7.2 Interfaces	Clarified	New screens and endpoint families for verticals, SME and scoring, templates, creation modes, gates and checklists, packs and onboarding.

Section	Requirement IDs / Type	Change
Appendix A	3 rows	Conformance of the 1.3 additions in the reference implementation.
Appendix E (new)	Traceability	Mapping of every source requirement family to the requirements of this standard.

Appendix E — Traceability to the Verticals and SME Requirements

Each requirement family of the Application-Agnostic Requirements for Verticals and SME source is covered as follows. The 29 vertical instances of the source (REQ-VERT-{SECTOR}-...: Automotive, Aerospace & Defense, Industrial Machinery, Healthcare & Life Sciences, Electronics & High-Tech, AEC & Construction, Food & Beverage, Social & Solidarity Economy, Pharmaceutical & Process, Retail, Footwear & Apparel, Consumer Packaged Goods, Energy & Utilities, Telecommunications, Process/Petrochemical, Shipbuilding, Consumer Products Extended, Process Industries, Infrastructure, Fabrication & Assembly, Process/Packaged Goods, Semiconductor & Electronics, Retail, Distribution, Logistics, Pharma, Manufacturing (Generic), Private Education, Public Education, Healthcare Providers) are content delivered as Vertical Instances under Sections 4.25 to 4.28; they stay in the vertical packs, not in this standard.

Source family	Requirements in this standard	Note
REQ-VERT-FW-001 – 010	FR-DA-VRT-01 – 08	FW-005 and FW-008 merged into VRT-04 and VRT-05.
REQ-VERT-GOV-001 – 005	FR-DA-VRT-04, 08, 09, 10	GOV-002 and GOV-004 duplicate FW-005 and FW-010.
REQ-VERT-MP-001 – 010	FR-DA-VPR-01 – 05, 09	Tasks, steps and dependencies merged into VPR-05.
REQ-VERT-E2E-001 – 007	FR-DA-VPR-06 – 09	Triggers and terminal states merged into VPR-07.
REQ-VERT-DATA-001 – 005	FR-DA-VDT-01 – 03	Retention and archiving merged into VDT-03.
REQ-VERT-COMP-001 – 005	FR-DA-VCP-01 – 03, NFR-DA-COMP-05	Kept subject to FR-DA-CFG-09.
REQ-VERT-INT-001 – 005	FR-DA-VIN-01 – 02	Builds on FR-DA-CFG-10 – 14.
REQ-VERT-CONF-001 – 005	FR-DA-VCF-01 – 04	Audit merged into VCF-04.
REQ-SME-FW-001 – 005	FR-DA-SME-01, 02, 05	
REQ-SME-MP-001 – 005, REQ-SME-E2E-001 – 005	FR-DA-SME-04	Reuses the vertical process rules.
REQ-SME-TRACK-001 – 005	FR-DA-SME-03, 06, 07, 08	
REQ-SME-SCORE-001 – 005	FR-DA-SCO-01, 02, 04	
REQ-SME-ONB-001 – 005	FR-DA-ONB-01 – 03, 05	
REQ-SME-PRICE-001 – 005	FR-DA-PKG-01 – 04	Packs are Solution Pack variants.
REQ-SME-UX-001 – 005	FR-DA-ONB-03, 04, NFR-DA-UX-02, 03	WCAG 2.1 AA and mobile use already required in Section 5.5.
REQ-SME-GOV-001 – 005	FR-DA-SME-08	
REQ-VS-ACT-001 – 003	FR-DA-SME-02, FR-DA-VRT-07, 08, Section 3.14	Vertical processes first, SME processes second.
REQ-VS-SCORE-001 – 003	FR-DA-SCO-03	
REQ-VS-CHK-001 – 003	FR-DA-CHK-01 – 07, FR-DA-GTE-05, 06	Checklists right-sized per vertical and track.
REQ-VS-GATE-001 – 003	FR-DA-GTE-01, 02, 04, 06	
REQ-VS-PKG-001 – 003	FR-DA-PKG-01 – 04	

Source family	Requirements in this standard	Note
REQ-NFR-PERF, SCALE	NFR-DA-PERF-06 – 08, NFR-DA-SCALE-04 – 05	Production targets; existing reference targets unchanged.
REQ-NFR-SEC, REL	NFR-DA-SEC-14 – 16, NFR-DA-REL-06	Audit and tenant isolation already covered (Section 4.11, NFR-DA-SEC-11).
REQ-NFR-UX, LOC	NFR-DA-UX-09 – 10, NFR-DA-I18N-02	RTL already covered (NFR-DA-I18N-01).
REQ-NFR-COMP, MNT; Part 7 governance	NFR-DA-COMP-05 – 06, NFR-DA-MAINT-08	

Product-specific material — the content of each vertical instance, the phase-gate vocabulary of each application and its demonstration data — stays in the documentation of that application.

Appendix F — Revision 1.4 Change Log

Revision 1.4 keeps every earlier requirement ID and is purely additive. Change request CR1 named the new section 4.22 and the new template requirements FR-DA-TPL-05 to 08, and asked for a revision 1.2; because revisions 1.2 and 1.3 already use Section 4.22 (Portfolio Overview) and FR-DA-TPL-05 (Project Template Catalog), the new section is 4.36, the new template requirements are FR-DA-TPL-06 to 09, and the revision is 1.4.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.4	Version raised to 1.4; purpose, scope and overview describe the revision; CR1 added to the references.
1.4 Definitions	4 terms added	Questionnaire Template, Focus, Distribution Channel, Stakeholder.
2.3 Product Functions	1 function added	Questionnaire design, multi-channel distribution, consented collection and consolidation.
4.13 Template Libraries	FR-DA-TPL-06 – 09 (new)	Seeded questionnaire template library by focus and sector; organization-authored templates; versioning through Section 4.16; export and re-import.
4.17 Multi-Channel Communication	FR-DA-COMM-06 – 08 (new)	Questionnaire dispatch channel abstraction; reminder cadence per channel and questionnaire; audit log of dispatches, reminders and responses.
4.36 (new) Questionnaire & Survey Management	FR-DA-QLT-01 – 12 (new)	Template CRUD with tags; Load / AI / Manual elaboration; human validation of AI content; RACSI per section; channels; consent; reminders; offline collection; completeness; tenant-scoped consolidation; downstream feed; response reporting.
5.12 (new) Questionnaire Data	NFR-DA-QLT-01 – 03 (new)	Response stored within 2 s; retention per tenant policy, default 5 years; identity stripped for anonymized questionnaires.
6 Data Model	5 entities	QuestionnaireTemplate, Questionnaire, QuestionnaireDistribution, QuestionnaireResponse, Stakeholder.
Appendix A	1 row	Conformance of the 1.4 additions.
Appendix F (new)	Change log	This change log.

Appendix G — Revision 1.5 Change Log

Revision 1.5 keeps every earlier requirement ID and is purely additive. Its requirements are application agnostic: they come from a reference implementation and from the product direction received with its user feedback, restated so that any application can meet them.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.5	Version raised to 1.5; purpose, scope and overview describe the revision; a reference implementation added to the references.
1.4 Definitions	12 terms added	Function, OBS Role, Role Assignment, Process Design, Process Element, Design Release, Current Version, Prompt Specification, Step Form Kind, Readiness Checklist, Documented Information Template, Global Search.
2.2, 2.3	3 principles, 4 functions	Versioned process design; accountability through roles; explicit prompts. Global search; process design management; OBS role model; generated documents.
3.15 – 3.18 (new)	Architecture	Process design and versioning; OBS role model; AI prompt specification; global search.
4.4 AI Use Case Library	FR-DA-AI-16 – 18 (new)	Organization-level live model stored encrypted; model per use case; view of the assembled prompt and of the engine used.
4.6 Governance	FR-DA-GOV-09 – 10 (new)	RACSI assigned to OBS Roles; RACSI editor per macro process.
4.16 Version Management	FR-DA-VER-07 – 09 (new)	List and restore of any version; composite versions; named baselines.
4.19 Navigation Shell	FR-DA-NAV-12 – 13 (new)	Search button in the menu, in every dock position; menu filter.
4.24 Attachments	FR-DA-ATT-05 – 06 (new)	Attachment version chains; attachments on steps, documents and registers.
4.37 (new) Global Search	FR-DA-SRCH-01 – 07 (new)	Scope, matching, results, filters, rights, index freshness.
4.38 (new) Process Design Management	FR-DA-PDM-01 – 11 (new)	CRUD and versions on every process element; compare and restore; composite versions; Design Releases; binding of running records; impact check; sharing and customization; import and export; audit.
4.39 (new) OBS: Functions, Roles and People	FR-DA-OBS-01 – 10 (new)	Functions; roles linked to one or more functions; people playing one or more roles; views; assignment and resolution through roles; hand-over; pickers; separation from RBAC; history.

Section	Requirement IDs / Type	Change
4.40 (new) AI Use Case Mapping & Prompt Specification	FR-DA-AIP-01 – 11 (new)	Mapping to tasks and steps with fit check; separate prompt fields; field and whole versions with restore; completeness; full population per step; assembly; test; logging; suggestion of new use cases.
4.41 (new) Naming & Descriptions	FR-DA-NAM-01 – 04 (new)	Verb-object names; naming checks; brief and detailed descriptions; step context.
4.42 (new) Typed Step Forms & Readiness	FR-DA-SFM-01 – 07 (new)	Form kinds; record tables with sources; decision matrix; completion effects; KPI and OBS pickers; validation and lock; readiness checklist.
4.43 (new) Documented Information	FR-DA-DOC-01 – 11 (new)	Template library; data-bound sections; documents required by the standards; generation; versions and approval; downloads; layout; tenant copies; retirement; links; seed.
5.1, 5.2, 5.3, 5.5, 5.8	NFR-DA-PERF-09 – 10, SEC-17 – 18, REL-07, UX-11 – 12, MAINT-09 (new)	Search and restore times; encrypted Organization key; rights-safe search; complete versions; keyboard search; step descriptions; configuration data.
6 Data Model	16 entities	Function, OrgRole, RoleFunction, RoleAssignment, ProcessElement(Version), DesignRelease, RecordDesignBinding, PromptSpecification, PromptFieldVersion, AIUseCaseLink, StepForm(Field), ReadinessItem, DocumentTemplate, TemplateSection, Document(Version), DocumentLayout, AttachmentVersion, SearchIndexEntry.
7.1 User Interface	1 paragraph	Screens added by revision 1.5.
Appendix A	5 rows	Conformance of the 1.5 additions in the reference implementation.
Appendix G (new)	Change log	This change log.

Appendix H — Revision 1.6 Change Log

Revision 1.6 keeps every earlier requirement ID and is purely additive. Its requirements are application agnostic: they come from the second and third rounds of user feedback on that reference implementation, restated so that any application can meet them. The generated-document requirements of Sections 4.44 and 4.45 detail those of Section 4.43.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.6	Version raised to 1.6; purpose, scope and overview describe the revision; the feedback rounds 2 and 3 added to the references.
1.4 Definitions	12 terms added	Document Profile, Data Source Binding, Consistency Check, Section Override, Edit-through, Controlled List, Choice Library, Entry Origin, Record Link, Register Propagation, Row Editor, Engine Transparency.
2.2, 2.3	2 principles, 2 functions	One truth, many documents; usable before complete. Consistent documents with full CRUD; usable record capture with links and propagation.
4.4 AI Use Case Library	FR-DA-AI-19 – 20 (new)	Model capability profiles; plain-language connection test.
4.19 Navigation Shell	FR-DA-NAV-14 – 15 (new)	Search bar in the header of every screen; menu entries for managed catalogs.
4.40 AI Prompt Specification	FR-DA-AIP-12 – 16 (new)	Engine Transparency; useful built-in answers; clean inputs; insertion in the output field; multi-row suggestions with Entry Origin.
4.44 (new) Document Consistency & Content	FR-DA-DGC-01 – 13 (new)	Single source; trace and data date; fallback and notices; Document Profiles; finest grain; procedures with BPMN, SIPOC and Go / No-Go; Consistency Checks and findings; change detection; cross-document consistency; layout; languages; master list.
4.45 (new) Document Full CRUD	FR-DA-DCR-01 – 10 (new)	Documents; versions with compare and restore; sections; Edit-through; Section Overrides; regeneration keeping manual text; templates; layout; permissions and audit; entry points for customization.
4.46 (new) Data Entry Usability	FR-DA-DEU-01 – 06 (new)	Qualified titles; Controlled Lists with Custom; Row Editor; ID (name) pickers; explained fields; unsaved input kept.

Section	Requirement IDs / Type	Change
4.47 (new) Linked Records & Propagation	FR-DA-LNK-01 – 05 (new)	Many-to-many links; Manual, Library and AI entry; Choice Libraries; Register Propagation; consistency of steps, registers, pickers and documents.
4.48 (new) Audits & Graded Findings	FR-DA-AFP-01 – 05 (new)	Frequency list with Custom; audit plan; grading; detailed nonconformity record; audit report.
5.1, 5.2, 5.3, 5.5, 5.8	NFR-DA-PERF-11, SEC-19, REL-08, UX-13 – 16, MAINT-10 (new)	Document and check times; rights-safe documents; reproducible generation; scrollable dialogs; ID (name); computed fields; field sizes; configuration data.
6 Data Model	12 entities	DocumentProfile, DocumentSection, DataSourceBinding, SectionOverride, DocumentSourceTrace, ConsistencyFinding, ChoiceList / ChoiceValue, RecordLink, EntryOrigin, ModelProfile, AuditPlan, Finding.
7.1 User Interface	1 paragraph	Screens added by revision 1.6.
Appendix A	6 rows	Conformance of the 1.6 additions in the reference implementation.
Appendix H (new)	Change log	This change log.

Appendix I — Revision 1.7 Change Log

Revision 1.7 keeps every earlier requirement ID and is purely additive. Its requirements come from the AI Value Graphical Chart & Brand Guidelines 1.2 (5 October 2026), restated so that any application can meet and test them. The document has also been re-set in the AI Value style and given a generated table of contents.

Revision 1.7 also absorbs the (New) SRS revision 1.7, which states the fourth-round user feedback and the same revision-1.7 content under different section numbers, and the Universal Field-Typing Fix. The rows after the first block below record that absorption; Appendix I.1 gives the numbering reconciliation.

Section	Requirement IDs / Type	Change
Cover, 1.1, 1.2, 1.5, 1.6	Version 1.7	Version raised to 1.7 (October 2026); purpose, scope and overview describe the revision; the Graphical Chart 1.2 added to the references.
Whole document	Presentation	Re-set in the AI Value style: Montserrat headings, Open Sans 11-point body in Ink, Navy and Azure accents, Navy table headers with alternating rows, 2-centimeter margins; generated, hyperlinked table of contents.
1.4 Definitions	11 terms added	Design Token, Status Scale, Component State, Inline Editable Table, Master–Detail, Focus Contract, Layout Panel, Density Mode, Command Palette, KPI Tile, System State.
2.2 Design Principles	1 principle	Evidence-first, light and precise.
4.19 Navigation Shell	FR-DA-NAV-16 – 20 (new)	Top bar; menu items; breadcrumbs; Command Palette; tabs and segmented controls.
4.49 (new) Multi-Record Data Entry	FR-DA-MRE-01 – 14 (new)	Pattern by number of fields; table layout; row status; keyboard map; row actions with Undo; carry-over and lists; paste from a spreadsheet; validation on blur; per-row autosave; feedback and count; column control; bulk operations; toolbar and footer; patterns not to use.
4.50 (new) Resizable Layout Panels	FR-DA-PNL-01 – 08 (new)	Resizable side panels; limits; drag handles; keyboard and ARIA; reset and collapse; server-side preference; mobile; right-to-left.
4.51 (new) Charts & Data Visualization	FR-DA-VIZ-01 – 08 (new)	Chart types; series colors; labels, legend and caption; responsive, loading and empty; KPI Tiles; heatmaps; export; accessibility and right-to-left.

Section	Requirement IDs / Type	Change
4.52 (new) Feedback & System States	FR-DA-STA-01 – 07 (new)	Loading; empty; error; success; notifications; offline; permission.
5.5 Usability & Accessibility	NFR-DA-UX-17 – 19 (new)	Screen readers; color never alone; brand color contrast.
5.6 Localization	NFR-DA-I18N-03 – 06 (new)	Text expansion; full mirroring; locale formats; no text in images.
5.13 (new) Visual Design System	NFR-DA-VDS-01 – 21 (new)	Tokens; light surfaces and gradient; text colors; Status Scale; fonts; type scale; spacing and grid; radii; elevation; icons; component states and buttons; fields and forms; dropdowns and controls; dialogs, drawers and tooltips; tables and density; motion; breakpoints; voice; logo; documents and exports; shared design system.
5.14 (new) Focus & Input Integrity	NFR-DA-FOC-01 – 05 (new)	Focus Contract; no input lost; focus ring and keyboard; code review rules; release tests.
6 Data Model	2 entities	UiPreference, TableViewPreference.
7.1 User Interface	1 paragraph	Screens and components added by revision 1.7.
Appendix A	8 rows	Assessment status of the reference builds and alignment between this standard and the Graphical Chart.
Appendix I (new)	Change log	This change log.
Appendix J (new)	Reference	Design token reference: colors, contrast, typography, spacing, shape, elevation, motion, breakpoints, panels and layers.
Appendix K (new)	Checklist	UI release checklist.
Cover	Document control	Supersedes line states that this document contains and extends the (New) SRS revision 1.7, whose IDs remain valid as aliases.
1.1, 1.2, 1.5, 1.6	Text	Purpose, scope, references and overview describe the fourth-round feedback, the absorption of the (New) SRS and the Universal Field-Typing Fix.
1.4 Definitions	10 terms added; 2 synonyms	Project Template Blueprint, Custom Element, Reporting Plan, Document Formatting, Section Formatting, Picture Section, Page Size (from the (New) SRS); Active Draft, Input-Integrity Defect, Typeable Field Inventory (from the Universal Field-Typing Fix). Density Mode also called Density; Layout Panel also called Resizable Panel.

Section	Requirement IDs / Type	Change
4.0 (new)	Reconciliation	Section-number reconciliation with the (New) SRS revision 1.7 (Table 4.0-1).
4.33 Project Template Catalog	FR-DA-PTC-07, 08 (new)	Blueprint view from project creation; organization templates offered first. Wording of the (New) SRS, the reference to its Section 4.49 restated as FR-DA-PTB-01 to 10. Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
4.33 Project Template Catalog	FR-DA-PTB-01 – 10 (new here, same IDs as New 4.49)	Project template blueprints. FR-DA-PTB-05 and 06 carry the exact wording of the (New) SRS; FR-DA-PTB-01 – 04 and 07 – 10 are stated from the fourth-round feedback and the content summary of the (New) SRS and are to be confirmed against its text. Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
4.43 Documented Information	FR-DA-DFP-01 – 08 (new IDs)	Document formatting and pictures, previously within FR-DA-DOC-06, 07; FR-DA-DFP-05, 06 carry the exact wording of the (New) SRS (New 4.50). Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
4.49 Multi-Record Data Entry	Note; FR-DA-MRE-01 – 08 of New (alias)	Cross-reference to New 4.49; New's FR-DA-MRE-01 – 08 are a subset of FR-DA-MRE-01 – 14, which govern (New 4.52). Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
4.50 Resizable Layout Panels	FR-DA-PNL-01, 08 (reworded); FR-DA-PNL-01 – 05 of New (alias)	Cross-reference to New 4.50. FR-DA-PNL-01 takes the wording of the (New) SRS (drag or keyboard, limits, reset by double-click). The RTL wording of New's FR-DA-PNL-05 is added to FR-DA-PNL-08, which already covered right-to-left mirroring; FR-DA-PNL-05 (reset and collapse) is kept. New's FR-DA-PNL-01 – 05 are a subset of FR-DA-PNL-01 – 08 (New 4.53). Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
4.51 Charts & Data Visualization	FR-DA-TBL-01 – 07 (new IDs)	Data tables and pagination, previously within FR-DA-MRE-11 – 13; FR-DA-TBL-01 carries the exact wording of the (New) SRS (New 4.51). Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
4.51, 4.52	Notes	Cross-references to New 4.51 and New 4.52.

Section	Requirement IDs / Type	Change
4.53 (new)	Alias section	Pointer to Section 4.50 and FR-DA-NAV-16 – 20; no requirement.
5.1 Performance	NFR-DA-PERF-12 (new)	Page Size rendering; 1,500-step blueprint opened in under 1 s on the server and 2 s on screen. Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
5.2 Security	NFR-DA-SEC-20 (new)	PNG and JPEG uploads up to 5 MB, generated per-tenant names, validated paths, formatting values checked against allowed lists. Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
5.3 Reliability	NFR-DA-REL-09, 10 (new)	A page never blank; inline data entry never loses a row. Source: fourth round of user feedback (1 October 2026), via the (New) SRS revision 1.7.
5.14 Focus & Input Integrity	NFR-DA-FOC-06 – 13 (new)	Field identity; primitive and controlled values; validation and save never reset a draft; stale asynchronous results discarded; draft precedence; event order; fixes not accepted; remediation audit. Source: Universal Field-Typing Fix.
6 Data Model	9 entities; Table 6-1	PanelWidths, DesignTokenSet, ProjectTemplateBlueprint, CustomElement, TemplateListRow, ProjectBlueprintTrace, DocumentFormatting, SectionFormatting, DocumentPicture; entity-alias table; TableViewPreference also called TablePreference.
7.1, 7.2	1 paragraph; 2 endpoint families	Screens of the (New) SRS mapped to this document; document formatting and pictures, and data tables and pagination preferences endpoints.
Appendix A	4 rows	ID aliases; new ID families; fourth-round feedback status; input-integrity audit status.
Appendix I.1 (new)	Reconciliation	Revision 1.7 numbering reconciliation.
Appendix K	1 row	Input integrity release check.
Appendix L (new)	Catalog and audit	Input-integrity defect catalog (C1 – C23), remediation audit and report. Source: Universal Field-Typing Fix.

I.1 Revision 1.7 numbering reconciliation

The (New) SRS revision 1.7 and this document describe the same revision-1.7 content under different section numbers. Where they differ, this document's numbering governs. New's IDs remain valid as aliases; no requirement of the New file is absent from this document.

Table I.1-1 — Revision 1.7 numbering reconciliation

New SRS section	Topic	Section in this document	Requirement IDs
4.49	Project Template Blueprints	4.33 (cross-reference at 4.49)	FR-DA-PTC-07, 08; FR-DA-PTB-01 – 10
4.50	Document Formatting & Pictures	4.43 (cross-reference at 4.50)	FR-DA-DFP-01 – 08 (with FR-DA-DOC-06, 07)
4.51	Data Tables & Pagination	4.51, with 4.49	FR-DA-TBL-01 – 07 (with FR-DA-MRE-11 – 13)
4.52	Inline Multi-Record Entry	4.49	FR-DA-MRE-01 – 14
4.53	Layout Panels & Navigation Robustness	4.50 (alias Section 4.53)	FR-DA-PNL-01 – 08; FR-DA-NAV-16 – 20

Appendix J — Design Token Reference

The values below are those of the AI Value Graphical Chart 1.2, §3 to §6, §17 and §22. Components reference the token name, never the value (NFR-DA-VDS-01); the CSS block of the Graphical Chart §17 is the file to copy into an application unchanged.

J.1 Colors

Table J-1 — Core palette and tints

Swatch	Token	Name	Hex	Usage
	`--aiv-navy`	Deep Navy	#123A5F	Headings, primary text, dark fills, table headers
	`--aiv-navy-dark`	Navy Dark	#0D2A47	Pressed states, deep backgrounds
	`--aiv-azure`	Azure	#1876C6	Primary action, links, active states, primary series
	`--aiv-green`	Green	#28C87C	Success, call to action, growth
	`--aiv-teal`	Teal	#17A2B8	Overlay accent, secondary metric
	`--aiv-ink`	Ink	#2C3E50	Body text
	`--aiv-muted`	Muted	#5A6B7B	Secondary text, captions, axis labels
	`--aiv-white`	White	#FFFFFF	Surfaces
	`--aiv-bg`	Background	#F5F8FB	Page background, alternating rows
	`--aiv-line`	Line	#E1E8F0	Borders, dividers, gridlines
	`--aiv-azure-tint`	Azure tint	#E8F1FB	Callouts, icon badges, row hover
	`--aiv-green-tint`	Green tint	#E7F9F0	Success callouts, positive metric cards
	`--aiv-teal-tint`	Teal tint	#E6F6F8	Secondary icon backgrounds

Table J-2 — Status Scale (always in this order)

Swatch	Token	Level	Hex	Meaning
	`--aiv-status-1`	1 — Critical	#F4C7C3	Blocker / failing
	`--aiv-status-2`	2 — Emerging	#FBE0B5	At risk
	`--aiv-status-3`	3 — Developing	#FFF3B0	Progressing
	`--aiv-status-4`	4 — Managed	#D9EAD3	On track
	`--aiv-status-5`	5 — Optimized	#B6D7A8	Leading

Table J-3 — Contrast on white and permitted text use

Color	Contrast on white	Text use
Navy #123A5F	12.6 : 1	Any text
Ink #2C3E50	10.9 : 1	Any text
Muted #5A6B7B	5.1 : 1	Any text (AA)
Azure #1876C6	3.9 : 1	Large bold text and interface components only
Teal #17A2B8	2.9 : 1	Fills only
Green #28C87C	2.2 : 1	Fills only

Chart series: primary or current in solid Azure; comparison or prior period in dashed Muted; growth in solid Green; overlay in solid Teal; gridlines in Line; axis labels in Muted (FR-DA-VIZ-02). The gradient linear-gradient(90deg, #1876C6 0%, #17A2B8 50%, #28C87C 100%) is reserved for the logo swoosh and one hero element per page.

J.2 Typography

Table J-4 — Type hierarchy (sizes in pixels)

Element	Font	Weight	Size	Color and use
Display	Montserrat	900	48	Navy — hero only
H1	Montserrat	900	36	Navy — page titles
H2	Montserrat	800	30	Navy — section titles
H3	Montserrat	800	24	Navy — sub-sections
H4	Montserrat	700	20	Navy — card titles
Body	Open Sans	400	16	Ink
Body small	Open Sans	400	14	Ink
Caption	Open Sans	400	12	Muted
Label	Open Sans	600	13	Navy — form labels
Button	Open Sans	600	14	Per variant
Eyebrow	Open Sans	700	12	Uppercase, tracked (see Appendix A)
Data	JetBrains Mono	400	12–14	Navy or Ink, tabular figures

Table J-5 — Font fallbacks by platform

Platform	Primary	Fallback
Web application	Montserrat / Open Sans	sans-serif
Word and Outlook documents	Montserrat / Open Sans	Calibri
Power BI reports	Montserrat / Open Sans	Segoe UI
Teams and SharePoint	Segoe UI	system-ui
HTML email	Open Sans	Helvetica, Arial, sans-serif

J.3 Spacing, shape, elevation, motion and layout

Table J-6 — Other tokens

Family	Token	Value
Spacing	`--aiv-space-0 … -9`	0 · 4 · 8 · 12 · 16 · 24 · 32 · 48 · 64 · 96 px
Radius	`--aiv-radius-btn / -input / -card / -modal`	8 · 8 · 12 · 16 px (badges 6 px or circle)
Elevation	`--aiv-shadow-card`	0 2px 8px rgba(18,58,95,0.06)
Elevation	`--aiv-shadow-hover`	0 8px 28px rgba(18,58,95,0.12)
Elevation	`--aiv-shadow-modal`	0 20px 60px rgba(18,58,95,0.18)
Focus	`--aiv-focus-ring / -offset`	0 0 0 3px rgba(24,118,198,0.35) · 2px
Fonts	`--aiv-font-heading / -body / -mono`	'Montserrat' · 'Open Sans' · 'JetBrains Mono', with sans-serif / monospace fallback
Motion	`--aiv-duration-fast / -base / -slow`	150 · 200 · 300 ms
Motion	`--aiv-easing-standard / -emphasis`	cubic-bezier(0.2,0,0,1) · cubic-bezier(0.3,0,0,1.2)
Panels	`--aiv-panel-left-default / -min / -max`	260 · 180 · 480 px
Panels	`--aiv-panel-right-default / -min / -max`	320 · 220 · 560 px
Panels	`--aiv-panel-center-min`	400 px
Row status	`--aiv-row-hover / -dirty / -saved / -error`	Azure tint · Azure · Green · Status 1
Breakpoints	`--aiv-bp-sm / -md / -lg / -xl / -2xl`	480 · 768 · 1024 · 1280 · 1536 px
Layers	`--aiv-z-base / -sticky / -dropdown / -drawer / -modal / -toast / -tooltip`	0 · 10 · 100 · 200 · 300 · 400 · 500

Appendix K — UI Release Checklist

Before a release of an application, the product owner confirms each line below for the screens and documents the release changes, and records the result with the release notes (NFR-DA-VDS-21).

Area	Check	Requirements
Tokens	No raw color, size or spacing value outside the token file; the CI token check passes.	NFR-DA-VDS-01
Color and contrast	Body text in Ink or Navy; no Azure, Teal or Green text below 18.66 px bold; automated accessibility scan clean.	NFR-DA-VDS-03, NFR-DA-UX-06
Status	Status Scale only, in fixed order, always with a label.	NFR-DA-VDS-04, NFR-DA-UX-18
Typography	Montserrat, Open Sans and JetBrains Mono served with the application; sizes from the scale.	NFR-DA-VDS-05, -06
Shape and elevation	Radii 8 / 12 / 16 px; no pill buttons; card border or shadow, never both.	NFR-DA-VDS-08, -09
Icons	Lucide line icons only; no emoji; icon-only buttons named.	NFR-DA-VDS-10, NFR-DA-UX-17
Component states	Hover, focus-visible, active, disabled, loading, error and success designed.	NFR-DA-VDS-11
Record tables	Pattern chosen by field count; keyboard map, row status, autosave, Undo, paste from spreadsheet.	FR-DA-MRE-01 – 14
Focus	Focus kept while typing and on dropdown change; focus tests pass; no new console warning.	NFR-DA-FOC-01 – 05
Input integrity	Remediation audit of Appendix L recorded; every cause C1 – C23 checked; no fix of NFR-DA-FOC-12 used.	NFR-DA-FOC-06 – 13
Panels	Side panels resizable by mouse and keyboard; widths kept; disabled below 768 px.	FR-DA-PNL-01 – 08
Charts	Series colors, caption, legend below, no pie or 3D, export, data-table view.	FR-DA-VIZ-01 – 08
System states	Loading, empty, error, success, offline and permission states designed.	FR-DA-STA-01 – 07
Responsive	Layout checked at 390, 768, 1024 and 1280 px.	NFR-DA-VDS-17, NFR-DA-UX-08
Languages	English, French and Arabic checked; full mirroring under Arabic; no clipped label.	NFR-DA-I18N-03 – 05
Motion	Durations 150–300 ms; reduced motion respected.	NFR-DA-VDS-16

Area	Check	Requirements
Copy	Evidence-first microcopy; brand and product names written exactly.	NFR-DA-VDS-18
Documents	Generated documents and exports in the AI Value style, fonts embedded, A4 or Letter.	NFR-DA-VDS-20
Version	Release notes state the Graphical Chart version conformed to.	NFR-DA-VDS-21
Full-screen entry	Every Multi-Entry opens full screen; zoom by Ctrl/Cmd + scroll, slider, Zoom in, Zoom out, Fit and Reset; add, save, change and delete work there; “Return to page” and Escape return with nothing lost and focus restored.	FR-DA-MRE-15 – 18, NFR-DA-UX-20
Paste of records	Records copied from Word, Excel, a CSV file and a web page paste as rows and columns; Paste Preview when columns differ; conversion, validation and Undo; markup discarded and formulas neutralized; copy out with column titles.	FR-DA-MRE-19 – 23, NFR-DA-SEC-21
AI use cases	Each use case opens on its page from the library; created and edited in its form with its step and links; coverage view lists tasks without AI; quota shown.	FR-DA-AI-01, 13, 21, 22, 25
Governed AI	No AI action where no active use case serves the place; every proposal shows tier, use case, engine, reason, sources and prompt; each outcome logged.	FR-DA-AI-07, 23, 24, 26
Multi-Entry errors	No error on a new empty row; errors after a change; Issues list and “Rows with errors” filter; retry per row and for all.	FR-DA-MRE-08, 24
Multi-Entry saving	Save state always shown; Save only when pending; draft restored after a refresh; conflict offered on a stale save; offline queue retried.	FR-DA-MRE-25 – 27, NFR-DA-REL-10, 11
Large tables	10,000 rows scroll and type without delay; batches with progress, Cancel and Resume.	FR-DA-MRE-28, 29, NFR-DA-PERF-14
Import and export	Five-stage import with template, checks, mapping, repair and Undo; CSV and workbook export of the view or selection.	FR-DA-MRE-30, 31, NFR-DA-SEC-22
Leaving and phones	Leave guard with three choices; no question when returning from full screen; phone list with one-by-one editing; no horizontal scrolling with many columns.	FR-DA-MRE-32 – 34

Area	Check	Requirements
Undo and keys	Ctrl/Cmd + Z and Shift + Z on every change; fill down; Page Up/Down; Ctrl/Cmd + A; new rows stay in place while sorted.	Table 4.49-1, FR-DA-MRE-11, 35
Office files and upgrade	Generated Word, Excel and PowerPoint files open without repair; an upgrade migrates the data in place.	FR-DA-DOC-12, NFR-DA-MAINT-11

Appendix L — Input-Integrity Defect Catalog & Remediation Audit

This appendix restates the Universal Field-Typing Fix as the reference for NFR-DA-FOC-04 and NFR-DA-FOC-06 to 13. Table L-1 is the catalog of known causes of Input-Integrity Defects, with the correction each requires; Table L-2 is the audit a remediation pass follows; Table L-3 is the report it produces. The causes are written for a component framework such as React; for another framework, the audit applies the equivalent rules of component identity, lifecycle and state and records the translation.

Table L-1 — Causes of Input-Integrity Defects

Cause	Description	Required correction	Requirement
C1	Component defined inside another component (field, row, editor, dialog, panel, card, section or list item).	Move the component to module scope.	NFR-DA-FOC-04
C2	Unstable list, row or field key: array index, random or time value, serialized object, or a value that changes while the user types.	Use the stable record identifier; for form-library field arrays, the library's field identifier.	NFR-DA-FOC-04, 06
C3	Focus effect with no dependency list, or with dependencies that change on every keystroke or render.	Make genuine autofocus a one-time mount behavior or guard it on the active element; never refocus after renders.	NFR-DA-FOC-04
C4	Dialog, drawer, panel, wizard step, editor or form re-created on every render.	Hoist definitions, use stable keys and stable mount conditions; preserve identity while editing.	NFR-DA-FOC-06
C5	Form-library misuse: index keys in field arrays, arrays rebuilt or fields re-registered on render, unstable defaults or names, needless resets.	Use the library's stable field identifiers; preserve field identity and defaults; never reset while typing.	NFR-DA-FOC-10
C6	Controlled input receives an object or array as its value.	Pass the primitive representation without changing the public data shape; do not stringify to silence the problem.	NFR-DA-FOC-07
C7	Element type changes while in use (single-line and multi-line, select and input, picker and text).	Keep the same element; change attributes; defer a required type change to blur.	NFR-DA-FOC-04
C8	Context provider value re-created on every render.	Memoize the provider value with the project's existing mechanism.	NFR-DA-FOC-04
C9	Memoized child receives a callback re-created on every render.	Stabilize the callback only where its identity contributes to the defect.	NFR-DA-FOC-10

Cause	Description	Required correction	Requirement
C10	Ref-based autofocus runs again after the first display.	Make autofocus one-shot.	NFR-DA-FOC-04
C11	Validation or save effect calls focus or blur.	Remove the focus or blur side effect; no automatic refocus in its place.	NFR-DA-FOC-08
C12	Row key contains an edited or computed value.	Use the stable row identifier, unchanged while the row is edited.	NFR-DA-FOC-06
C13	Field name or identifier changes while focused.	Keep name and identifier stable for the field's lifetime; change them only at creation or remount.	NFR-DA-FOC-06
C14	Form region re-rendered through a changing key or injected HTML.	Remove the identity-changing behavior; render the form normally.	NFR-DA-FOC-06
C15	Route, tab, panel or wizard switch remounts an active form.	Preserve the form with the existing keep-alive, persistence or overlay mechanism; no new dependency; no routing change.	NFR-DA-FOC-10
C16	Validation or normalization on every keystroke rewrites the value or moves focus.	Validate on blur or at the existing lifecycle point; never overwrite the typed value.	NFR-DA-FOC-08
C17	Autosave, save or synchronization clears, resets or reinitializes the active form.	Save without touching the Active Draft; reset only on close, cancel, discard or save-and-close; keep autosave.	NFR-DA-FOC-08
C18	Field conditionally unmounted while typing (on its own value, another typed value or validation state).	Keep the field mounted and hide it where needed, unless existing behavior requires removal of an empty, unfocused field.	NFR-DA-FOC-06
C19	Asynchronous race: a stale result overwrites newer input.	Use cancellation, abort signals, request identifiers or sequence checks; only the current operation updates the field.	NFR-DA-FOC-09
C20	Debounce or throttle re-created on every render or keystroke.	Keep one instance for the component lifetime; cancel pending work on intentional unmount.	NFR-DA-FOC-09
C21	Parent, derived, subscribed or server state overwrites the Active Draft.	Preserve the draft while editing; synchronize at a lifecycle boundary.	NFR-DA-FOC-10
C22	Input switches between controlled and uncontrolled.	Keep it consistently controlled or uncontrolled; normalize nullable values.	NFR-DA-FOC-07

Cause	Description	Required correction	Requirement
C23	Blur, focus, pointer, click and change events in the wrong order cause a premature commit, reset, close or unmount.	Correct the event order so the interaction completes; no generic refocus; no global blur suppression.	NFR-DA-FOC-11

Scope of a remediation pass. A remediation pass is a focused correction of Input-Integrity Defects. It does not redesign the application, refactor unrelated code, change public component interfaces, routes, API contracts, database schemas or data shapes, or add dependencies. It adds no tests, lint rules or continuous-integration checks of its own; the release tests of NFR-DA-FOC-05 remain required before the release, and the existing test suite, linter and type checker are run on the result. A cause is never assumed present: every fix answers a confirmed instance, and an implementation that is already correct is left unchanged. The fixes listed in NFR-DA-FOC-12 are never used.

Table L-2 — Remediation audit

Phase	Work
1. Understand	Identify framework, language, UI, form, routing, state, data-fetching and debounce libraries, custom input components and coding conventions; translate C1 – C23 to the framework in use.
2. Inventory	Build the Typeable Field Inventory: every form, dialog, drawer, panel, wizard, record table, Row Editor, template editor, search and filter, and every input, select, rich-text, picker, combobox and schema-generated field, with the shared field components they use.
3. Audit causes	For each cause C1 – C23: search, confirm or reject, record every confirmed instance; report “Checked — no instance found” when none exists; report a problem outside the catalog as an out-of-catalog finding.
4. Trace components	For custom and third-party controls, trace the value from user input through the component, wrapper, form state, parent state and asynchronous state back to rendering, and locate where value or identity becomes unstable.
5. Fix minimally	Make the smallest correct change for each instance; preserve public props, routes, API contracts, data shapes, business logic, validation rules and visual design; add no dependency; perform no unrelated refactoring.
6. Work by directory	In a large repository, inventory, fix and re-scan one directory at a time, shared components first.
7. Re-scan	Re-check the touched areas for unstable keys, focus and blur calls in effects, components defined during rendering, field arrays, form resets, conditional fields, asynchronous writes, debounce creation, nullable controlled values and event ordering around editors and popovers.
8. Inspect the diff	Confirm the change is limited to input integrity; revert any unrelated or formatting-only change.

Table L-3 — Remediation report, in this order

#	Section	Content
1	Framework and architecture	Framework, language, UI library, form libraries, routing, state management and data libraries.
2	Typeable screen inventory	Screen or route, component, typeable fields, audited, findings, fixed.
3	Findings and fixes	File, component, screen type, cause, line fixed, fix applied — one row per instance.
4	Cause counts	For C1 – C23, including zero counts: instances found, fixed and deferred.
5	Totals	Confirmed instances, fixed, deferred and out-of-catalog findings.
6	Files touched	File, lines added and removed, net change, reason.
7	Deferred findings	File, line, cause, reason deferred, risk, proposed follow-up; or “No findings were deliberately deferred.”
8	Out-of-catalog findings	File, component, line, description, why it does not fit C1 – C23, recommended follow-up; or “No out-of-catalog findings.”
9	Causes not found	Each zero-instance cause listed as “Checked; no instance found.”
10	Final verification	Confirmation of the final re-scan, with any remaining suspicious pattern reported rather than claimed fixed.
11	Behavioral-scope confirmation	Confirmation that no behavior changed other than what corrects the typing, focus, cursor or input-integrity defects, or an exact explanation of why it cannot be given.

A remediation pass is complete only when every typeable screen and shared field component has been considered, C1 to C23 have all been checked, every confirmed instance has been fixed or explicitly deferred, every zero-count cause, changed file, deferred issue and out-of-catalog finding is reported, the final re-scan is done, and no fix of NFR-DA-FOC-12, new dependency, contract change, test, lint rule or unrelated refactoring has been introduced. The report states the actual result; it does not describe the application as fully fixed while confirmed instances remain unresolved.

Appendix M — Revision 1.8 Change Log

Revision 1.8 keeps every earlier requirement ID. It is editorial for applicability — every reference to a particular application, product family, reference build or source file has been removed or restated in general terms, without changing what any requirement asks — and additive for multi-record entry.

Section	Requirement IDs / Type	Change
Cover, header and footer	Title and version	Renamed Standard Application SRS, version 1.8 (October 2026), application-agnostic edition; product-family names removed; supersedes revision 1.7 and the (New) SRS revision 1.7.
1.1 Purpose	Rewritten	Purpose stated for any application; the history of revisions 1.1 to 1.7 restated without application names; revision 1.8 described.
1.2 Scope	Clarified; 1 paragraph	Out-of-scope statement made generic; revision 1.8 scope added.
1.3 Intended Audience	Clarified	Audience stated across the applications of an organization.
1.4 Definitions	1 reworded; 5 terms added	The product-family term for a conforming product replaced by Application. Added: Multi-Entry, Full-Screen Entry Workspace, Zoom Controls, Structured Paste, Paste Preview.
1.5 References	Rewritten	Product SRS documents and builds replaced by one non-normative source entry; the standards cited in the text listed (ISO/IEC/IEEE 29148, BPMN 2.0, WCAG 2.1 and 2.2, COSO).
1.6 Document Overview	2 paragraphs	Revision 1.8 described; FR-DA- and NFR-DA- prefixes kept as frozen identifiers.
2.3, 2.4, 3.2, 3.10, 4.31, 4.36, 6	Clarified	Domain examples, role mappings, the number of standard roles, the principal-record examples and FR-DA-QLT-11 restated generically.
2.3 Product Functions	1 function added	Full-screen multi-entry with zoom, paste of complete records and return by one action or Escape.
4 (introduction)	Rewritten	Provenance of each revision stated without application names; revision 1.8 requirements listed.
4.14 Multi-Language & Localization & Currency	FR-DA-I18N-09 (new)	The statement “the application should be multi-currency” formalized as a verifiable requirement.

Section	Requirement IDs / Type	Change
4.49 Multi-Record Data Entry	FR-DA-MRE-15 – 18 (new)	Full-Screen Entry Workspace from every Multi-Entry; Zoom Controls (Ctrl/Cmd + scroll or pinch, slider, Zoom in, Zoom out, Fit, Reset); every entry action available there (add, edit, Row Editor, duplicate, delete with Undo, Save all) on the same records; return by one action or Escape with nothing lost and focus restored.
4.49 Multi-Record Data Entry	FR-DA-MRE-19 – 23 (new); FR-DA-MRE-07 clarified	Paste of complete records from Word, Excel, CSV, web pages and other applications, as from Word into Excel; Paste Preview with column mapping; conversion, validation, single Undo and audit; limits, rights and quotas; copy out with column titles.
4.49, Table 4.49-1	8 rows added; 1 reworded	Open the workspace; zoom with scroll, plus and minus; reset; fit; copy; paste; undo. Escape also leaves the workspace when no cell is being edited.
5.1 Performance	NFR-DA-PERF-13 (new)	Workspace opening, zoom and paste times.
5.2 Security	NFR-DA-SEC-21 (new)	Pasted content treated as untrusted; markup discarded; formula injection neutralized; server-side checks on every pasted row.
5.5 Usability & Accessibility	NFR-DA-UX-20 (new)	Accessible workspace and Zoom Controls; browser zoom preserved; reduced motion; mobile; right-to-left.
5.13 Visual Design System	NFR-DA-VDS-18, 19 clarified	The list of product names and the product logo restated for any application.
6 Common Data Model	1 entity; 1 reworded	PasteOperation added; TableViewPreference holds the workspace zoom level and the last confirmed paste mapping.
7.1, 7.2 Interfaces	1 paragraph; 1 endpoint family	Workspace, Paste Preview and copy; bulk-entry preview, apply and undo endpoints.
Appendix A	Rewritten	Application-neutral conformance record (Table A-1) and resolution rules (Table A-2), including the workspace zoom, the order of the Escape key and the paste extension.
Appendices B to I	Clarified	Application and build names replaced by “reference implementation”; requirement content unchanged.
Appendix K	2 rows added; 1 reworded	Full-screen entry; paste of records; brand and product names.
Appendix M (new)	Change log	This change log.

Appendix N — Revision 1.9 Change Log

Revision 1.9 keeps every earlier requirement ID; nothing was removed or renumbered. It comes from the acceptance of a reference implementation and stays application agnostic. Amended requirements keep their ID and are restated in full in their section; the amendments only add obligations or narrow an exception.

Section	Requirement IDs / Type	Change
Cover, header, document control	Version	Version 1.9 (October 2026); supersedes revision 1.8.
1.1, 1.2, 1.6, 4	1 paragraph each	Revision 1.9 purpose, scope and overview; pointer to this appendix.
1.4 Definitions	9 terms (new)	AI Use Case Link, Coverage of the Process Design, Model Precedence, Draft Backup, Row Stamp, Edit Conflict, Batch Operation, Import Session, Leave Guard.
3.4 AI Governance Architecture	1 paragraph (new)	One governed path for every AI generation; model precedence.
4.4 AI Use Case Library	FR-DA-AI-01, 07, 13 (amended)	Stored derived fields and dedicated form; Proposed outcome and per-item logging with specification version, engine and model; several links and the coverage view.
4.4 AI Use Case Library	FR-DA-AI-21 – 26 (new)	Use case page; library figures and filters; every generation through a linked, active use case; model precedence; draft by default for new, duplicated and suggested use cases; review of multi-item proposals.
4.14 Localization	FR-DA-I18N-10 (new)	Language-independent references; translations kept on save.
4.40 Prompt Specification	FR-DA-AIP-07 (amended)	Seeded specifications 100 % complete, Examples built from the step form.
4.42 Typed Step Forms	FR-DA-SFM-08, 09 (new)	Form kind from the meaning of the step (Function × Level Matrix); reopening clears every completion mark.
4.43 Documents	FR-DA-DOC-12 (new)	Office files open without repair; validated automatically.
4.49 Multi-Record Data Entry	FR-DA-MRE-05, 08, 11, 14, 18, 22 (amended)	Undo period stated, Insert row below; validation only after a change; unsaved rows not moved by a sort; file import exempted from the no-wizard rule; return button named after its destination; server checks equal to the grid's.

Section	Requirement IDs / Type	Change
4.49 Multi-Record Data Entry	FR-DA-MRE-24 – 35 (new)	Issues list and error filter; save state and Discard; Draft Backup and Edit Conflicts; offline queue; virtual rows and optional pages; Batch Operations; five-stage import; export; Leave Guard; readability without horizontal scrolling; phone list; undo and redo of every change.
4.49, Table 4.49-1	7 rows (new); 1 row (amended)	Redo; fill down; left and right arrows; Page Up and Down; first and last row; select all; range selection; Ctrl/Cmd + Z for every change.
5.1, 5.2, 5.3	NFR-DA-PERF-14, NFR-DA-SEC-22, NFR-DA-REL-11 (new); NFR-DA-REL-10 (amended)	Performance with 10,000 rows and 1,000 imported rows; import as untrusted input; no overwrite of a newer version; no row lost after a crash.
5.5, 5.6	NFR-DA-UX-21, NFR-DA-I18N-07 (new)	ARIA grid pattern for every Multi-Entry; non-Latin text and file names kept intact.
5.7, 5.8	NFR-DA-PORT-05, NFR-DA-MAINT-11 (new); NFR-DA-MAINT-04 (amended)	Automated tests of multi-record entry run one suite at a time; in-place data migration on upgrade; no AI suggestion outside an AI Use Case.
6 Common Data Model	3 entities (new); 2 (amended)	ImportSession, BatchOperation, DraftBackup; AIUseCase and AIUseCaseLink extended.
7.1, 7.2	1 paragraph; 2 interface families	Screens of revision 1.9; AI use case and bulk import and export interfaces.
Appendices A, K	3 and 9 rows (new)	Conformance areas and release checks of revision 1.9.
Appendix N (new)	Change log	This change log.